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Hoffman et al.

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(54) **OPTIMAL SAMPLING WITH SOIL STRATIFICATION**

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Related U.S. Application Data

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(60) Provisional application No. 62/995,484, filed on Jan. 29, 2020, provisional application No. 62/995,674, filed on Feb. 7, 2020, provisional application No. 62/995,948, filed on Feb. 20, 2020, provisional application No. 63/100,545, filed on Mar. 17, 2020.

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G06F 11/30 (2006.01)

G01N 33/24 (2006.01)

G06Q 50/02 (2012.01)

(52) **U.S. Cl.**

CPC **G01N 33/24** (2013.01); **G06Q 50/02** (2013.01); **G01N 33/245** (2024.05)

(58) **Field of Classification Search**

CPC **G01N 33/24**
See application file for complete search history.

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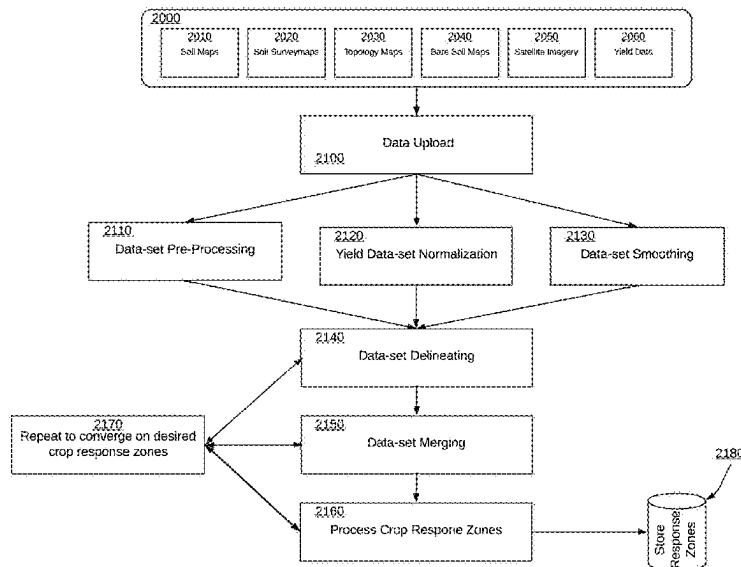
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(57) **ABSTRACT**

A system and method for determining optimal sampling parameters is described. The method gathers soil data from a soil data source, which is associated with a soil organic carbon (SOC) project area. The crop prediction engine then determines that the soil data is less than optimal, but that the soil data is sufficient to generate an optimal sampling plan. The method completes a Monte Carlo simulation, which generates an empirical sampling distribution. The optimal sampling plan is determined by defining a margin of error, which provides a deviation from a predictive analysis of measured soil chemistry for a plurality of collected soil samples, and performing Monte Carlo simulations that include one Monte Carlo simulation having a lowest sampling density that satisfies the margin of error. The optimal sampling plan has the lowest sampling density and includes one or more sampling locations.

20 Claims, 26 Drawing Sheets



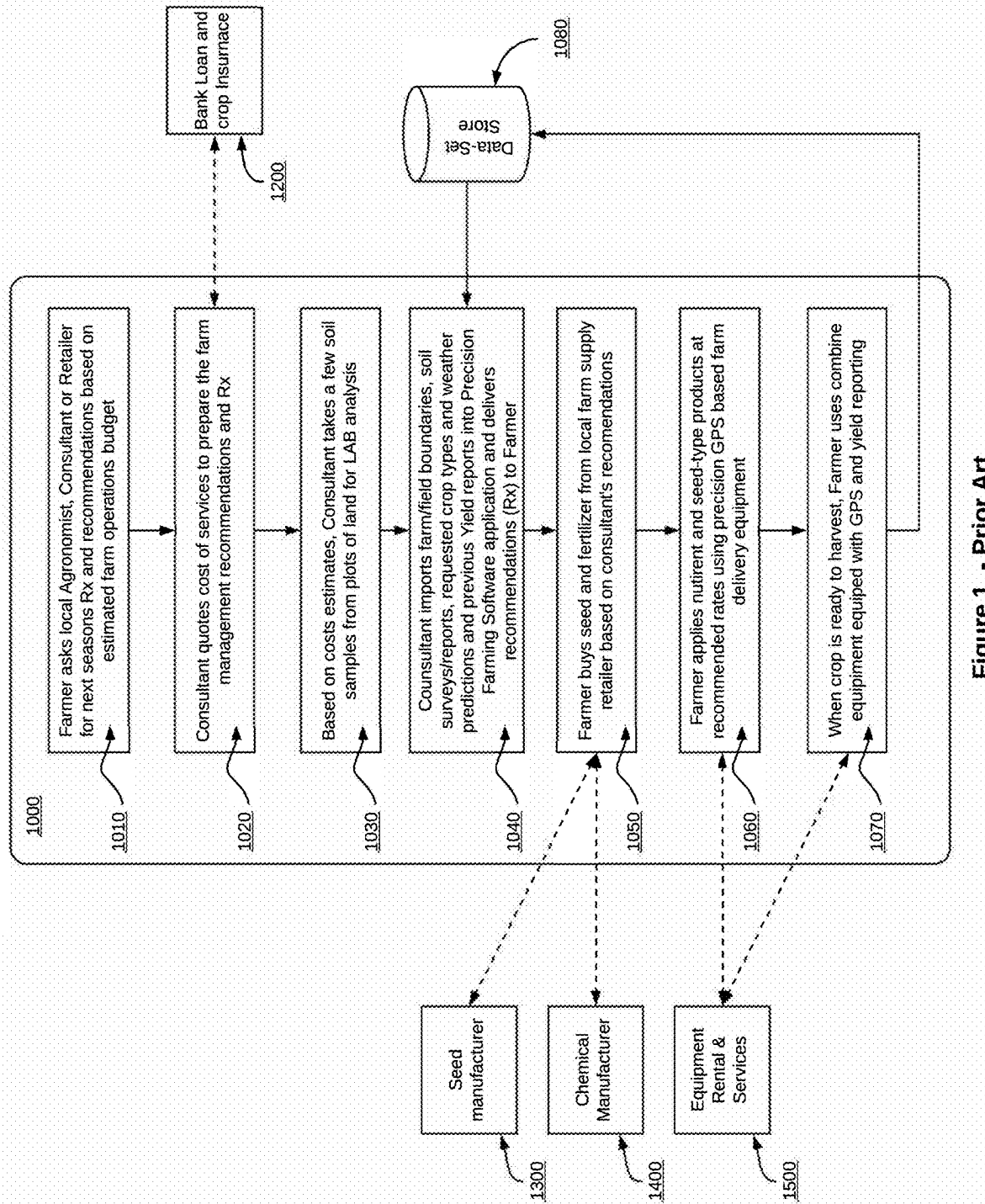
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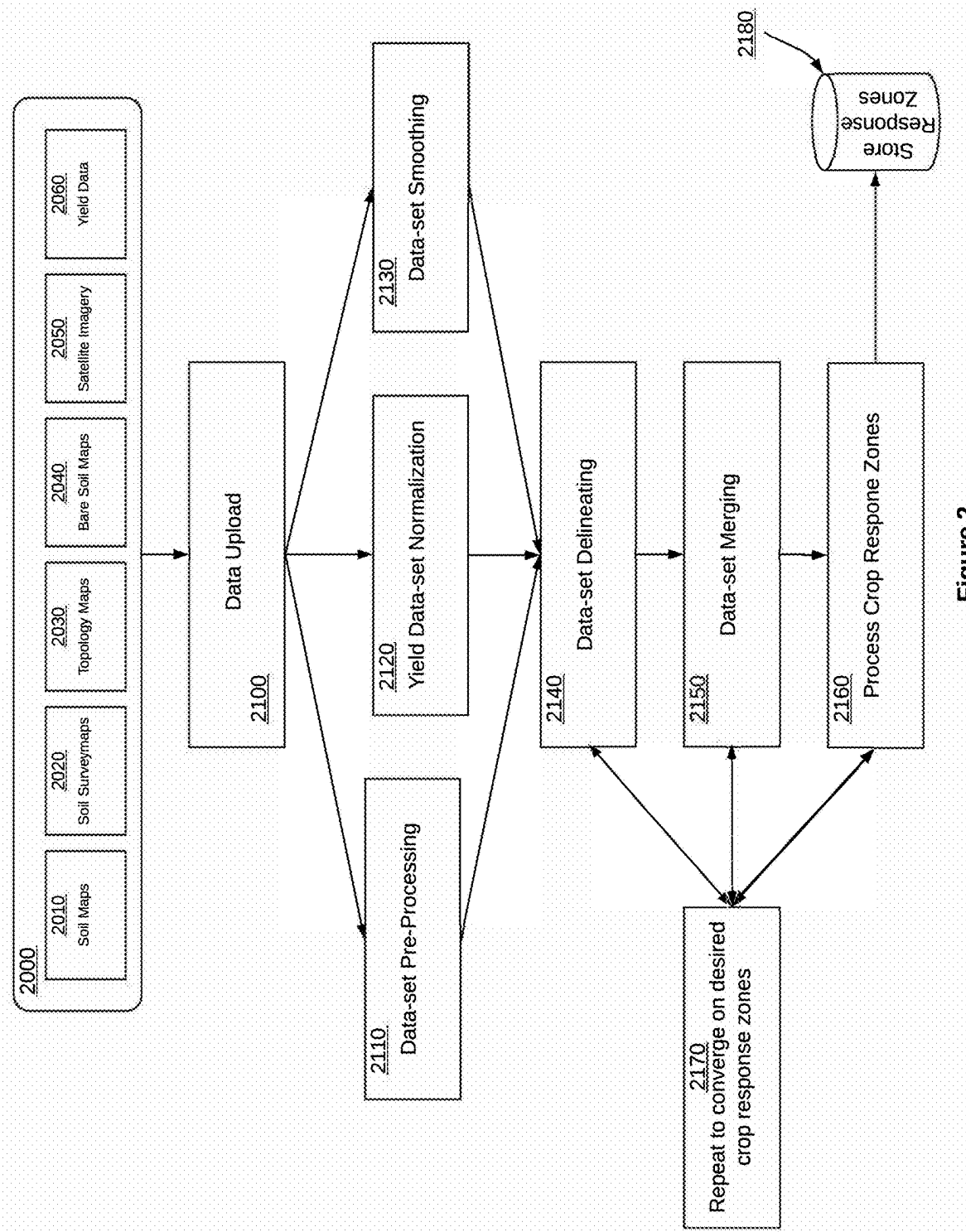


Figure 2

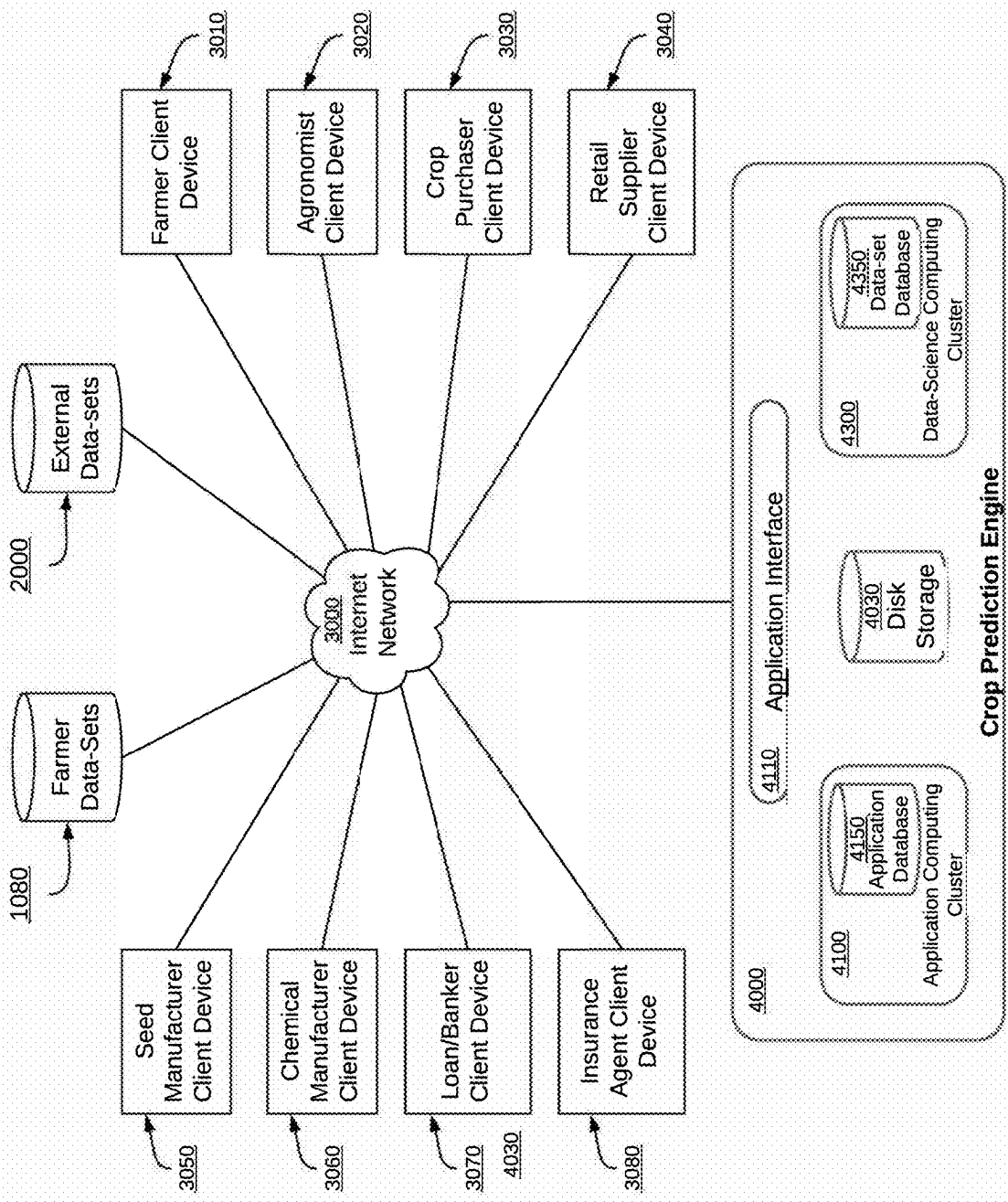


Figure 3a

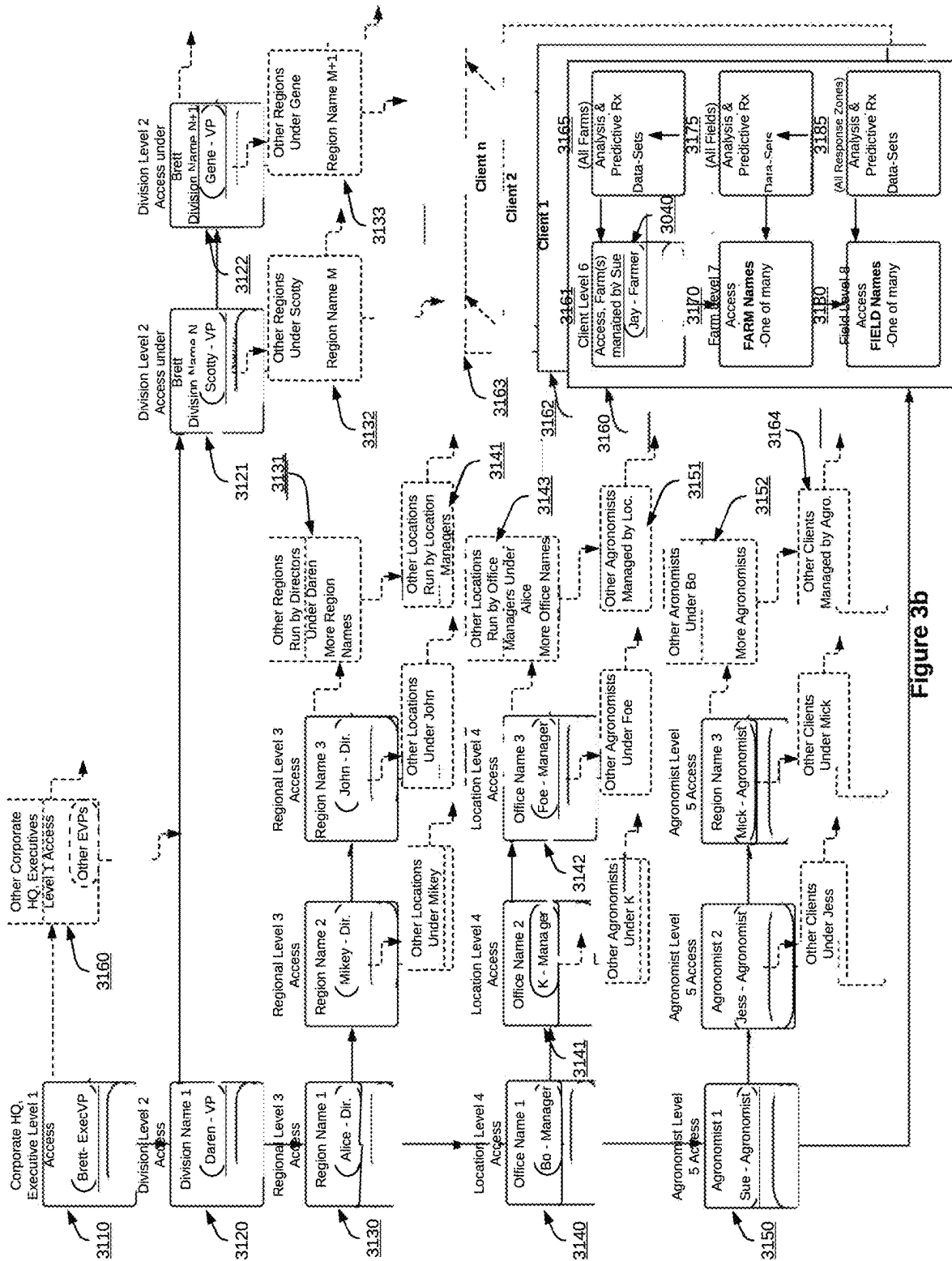


Figure 3b

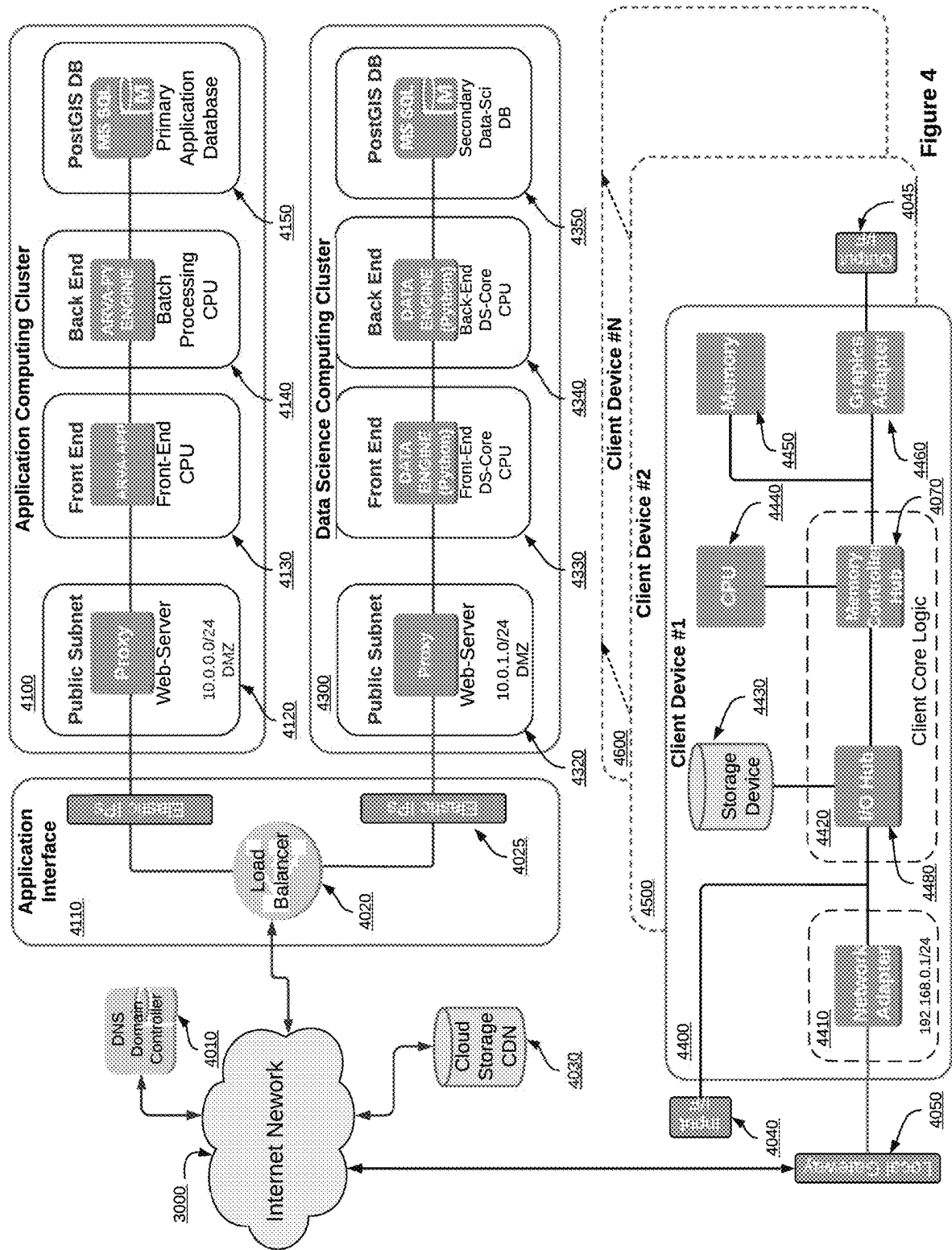


Figure 4

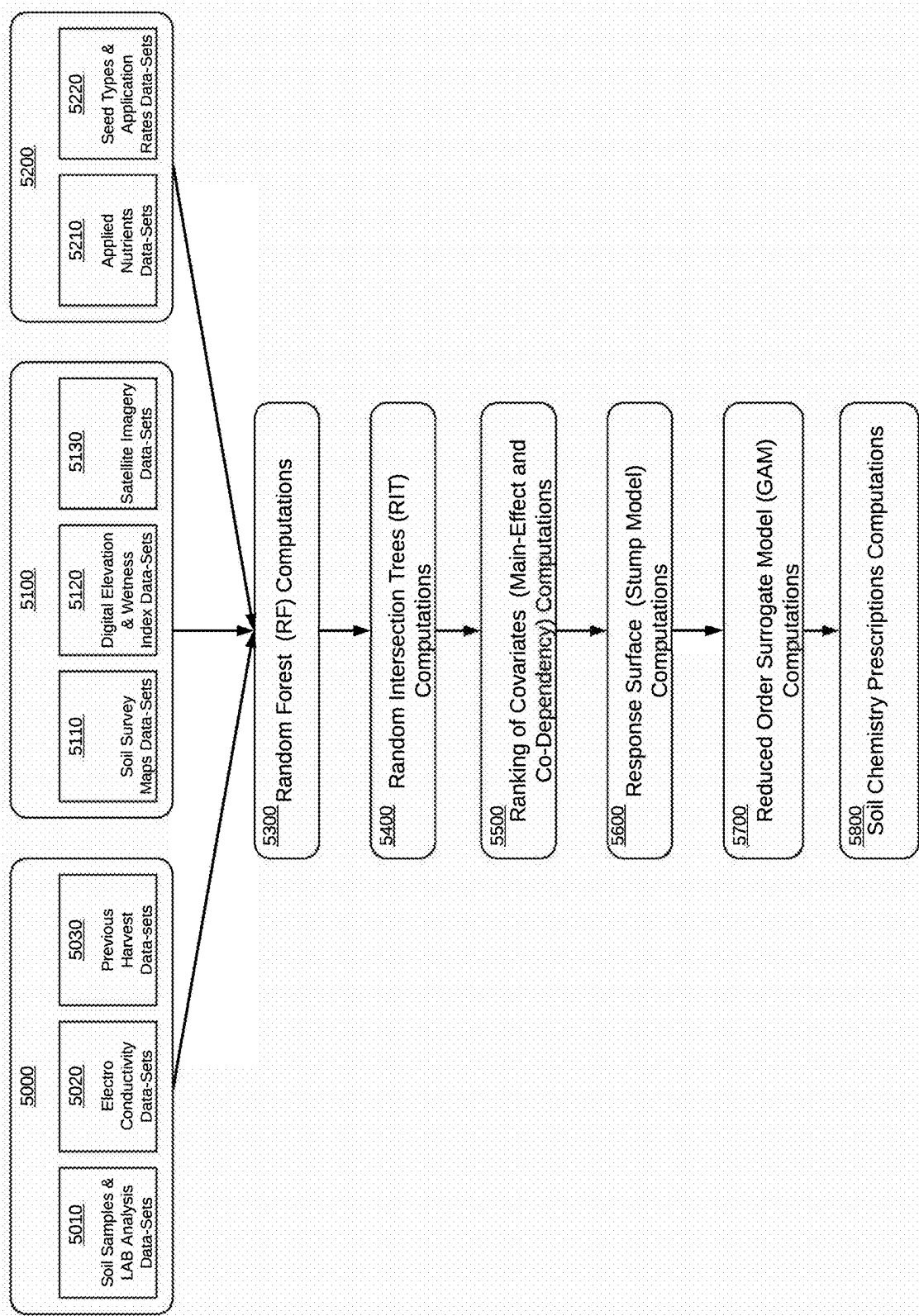


Figure 5

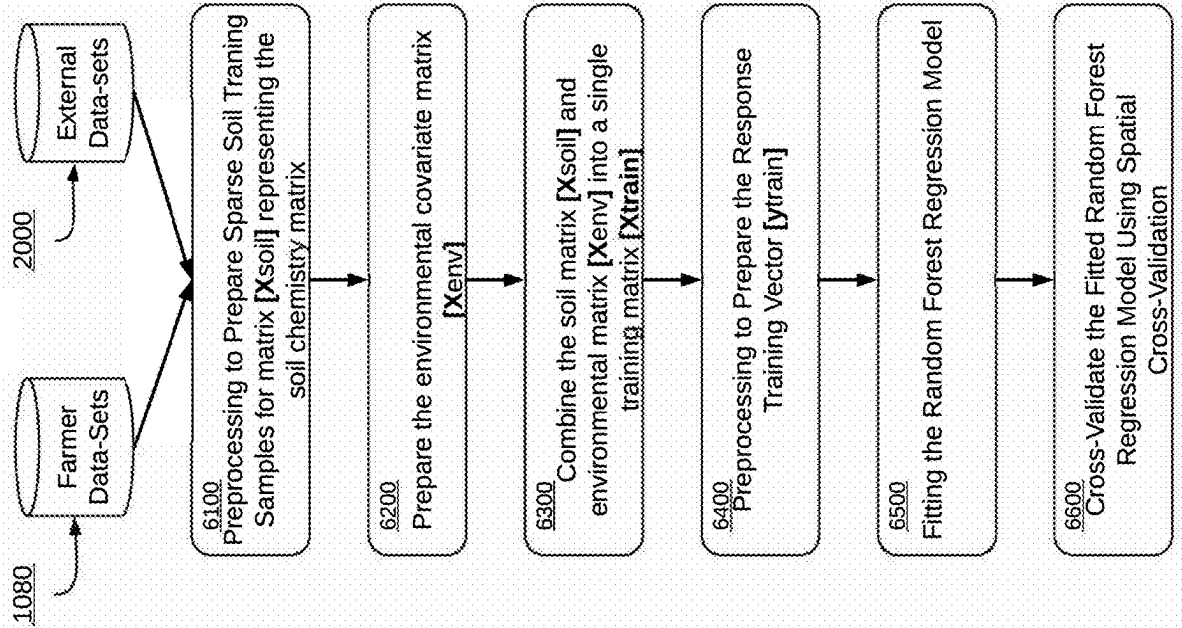


Figure 6

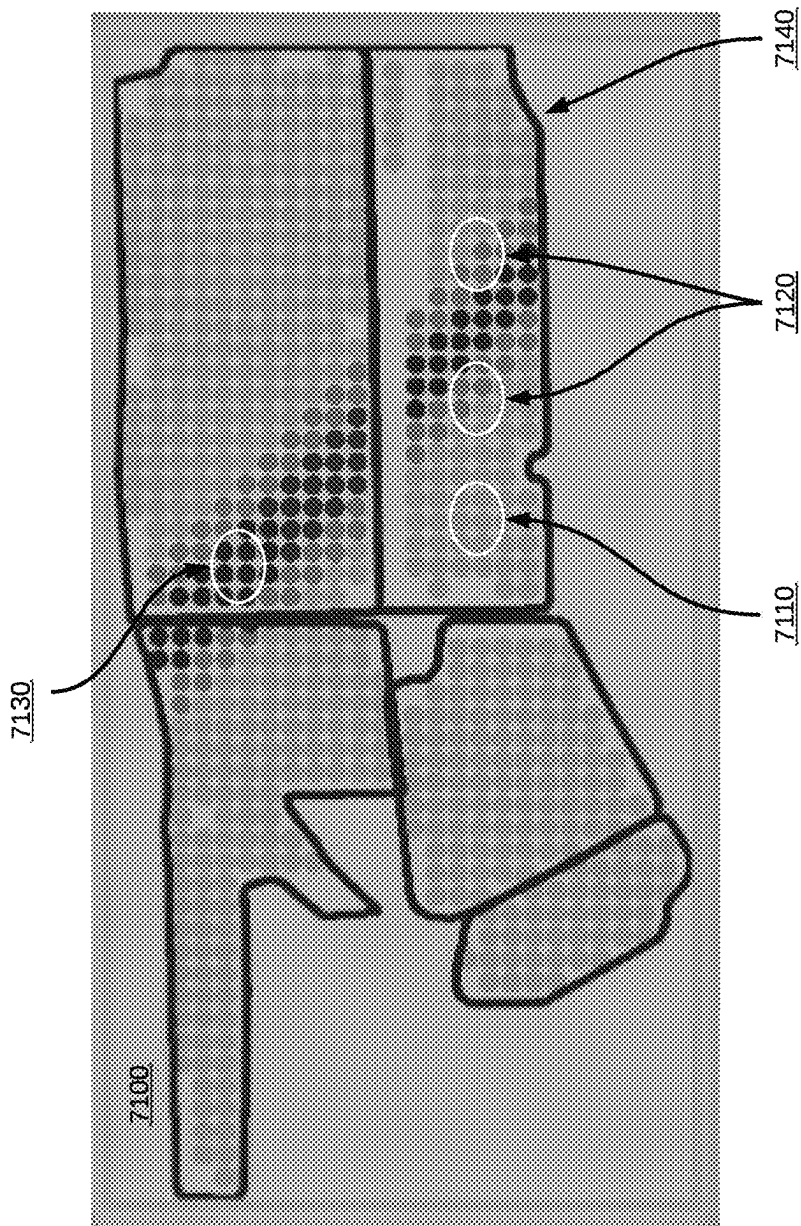


Figure 7

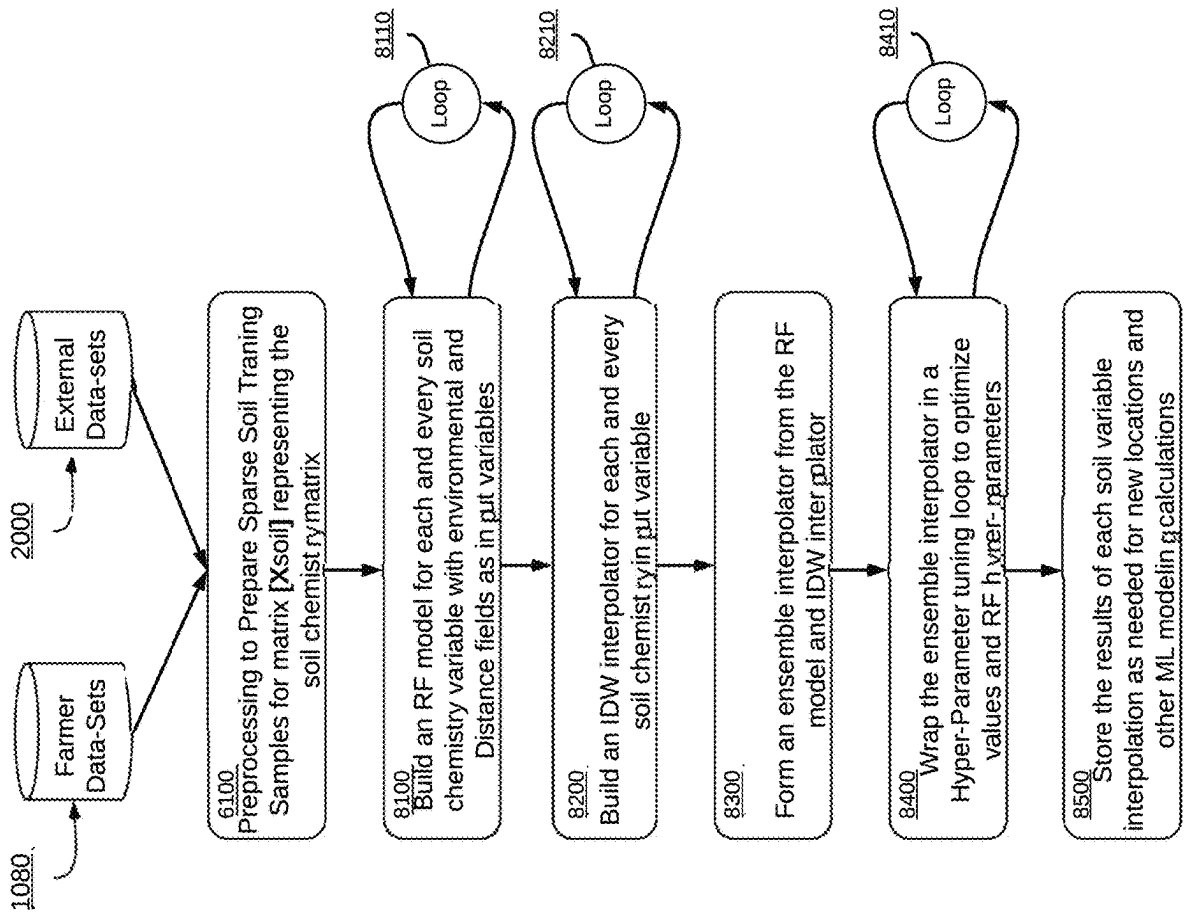


Figure 8

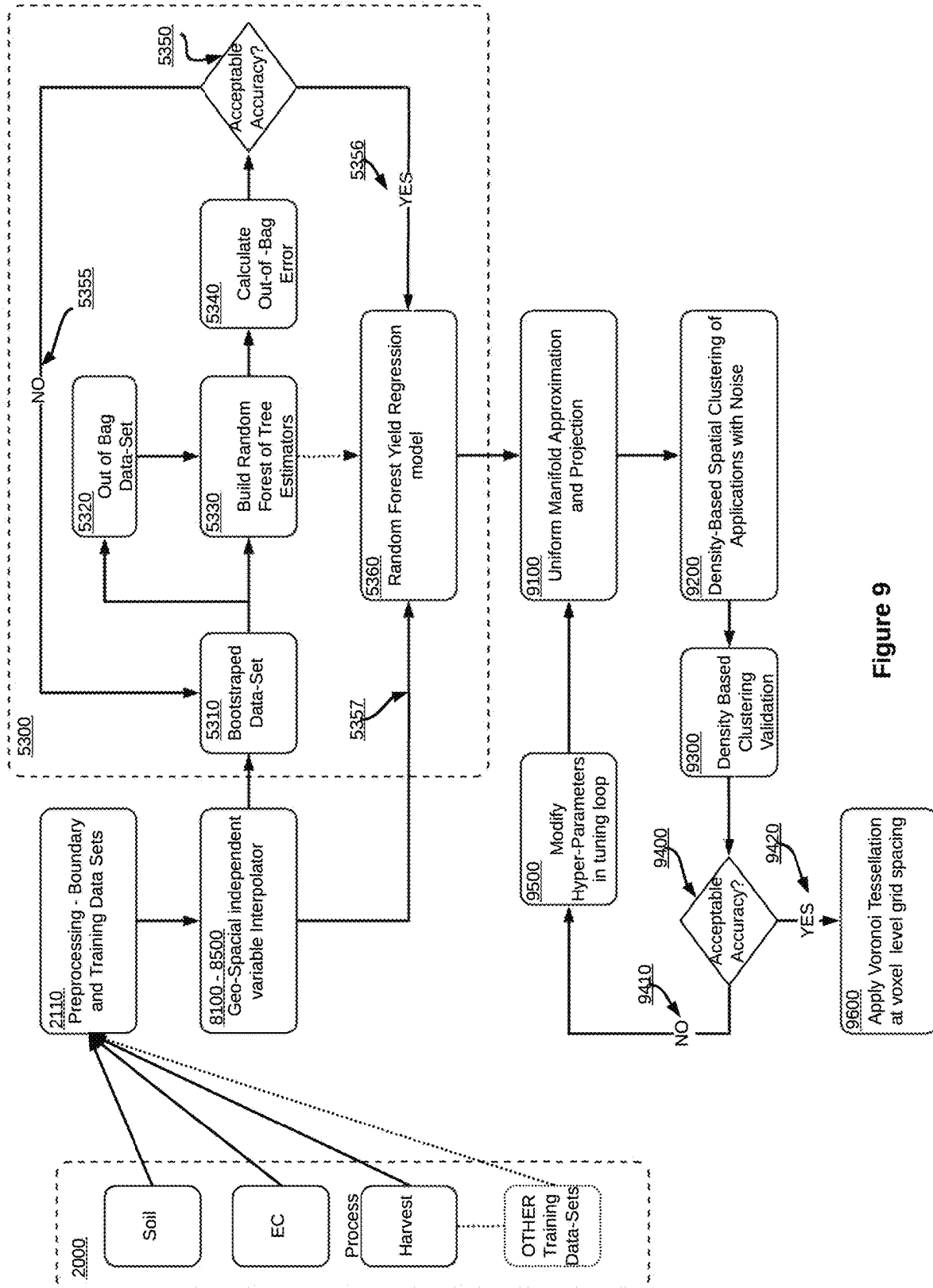


Figure 9

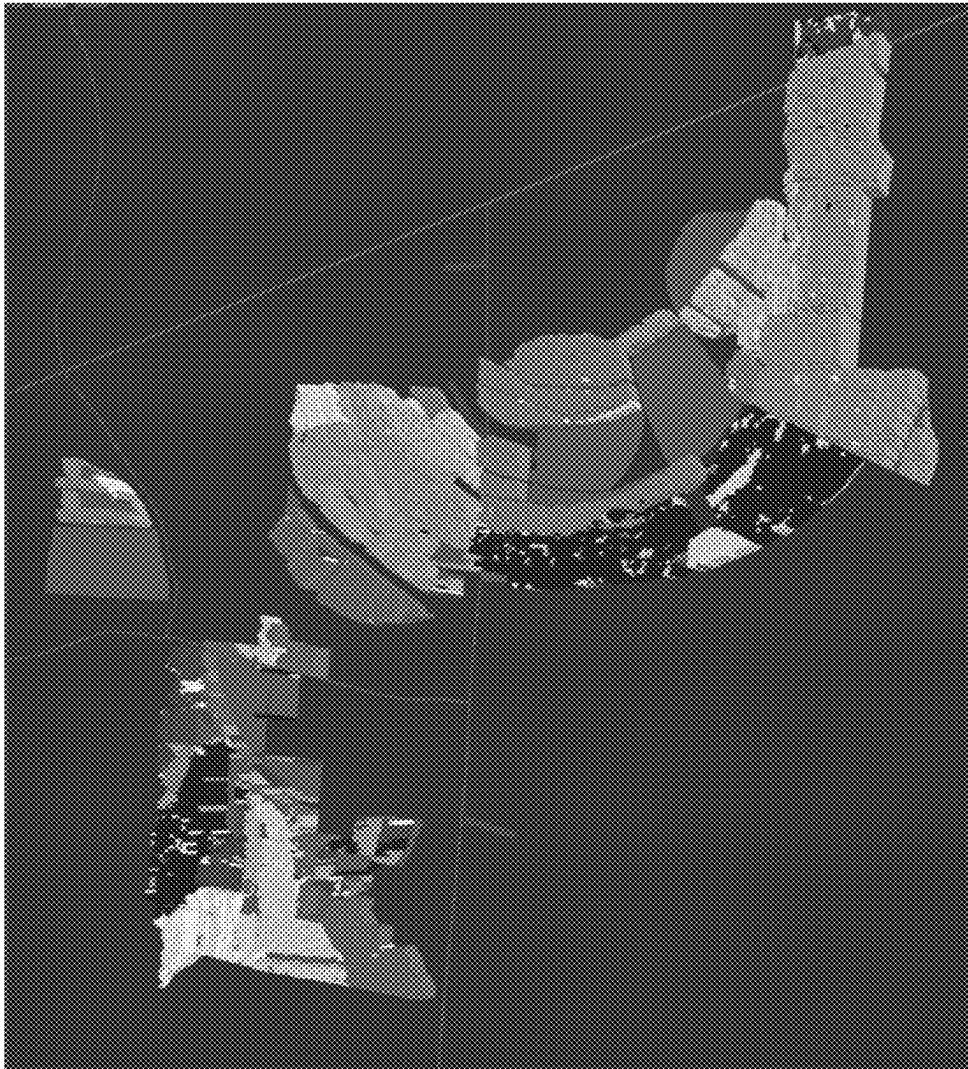


Figure 10

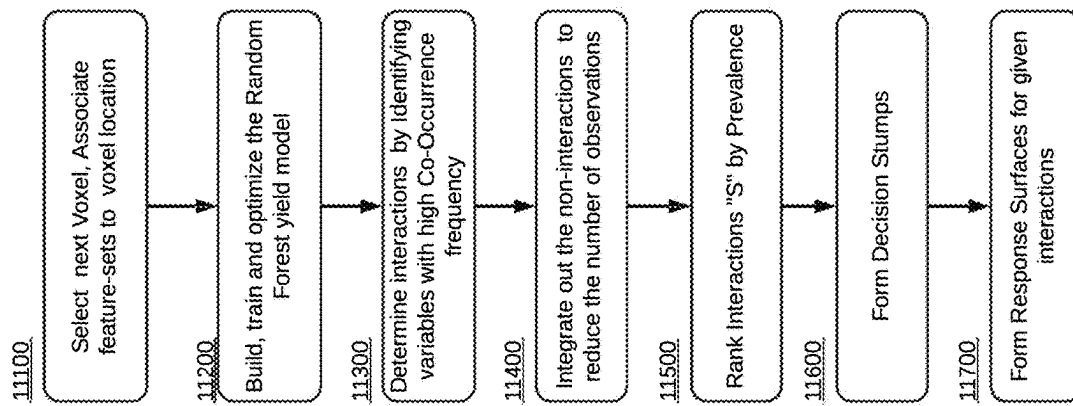


Figure 11

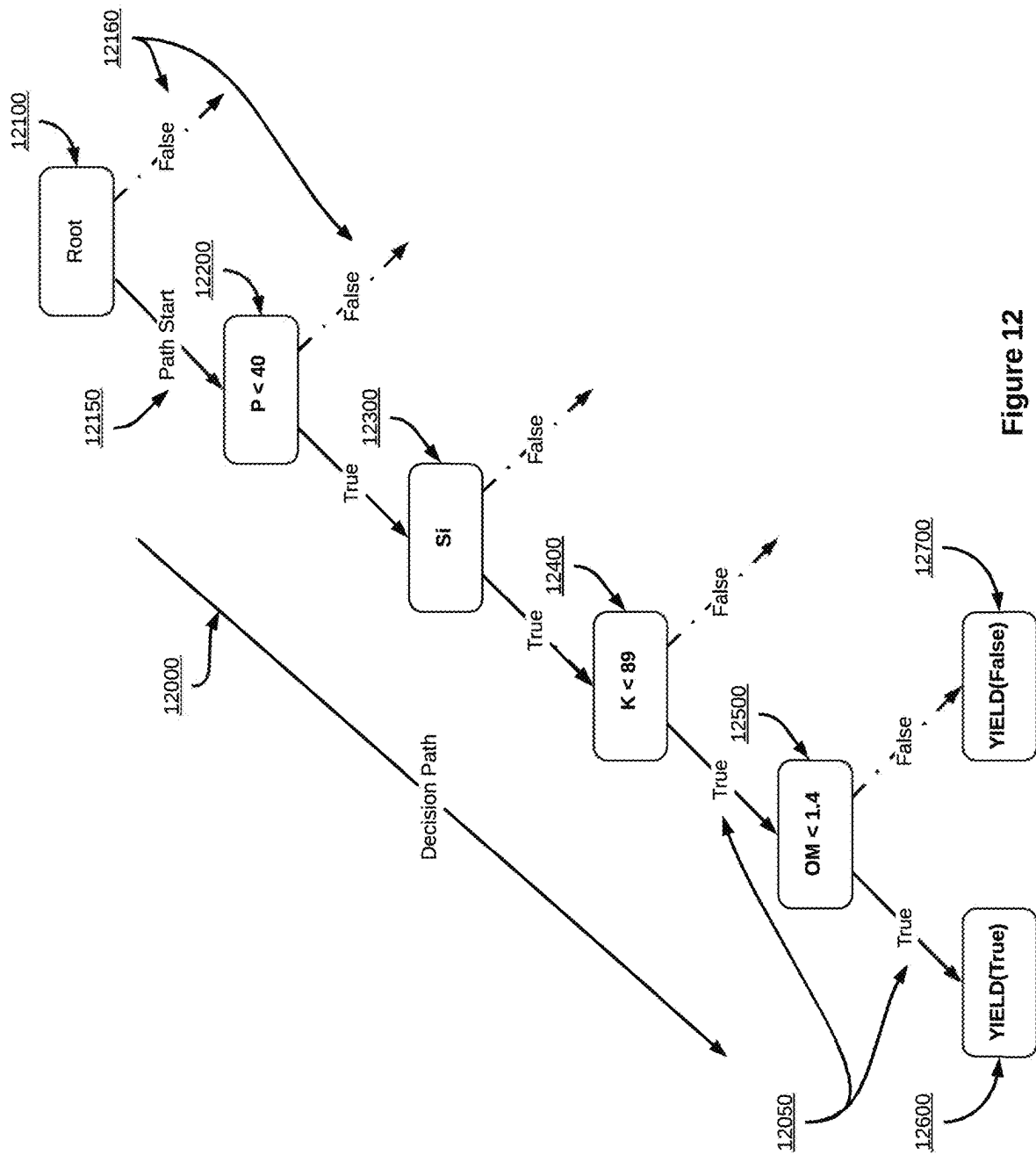


Figure 12

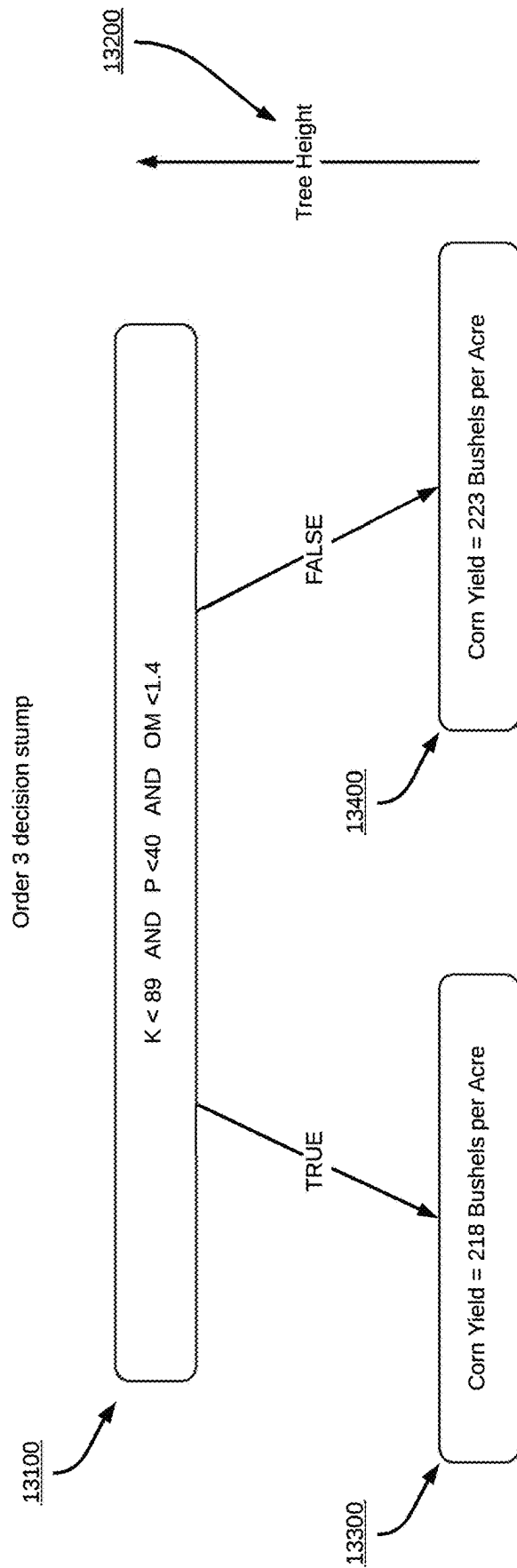


Figure 13

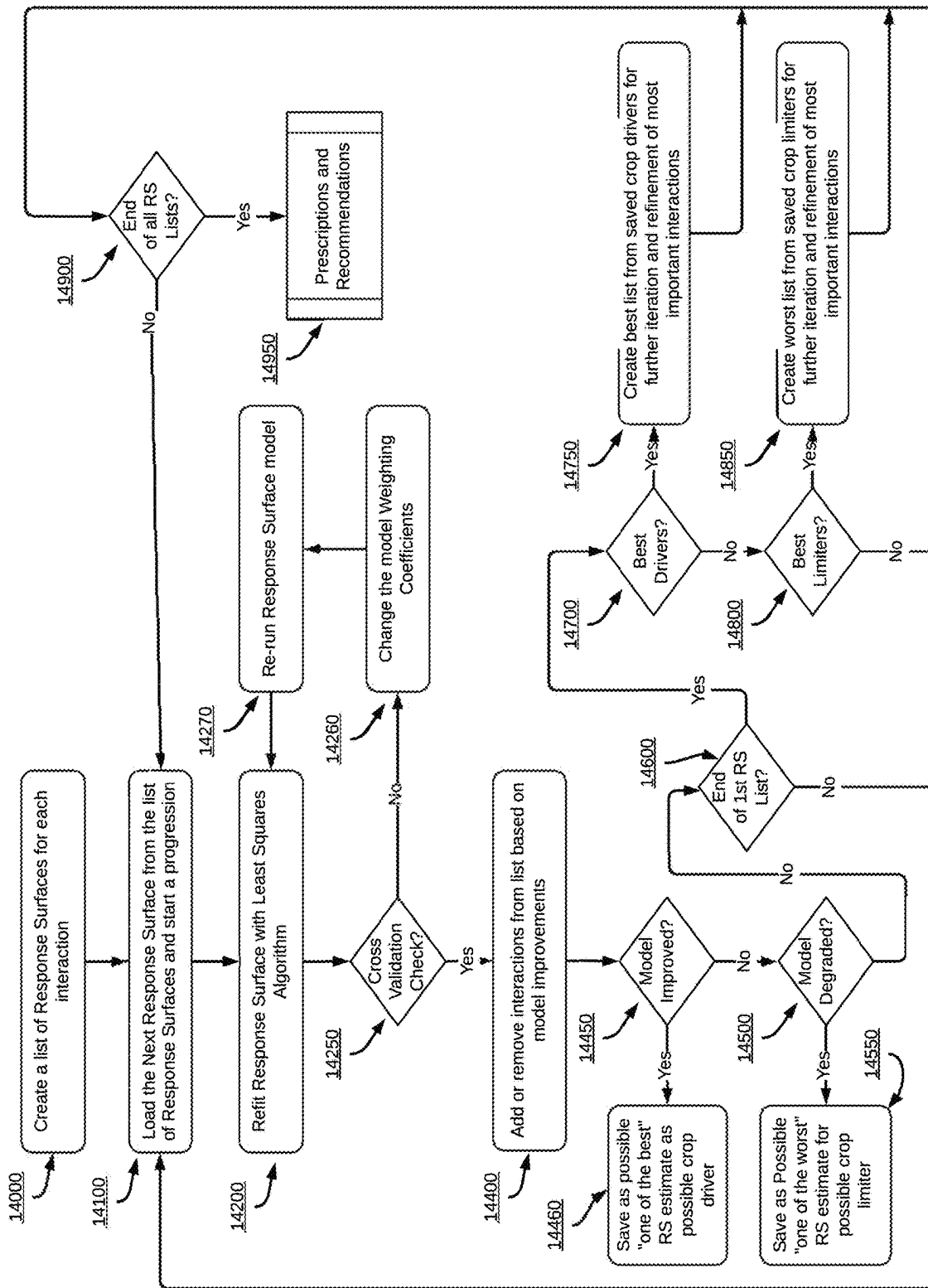


Figure 14

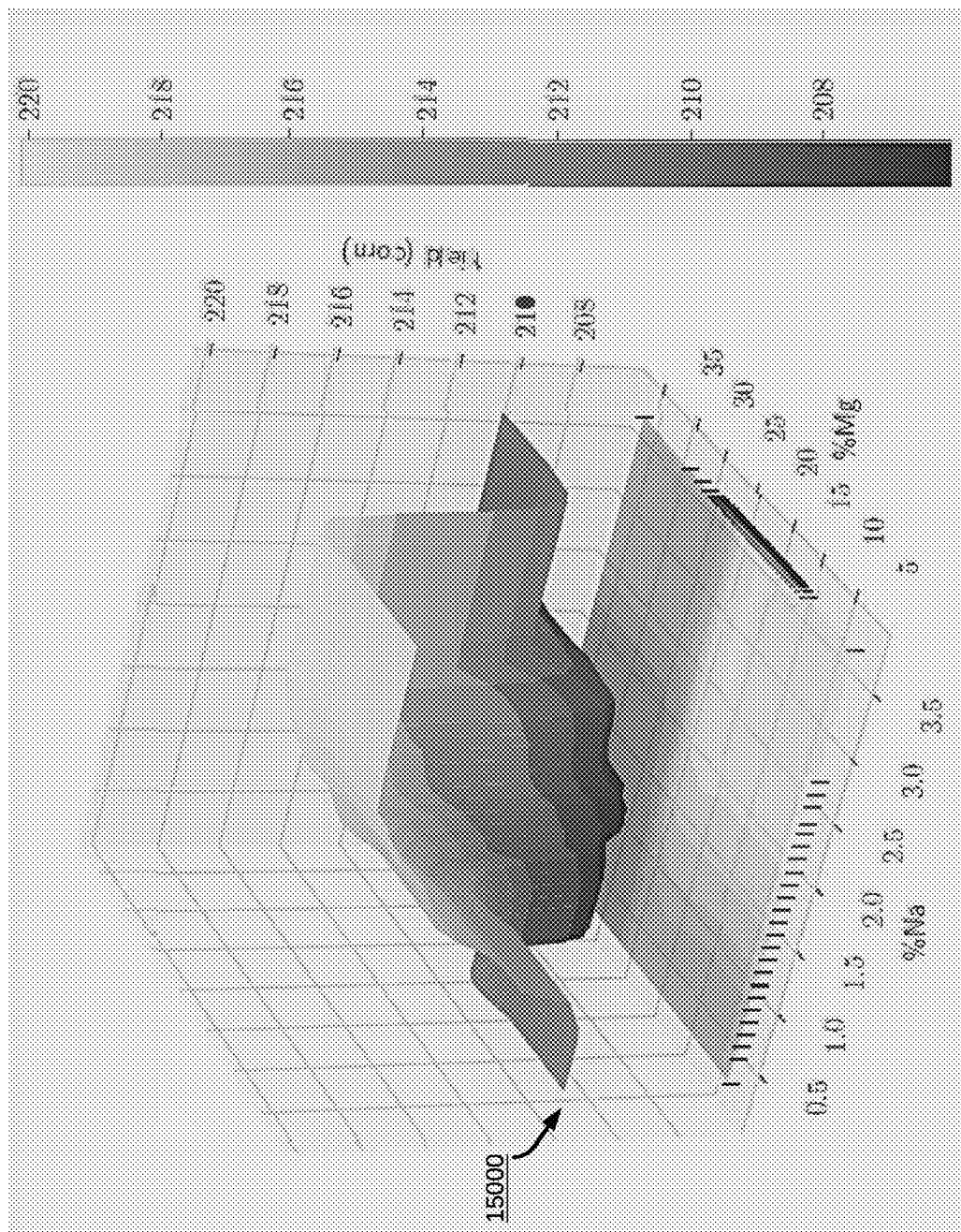
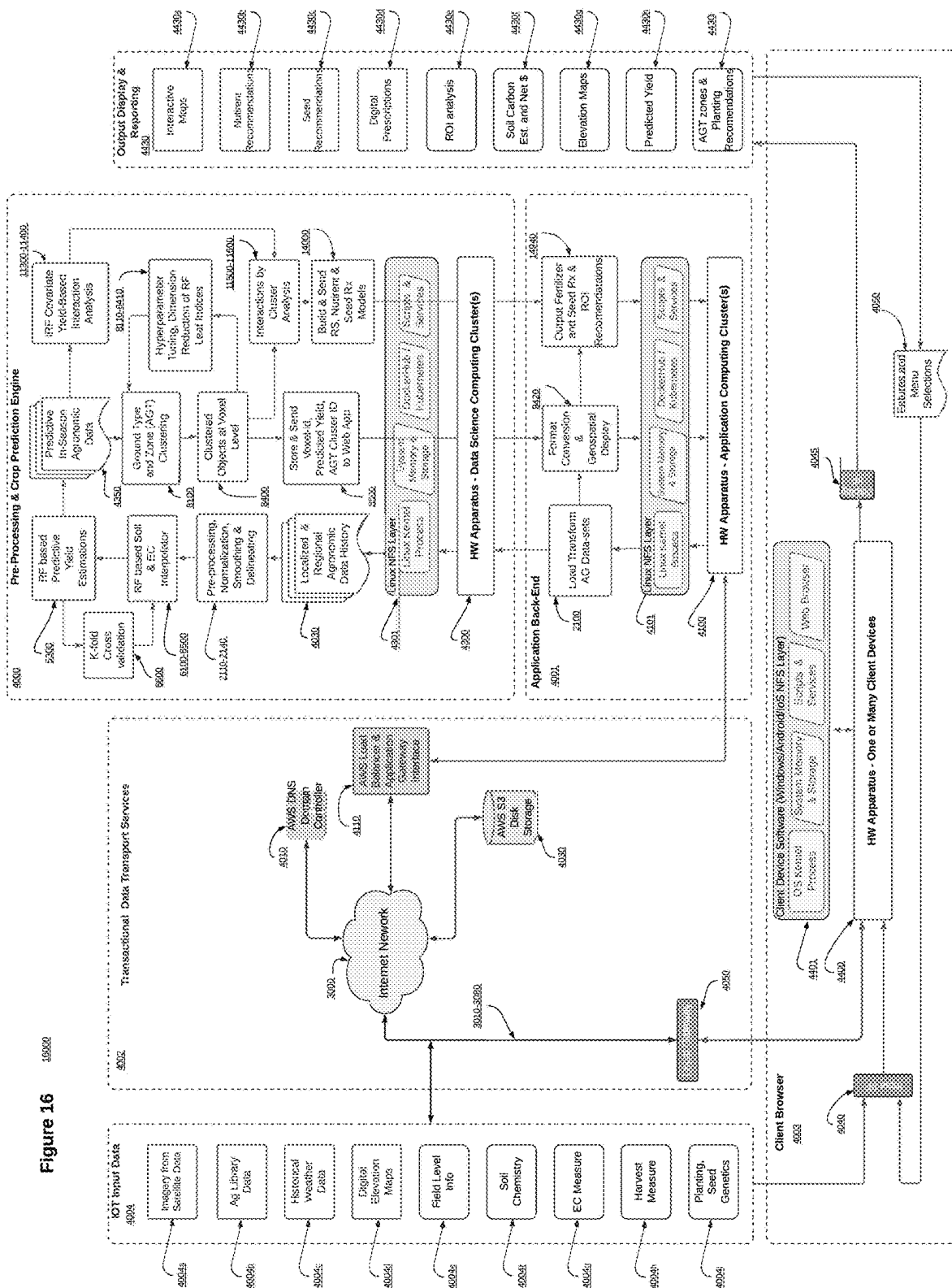


Figure 15



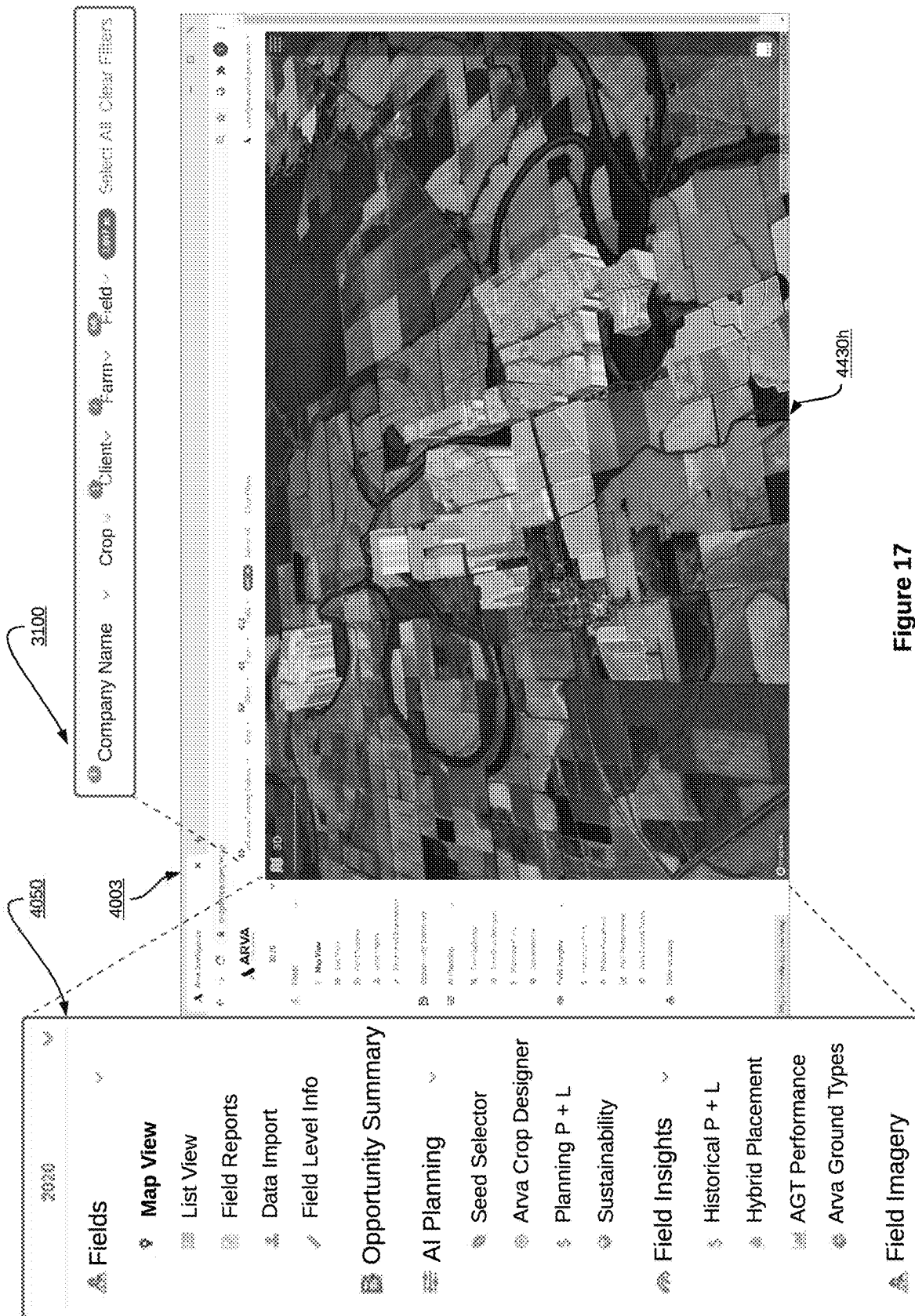


Figure 17

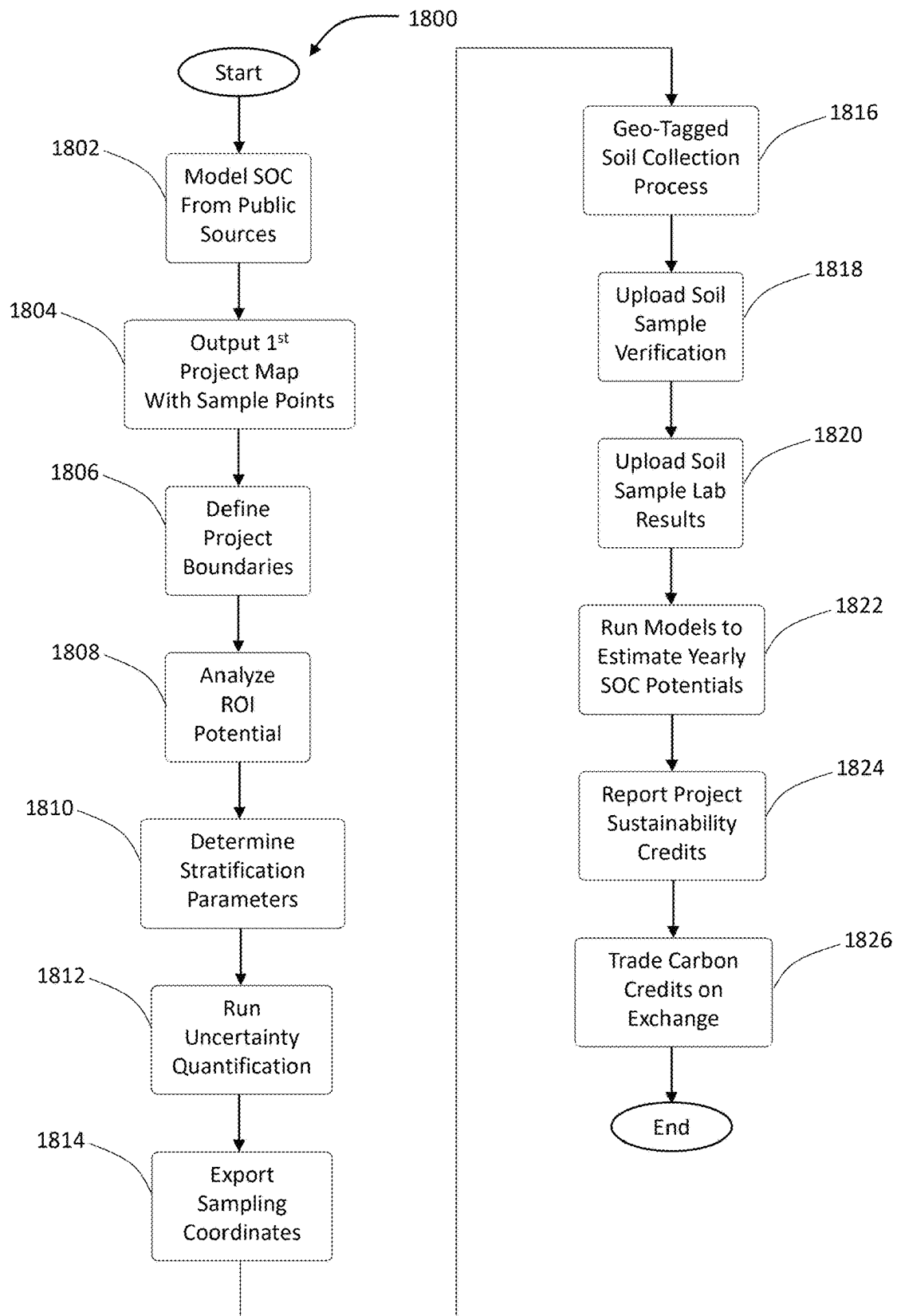


FIGURE 18

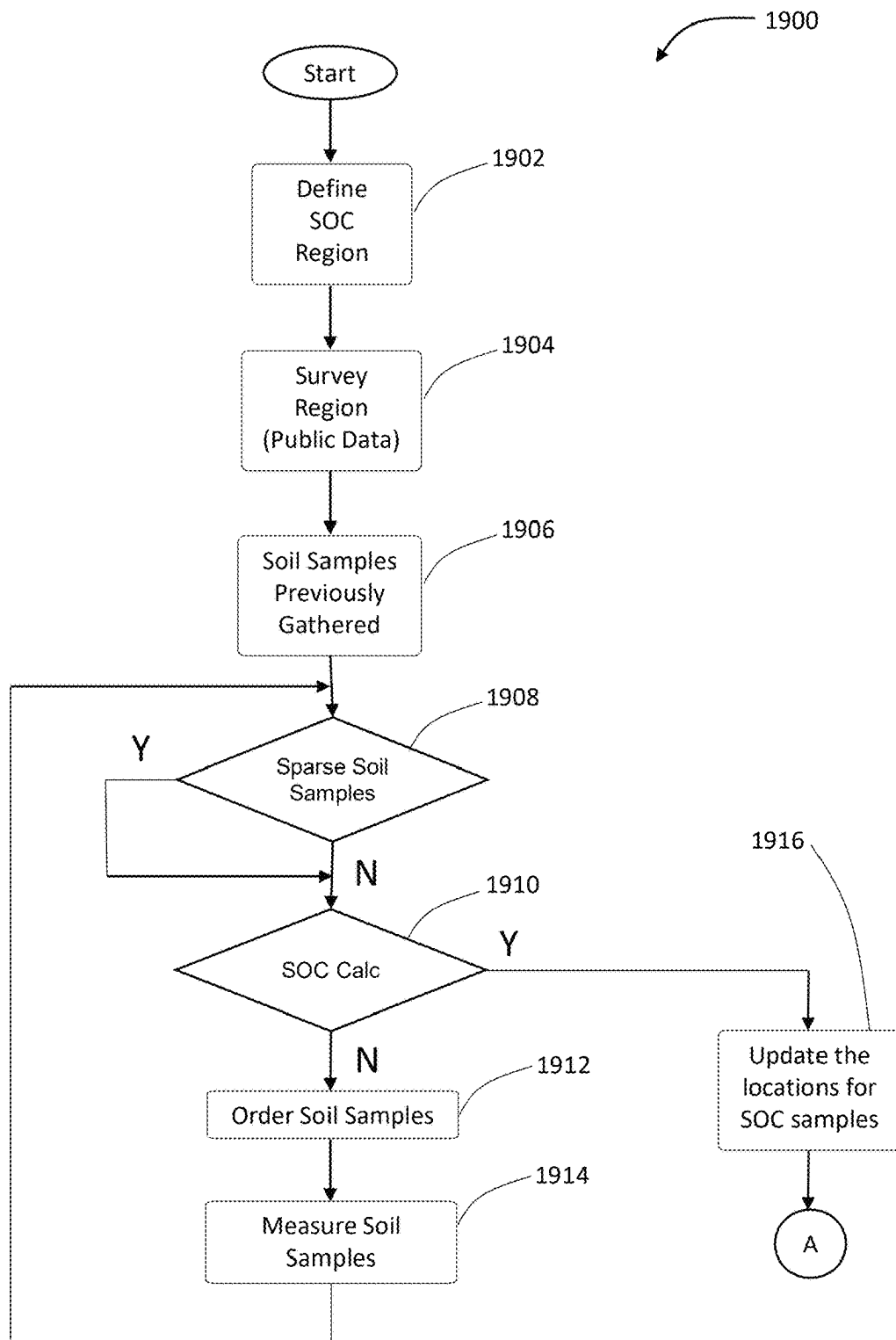


FIGURE 19

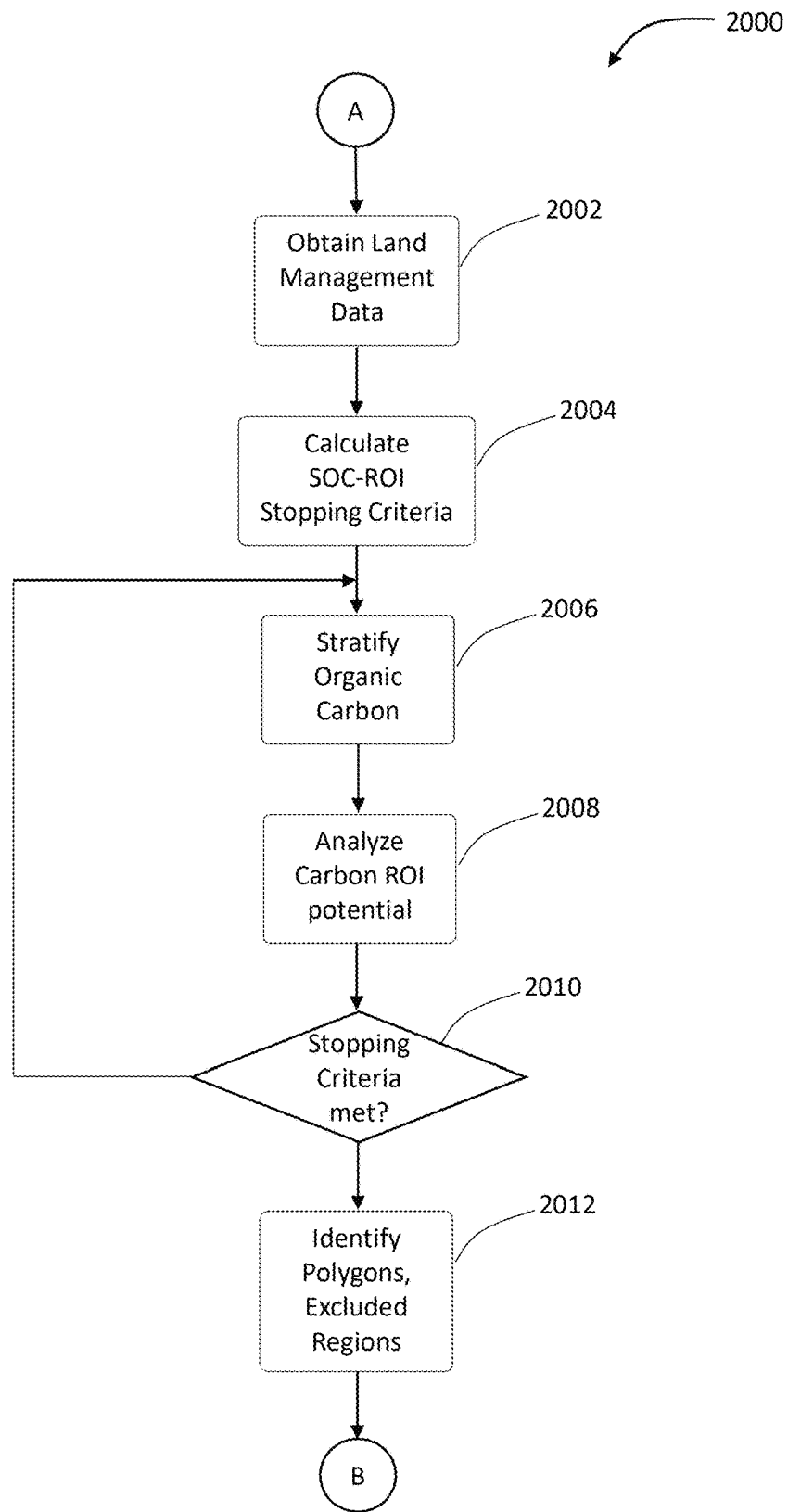


FIGURE 20

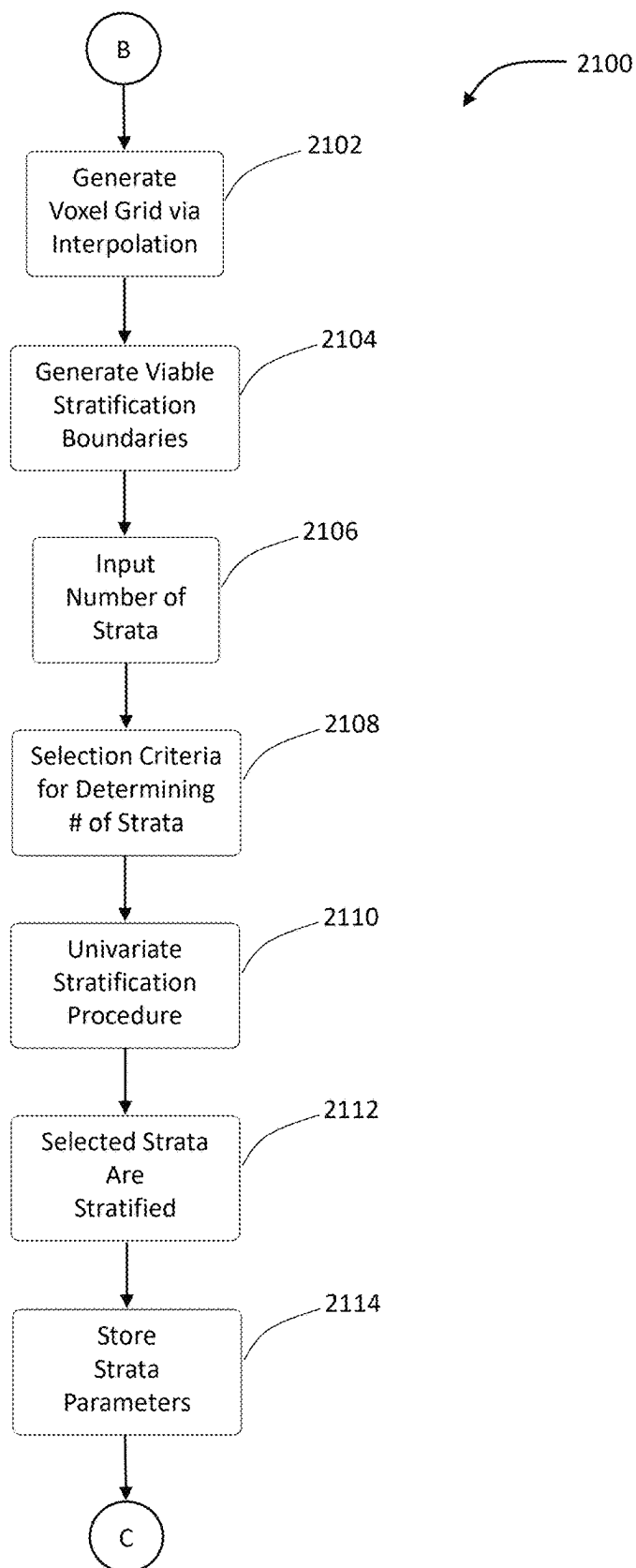


FIGURE 21

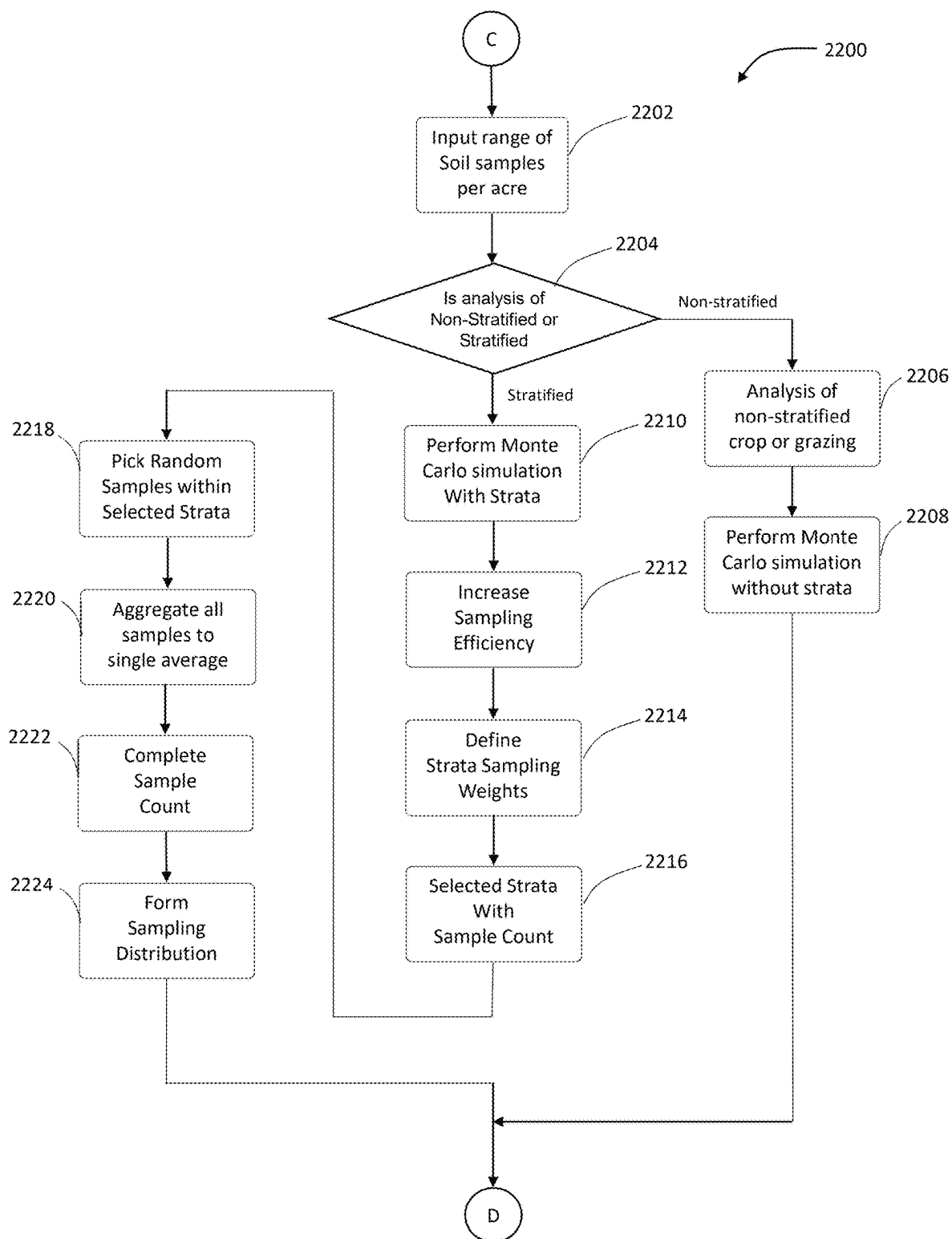


FIGURE 22

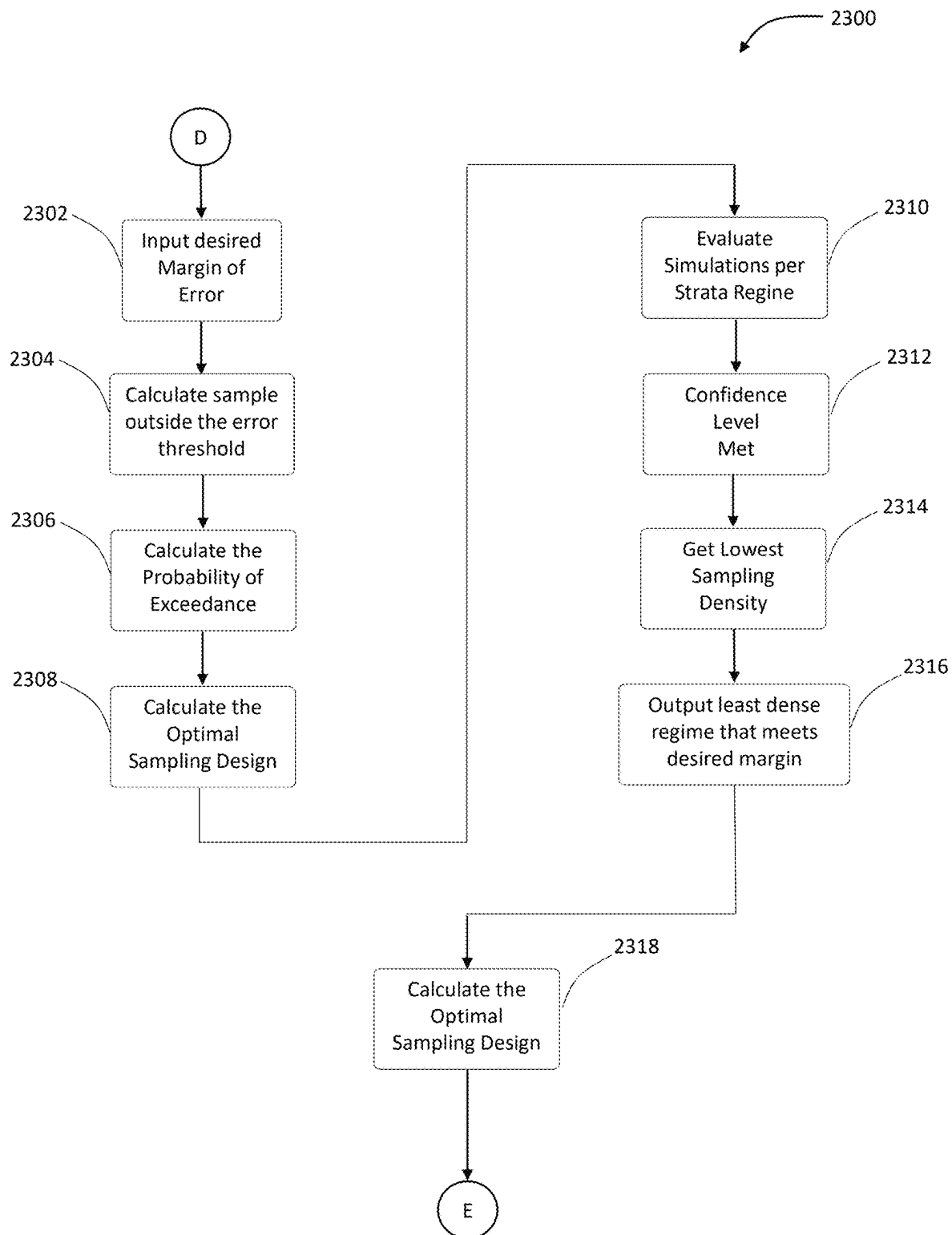


FIGURE 23

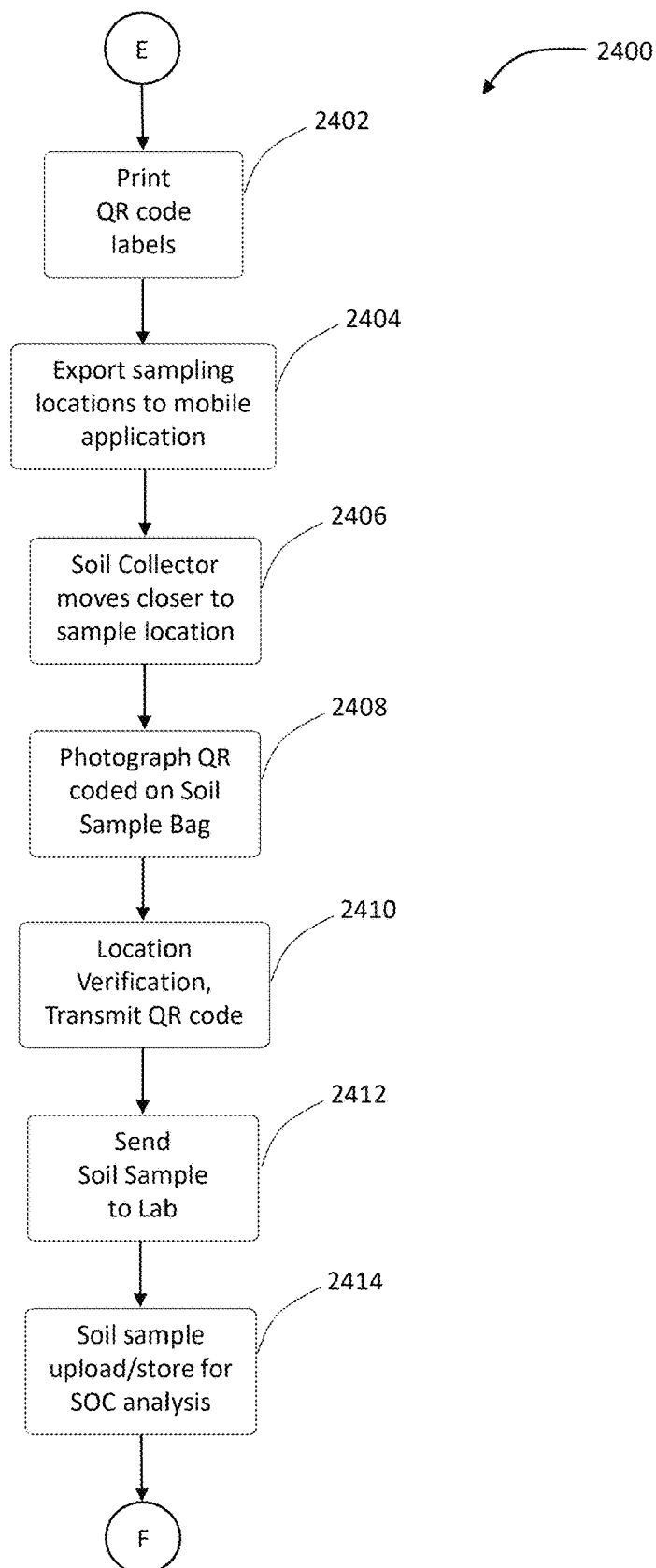


FIGURE 24

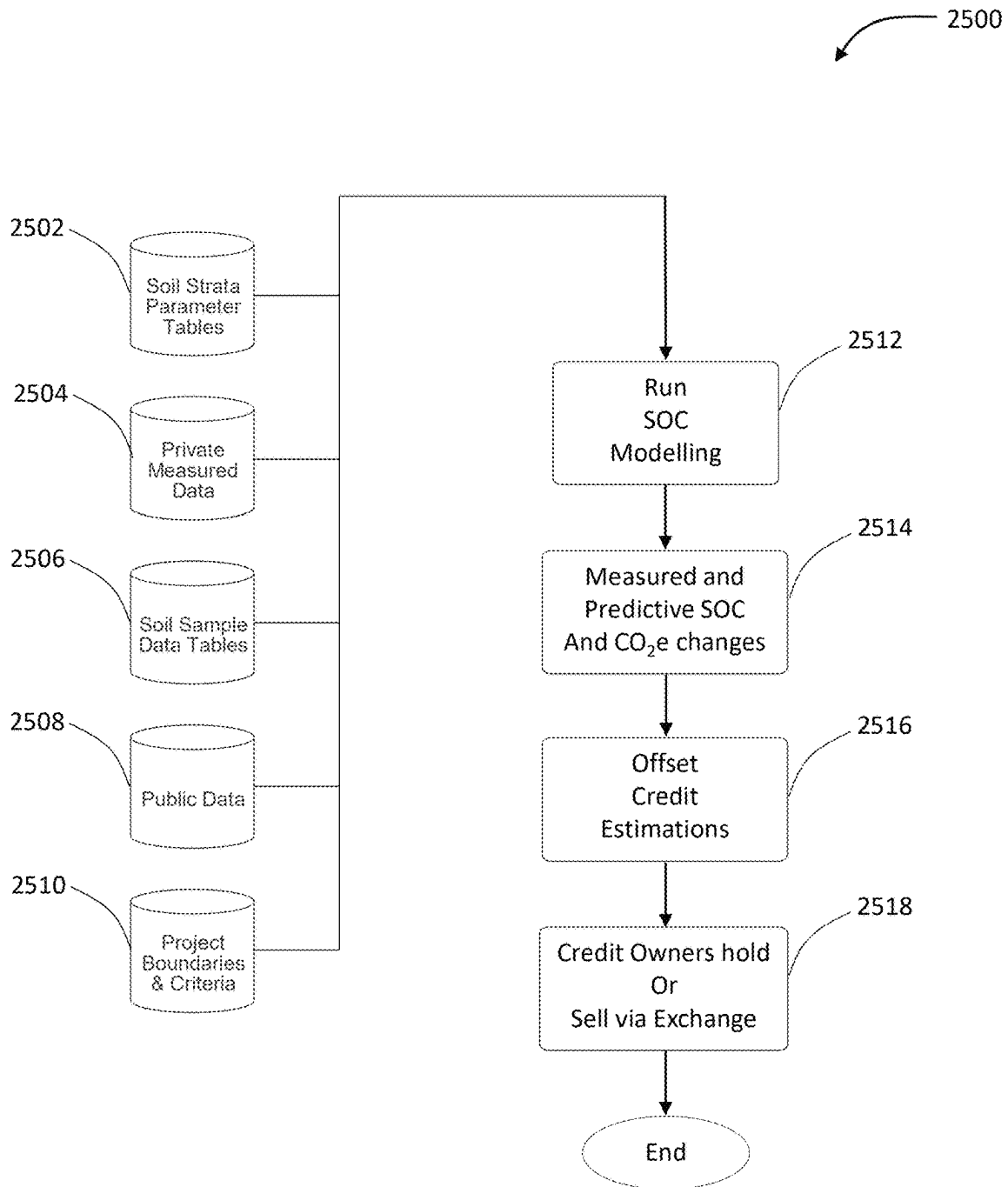


FIGURE 25

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**OPTIMAL SAMPLING WITH SOIL
STRATIFICATION****CROSS REFERENCE**

This patent application is a Continuation-In-Part of patent application Ser. No. 17/160,286 that was filed on Jan. 27, 2021, and entitled PREDICTING CROP YIELD WITH A CROP PREDICTION ENGINE, which claims the benefit of provisional patent application 62/995,484 that was filed on Jan. 29, 2020, and entitled METHOD AND APPARATUS FOR PREDICTING CROP-YIELD AS A FUNCTION OF ENVIRONMENTAL COVARIATES;

this patent application is a Continuation-In-Part of patent application Ser. No. 17/171,887 filed on Feb. 9, 2021, and entitled ESTIMATING SOIL CHEMISTRY AT DIFFERENT CROP FIELD LOCATIONS, which claims the benefit of provisional patent application 62/995,674 filed on Feb. 7, 2020, and entitled METHOD AND APPARATUS FOR ESTIMATING SOIL CHEMISTRY AT ANY CROP-FIELD LOCATION;

this patent application is a Continuation-In-Part of patent application Ser. No. 17/180,695 filed on Feb. 19, 2021, and entitled IDENTIFYING GROUND TYPES FROM INTERPOLATED COVARIATES, which claims the benefit of provisional patent application 62/995,948 that was filed on Feb. 20, 2020, and entitled METHOD AND APPARATUS FOR CLUSTERING GROUND TYPES FROM INTERPOLATED AND ENVIRONMENTAL COVARIATES; and

this patent application is a Continuation-In-part of patent application of patent application Ser. No. 17/203,670 filed on Mar. 16, 2021, and entitled DETERMINING CROP-YIELD DRIVERS WITH MULTI-DIMENSIONAL RESPONSE SURFACES, which claims the benefit of provisional patent application 63/100,545 filed on Mar. 17, 2020, and entitled METHOD AND APPARATUS FOR DETERMINING CROP-YIELD DRIVERS USING MULTI-DIMENSIONAL RESPONSE SURFACES;

all of the applications are incorporated by reference in this patent application.

FIELD

The present disclosure relates to a system and method for optimal sampling. More specifically, the system and method select a strata parameter, a sample count of strata parameters and a plurality of sample combinations for a Monte Carlo simulation, which determines an optimal sampling regime that includes a lowest sampling density and one or more sampling locations.

BACKGROUND

Recent international focus on climate change has caused industries around the world to focus on new methods to achieve global decarbonization and future carbon neutrality. Historically, companies that produce excess CO₂ based on greenhouse gas emissions have relied on technology or mechanical solutions to remove and artificially sequester carbon supply from the environment. These efforts have come only on a project-by-project basis and have very limited scalability. Recently, nature-based decarbonization, which is often referred to as agriculture, forestry, and other land use (AFOLU) has proven to be a highly scalable supply

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source for CO₂ sequestration. Nature-based CO₂ sequestration methods are cost efficient and scale when compared to historical mechanical decarbonization solutions.

Newly emerging ecosystem service markets represent a growing opportunity to combat climate change and improve environmental health by unleashing the power of private markets to encourage practices that reduce, sequester, or otherwise mitigate greenhouse gas emissions. Within the AFOLU industry, supply typically refers to the tons of carbon sequestered per acre and is often converted to monetary value through carbon offset credits. Once verified, the supply of carbon offsets may be purchased by demand buyers. Demand for carbon offset credits is generated mainly by corporations that are known to be greenhouse gas emitters and most have made public their wish to achieve carbon neutrality over time. Matching supply of carbon credits from nature based carbon suppliers with demand from greenhouse gas emitters, i.e., carbon offset buyers, has proven difficult because of complex markets, inefficient cost processes, and burdensome barriers to entry.

Determining the level of nature-based CO₂ sequestration based on agricultural process has proven complex. One method for determining the level of nature-based CO₂ sequestration relies on the measurement of a soil organic carbon (SOC) supply, which is performed with soil samples collected from a dense grid of cropland, rangeland, or timberland. Another method of determining the level of nature-based CO₂ sequestration includes analyzing the land for fertility opportunity zones. Regrettably, the cost of measuring SOC supply or analyzing fertility opportunity zones in a dense grid is often cost prohibitive because of the density of samples per acre.

Thus, there is a need for a system and process that reduces the costs of securing nature-based decarbonization, while at the same time, optimizing and unlocking the large-scale benefits of AFOLU carbon sequestration.

Additionally, there is a need to lower the cost of SOC measurements with a system and method that determines the amount of sequestered carbon while minimizing the cost by eliminating the requirement of more expensive soil grid soil sampling.

SUMMARY

A system and method for determining optimal sampling parameters is described. The method includes identifying, at a crop prediction engine, a soil organic carbon (SOC) project area on a map. The method then proceeds to gather a plurality of soil data from a soil data source. The soil data is associated with the SOC project area. The crop engine prediction engine then determines that the soil data is less than optimal, and that the soil data is sufficient to generate an optimal sampling plan. The crop prediction engine then determines a potential SOC per unit area for one or more polygons that are within the SOC project area. One or more polygons that satisfy an SOC requirement are determined. A voxel grid for the polygon that satisfies the SOC requirement is then generated. The method then proceed to complete a Monte Carlo simulation, which generates an empirical sampling distribution. The optimal sampling plan is determined by defining a margin of error, which provides a deviation from a predictive analysis of measured soil chemistry for a plurality of collected soil samples, and performing a plurality of Monte Carlo simulations that includes one Monte Carlo simulation having a lowest sampling density that

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satisfies the margin of error. The optimal sampling plan has the lowest sampling density and includes one or more sampling locations.

In one embodiment, after generating a voxel grid for the polygon that satisfies the SOC requirement, the method includes identifying a plurality of strata parameters, in which each strata parameter includes a stratification boundary associated with the SOC area. The illustrative method then identifies a number of strata within the stratification boundary and inputs a range of soil samples for the SOC area. The illustrative method continues by defining one or more strata sampling weights for the Monte Carlo simulation and selecting at least one strata parameter and a sample count from a plurality of strata parameters for the Monte Carlo simulation. The illustrative method may then select a plurality of random samples for the selected strata parameter. The illustrative method then aggregates the random samples for each selected strata parameter. Additionally, the illustrative method determines when the sample count is completed for each selected strata parameter. Furthermore, the illustrative method repeatedly selects random samples and aggregates the random samples until the sample count is completed for each selected strata parameter.

In another illustrative embodiment, the method exports the sampling locations to a mobile device. The method then validates, at the mobile device, a location, a date, and a time for collecting a soil sample with a GPS component. Furthermore, the method includes storing soil sample results that are accessible by the crop prediction engine. Further still, the method includes analyzing, at the crop prediction engine, the soil sample results with a soil prediction model to generate one or more SOC predictions. Further yet, the method includes estimating a greenhouse gas reduction. Further, the method converts the greenhouse gas reduction to one or more carbon credits that are traded on a carbon exchange.

FIGURES

The present systems and methods will be more fully understood by reference to the following drawings which are presented for illustrative, not limiting, purposes.

FIG. 1 shows a management flowchart for farmers running a typical precision farming operation.

FIG. 2 shows an illustrative flowchart for a crop modeling software application used by growers to estimate farm management operations and predictive yield.

FIG. 3A shows an illustrative system having distributed client devices that interface with a crop prediction engine.

FIG. 3B shows an illustrative system that supports access levels within corporate management in a typical organizational hierarchy.

FIG. 4 shows an illustrative high-level diagram of networked computing systems.

FIG. 5 shows an illustrative flowchart of the operations performed by a crop prediction engine, which is used to train a machine learning model.

FIG. 6 shows an illustrative flowchart for building and preparing the training vectors used to train a random forest (RF) model.

FIG. 7 shows an illustrative spatial image associated with a cross validation framework.

FIG. 8 shows a flowchart of the process for training covariates for estimation of soil characteristics and environmental characteristics at different geo-spatial resolutions.

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FIG. 9 shows a flowchart of a method for clustering ground types from interpolated covariates and environmental covariates.

FIG. 10 shows an illustrative image of the final output results from the method presented in FIG. 9.

FIG. 11 shows a high level flow diagram outlining the program instruction steps used to form a Response Surface of the illustrative embodiment.

FIG. 12 shows the decision path through one RF tree that builds the stump illustrated in FIG. 13.

FIG. 13 shows an example of a two value decision stump with height one for the illustrative embodiment.

FIG. 14 shows the program instruction blocks for tuning the number of interactions using a Generalized Additive Model for optimized nutrient and seed-type application recommendations.

FIG. 15 shows an order-2 response surface showing the effect of the Magnesium (Mg) percentage and Sodium (Na) percentage nutrient saturations on corn yield.

FIG. 16 shows a system architecture that supports the crop prediction engine's operation and generation of a multi-dimensional crop yield response surface.

FIG. 17 shows a client browser user interface displaying a multi-dimensional crop yield response surface.

FIG. 18 shows a high-level block flowchart for optimal sampling with soil stratification.

FIG. 19 shows a flowchart for collecting measured soil data from public and private sources.

FIG. 20 shows a flowchart to determine valid SOC boundaries and excluded areas that have no SOC potential within the project.

FIG. 21 shows a flowchart for determining the SOC project stratification parameters.

FIG. 22 shows a flowchart of the formation of soil sampling distribution in an SOC project.

FIG. 23 shows a flowchart to determine the project SOC uncertainty quantification.

FIG. 24 shows a flowchart for the use of QR codes to validate soil sample collection locations and dates.

FIG. 25 shows a flowchart for estimating SOC project carbon offset credit values.

DESCRIPTION

Persons of ordinary skill in the art will realize that the following description is illustrative and not in any way limiting. Other embodiments of the claimed subject matter will readily suggest themselves to such skilled persons having the benefit of this disclosure. It shall be appreciated by those of ordinary skill in the art that the systems and methods described herein may vary as to configuration and as to details. The following detailed description of the illustrative embodiments includes reference to the accompanying drawings, which form a part of this application. The drawings show, by way of illustration, specific embodiments in which the invention may be practiced. It is to be understood that other embodiments may be utilized, and structural changes may be made without departing from the scope of the claims. It is further understood that the steps described with respect to the disclosed process(es) may be performed in any order and are not limited to the order presented herein.

The optimal sampling with soil stratification described herein overcomes the limitations associated with calculating soil organic carbon (SOC) using conventional grid or zone sampling. SOC is a measurable component of soil organic matter. Organic matter makes up 2-10% of most soil's mass and serves an important role in the physical, chemical, and

biological function of agricultural soils. Organic matter contributes to nutrient retention and turnover, soil structure, moisture retention and availability, degradation of pollutants, and carbon sequestration. SOC only refers to the carbon component of organic compounds. Soil organic matter (SOM) is difficult to measure directly, so laboratories measure and report SOC as a proxy for SOM. Sequestering carbon as SOC can mitigate climate change by reducing atmospheric carbon dioxide because small increases of SOC over very large areas reduces atmospheric carbon dioxide, i.e., CO₂.

Also, the optimal sampling with soil stratification described herein reduces the costs of securing nature-based decarbonization by reducing sampling costs, which optimizes and unlocks the large-scale benefits of AFOLU carbon sequestration.

In summary, optimal sampling with soil stratification minimizes the required sampling density. The optimal sampling with soil stratification systems and methods described herein is based on the analysis and optimization of various soil strata environments. Soil strata refers to layers of soil. A singular layer is referred to as a "stratum." The optimal sampling with soil stratification described herein lowers the cost of SOC measurements by eliminating expensive grid and/or zone soil sampling.

More specifically, optimal sampling with soil stratification determines the appropriate sampling density that matches the soil organic carbon variance in the region. This approach identifies the appropriate soil strata parameters for modeling and determines the number and location of collected soil samples within a certain degree of statistical confidence.

Soil strata parameters are selected variables that are associated with the layers of soil. Soil strata parameters may also be referred to as "strata parameters" or "strata." Soil strata parameters are based on raw soil sample data that may include soil chemistry, soil texture, soil type (silt, sand, clay, loamy), soil climate (arid, dry, tropical), and other such laboratory generated and/or measured parameters. For example, soil chemistry may include measuring Nitrogen, Phosphorus, and Calcium from the collected soil sample.

The systems and methods described herein enable a project developer to factor in project costs and varying degrees of uncertainty in order to determine the optimal soil stratification, at minimum sample cost. Since project sampling costs are a barrier for landowners to enter into the carbon market, the optimal sampling with soil stratification can identify the least dense sampling regime that ensures the same quantification of SOC within the project area as a prohibitively expensive sampling regime.

Additionally, the optimal sampling with soil stratification systems and methods presented herein define project boundaries for plots of land and provide input soil data to determine the optimal sampling locations for an SOC project. In one embodiment, project design may be used to better define project boundaries, which determine the optimal location for performing a SOC project.

Furthermore, the optimal sampling with soil stratification systems and methods described herein determine where, within a set of soil strata, the best representation of soil chemistry may be found for a general area of the same soil environment. More specifically, the systems and methods determine the optimal geo-spatial point locations for sampling soil for one or more set of soil-strata representing a large area of possible soil chemistry data.

Further still, the optimal sampling with soil stratification includes a process for verifying that collection of such

samples was performed at a particular geo-spatial location and on a specific date. In operation, the collection of soil samples are geo-spatially verified and sent to at least one soil laboratory for soil analysis.

Each strata parameter includes a stratification boundary associated with an area. By way of example, but not of limitation, the area may be a Soil Organic Carbon (SOC) area. More generally, a stratification boundary includes an area with similar strata parameters. During the optimal sampling analysis described herein, the stratification boundary remains a dynamic value. However, the stratification boundary becomes fixed when the optimal sampling analysis is complete and the optimal sampling parameters are determined.

The optimal sampling analysis described herein determines the optimal, i.e., minimal, number of samples needed for a plot of land for one or more strata parameters within a stratification boundary. More specifically, the optimal sampling analysis includes soil stratification systems and methods to reduce soil sample density requirements by optimizing sampling density through a unique machine learning (ML) methodology and process described herein.

As used herein, a "project" is defined as understanding and quantifying the underlying soil organic carbon content within a specified region, parcel, or plot of land. "Projects" are typically defined by landowners, third party commercial companies or anyone with interest in the rights to the amount or units of Soil Organic Carbon (SOC) within the project.

A "project area" can be a large parcel of land or multiple parcels of land. Generally, one or more soil samples are collected from the project area.

After determining the variables/strata, the stratification boundary, the method proceeds to define one or more strata sampling weights. These sampling weights are defined before conducting a Monte Carlo simulation. The strata sampling weights are associated with the strata parameters and may be modified before conducting the Monte Carlo simulation. The process of modifying the strata sampling weights is referred to as "hyper tuning." For example, a first strata sampling weight is a value that is associated with a particular variable/strata and may be repeatedly modified.

The process of hyper tuning is performed to generate the strata sampling weights and determine the strata parameters/variables. By way of example and not of limitation, the hyper tuning process is performed by data fusion in a supervised framework when and/or where measured soil chemistry data is dense in a semi-supervised framework or a self-supervised framework, in which the measured data was "sparse." For the sparse instance, soil chemistry measurements are applied to one or more SOC maps. The system's internal SOC map is continuously updated as new data become available. In regions where there may be only dozens of soil samples, a learning framework based on iterative Random Forests (iRF) is used. In regions where there are at least one thousand soil samples, the method may use a feed-forward deep learning framework based on dense, multiscale networks. A univariate stratification optimization procedure is performed with a Kozak's random search optimization method that is used to create the stratification boundaries from the illustrative grid-wise SOC estimates.

After the hyper tuning process, the Monte Carlo simulation is performed. A Monte Carlo simulation is used to model the probability of different outcomes in a process that cannot be easily predicted because of random variables. A Monte Carlo simulation may also be referred to as a multiple probability simulation. The Monte Carlo method acknowledges that the probability of varying outcomes cannot be

determined because of random variable interference. The Monte Carlo simulation takes a random variable that has uncertainty and assigns the random variable a random value. The model is run and a result is provided. This process is repeated while assigning different values to the variable in question. Once the simulation is completed, the results are averaged to arrive at an estimate. The frequencies of the different outcomes form a normal distribution, which is also referred to as a “bell curve.” The most likely outcome is in the “middle” of the “bell curve.”

The Monte Carlo sampling distribution outcome is then used for Uncertainty Quantification (UQ). The UQ is initiated by requesting that the project manager input an acceptable margin of error for an estimated SOC and a measured SOC. The UQ process calculates the number of samples outside the desired error threshold. The probability of exceedance is then calculated and the optimal sampling design is determined. The optimal sampling design, the sample count, and the simulations per strata regime are evaluated using the Monte Carlo simulation. The simulations are run for each selected strata count in order to select the optimal number of strata. The optimal number of strata is determined so that the selected sampling design, having a minimal number of samples, satisfies the target risk criteria.

With respect to carbon offsets, and the associated carbon credit assignments, the optimal sampling analysis and soil stratification systems and methods described herein model the amount of sequestered CO₂. The optimal sampling with soil stratification validates the tonnage and issue carbon credits for third party verifiers, which require proof of both the geo-spatial coordinates and associated date and time of sample collection. Thus, the optimal number and type of soil samples required to achieve carbon supply credits pertaining to one or more environmental asset value types in one or more plots of land can be effectively modeled using the system and methods described herein.

Additionally, the illustrative systems and methods presented herein for building crop response surface models are based on Random Forest (RF) and Random Intersection Tree (RIT) Machine Learning (ML) training models. The ML training models are used to generate crop response surfaces that indicate the main chemical and environmental contributors to crop-yield production at large scale. Thus, the systems and methods described herein provide choices for nutrient and seed applications by determining the ranked importance of soil chemistry and environmental characteristics to better predict farm and/or field crop-yield performance.

The illustrative system and method presented in FIG. 11 through FIG. 15 uses Explainable Artificial Intelligence (XAI) and Machine Learning (ML) to build a generalized model running as a tool on scalable, client/server web based computing platforms in order to optimize agricultural efficacy is described. The illustrative method accesses crop related information and then proceeds to collect soil and environmental data typically from sparse ground samples. Additionally, external information from satellite (NDVI and Weather information) is also collected. Desired crop types to be grown and previous years plantings and crop cover history are also collected. The collection of data is cleaned and normalized prior to storage in digital format. An iterative Random Forest (RF) ML training model is then used to generate one or more response surfaces to determine crop yield and grower return on investment to make recommendations to improve crop and farm production efficiency.

An agriculture management system and method is described that integrates a supervised machine learning

architecture using one or more multi-dimensional input data-sets, i.e., covariates or feature sets, including past crop yield performance to define soil chemistry characteristics and farmland environment. Input data-sets (also referred to as “features”) are ingested, pre-processed, and run through at least one or more Machine Learning (ML) training model that is used to predict outcomes of yield performance based on one or more predicted response surfaces as described by the present embodiment. The multidimensional input data-sets (also referred to as “covariates”) are applied as independent variables, which are then applied to the ML model which, in turn is used to output one or more estimated crop production response surfaces.

Generally, the term “covariate” refers to a dependent input variable data-set or data-sets that are used as input data to train or tune a Machine Learning (ML) training model. Sometimes, the term “covariate” may be used to refer to an “independent variable” that may also be used to train or tune one more ML training models.

The illustrative ML training model uses at least one form of the Random Forest yield response model to generate response surface predictions. This method uses the Random Forest for ML training model to determine the most important interactions between soil chemistry variables (which are also referred to as “control parameters”) and the other environmental covariates (which are also referred to as “conditioning parameters”). Crop-yield response surfaces may be ranked in order of importance and subsequently interpreted as showing the effect of specific features on crop-yield production. Thus, the training model generated by the illustrative system and method may produce one or more response surfaces for analysis and understanding of how different nutrients may correspond to both the top yield limiters and top yield drivers.

The system and method also use an iterative Random Forest (iRF), a Machine Learning training model, and various other training models to predict crop-yield as a function of soil chemistry, seed genetics and environmental inputs. The illustrative method further describes the definition of a distance metric between pairs of feature vectors which serves as training inputs to the Random Intersection Trees (RIT) training model for further reduction of the data-set complexity. By identifying common predictors and interactions within the input feature-set space, the dimensional-order of the predictive model can be reduced allowing the signal to more accurately be defined.

The system and method further uses a Generalized Additive Model (GAM), derived by a form of Reduced Order Surrogate Modeling (ROSM), to produce a predictive linear additive model that includes non-linear functions. The training model having the predictive functions is then used to determine a series of response surfaces that learn the top contributors and/or limiters responsible for crop-yield productivity. The response surfaces produced by the learned GAM are further used to define nutrient prescriptions, seed-type genetics and product recommendations that improve crop production efficiency at large scale within one or more plots of land based on the sample-sets of multi-dimensional input variables described herein.

The illustrative system and method apply parallel computing resources and machine learning modeling to “cluster” ground types of similar chemical and nutrient composition on large scale plots of land. Subsequently, the illustrative system and method finds correlations of observations that co-associate similar ground types in geo-spatial areas. The discovery of “clusters” of similar ground types is introduced, which allows farm managers and their suppliers to

make decisions regarding where and what to apply for optimal seed selection and crop-yield production in different ground types. By understanding the composition of different ground types through the use of predictive modeling, the characteristics of soil chemistry clustering at high spatial resolution may be understood and farm managers, agronomists and crop consultants can predict and recommend correct field-application quantities, rates, and product-types to apply for optimization of crop production.

Co-associate refers to the likelihood that one or more inputs (independent variables) end up in the same leaf node (or traverse through the same branches) of one or more ML trees in the forest of trees. The observations of these co-associations define the term “Explainable Artificial Intelligence” (XAI) from normal Artificial Intelligence (AI). For example, the strength of the association between two or more independent variables in one or more input data-set(s) is determined by the variables that end up in the final leaf-node of a decision tree and by the observed path of the variables, i.e., do they travel together, as they traverse the different branches of the trees in the forest.

The illustrative system and methods presented herein enable managers to know what products to apply and precisely where to apply them without risk of additional costs or over/under application of nutrient chemicals. Additionally, by understanding the clustering of soil regimes within any farmland location, insurance companies, land brokers and agriculture consultants can better understand and predict the value of the land and the respective crop-production at large scale.

The illustrative system and method apply at least one data-set of soil chemistry, spatial boundaries, previous planted crop-type, cover crops and previously recorded crop-yield data-sets as independent input variables to a machine learning (ML) training model. The illustrative system and method include at least one learned model for soil classification and soil fertility zonation, which is also referred to interchangeably as “crop response zones,” “clusters” or “ground types.”

The classification of clusters and ground types as described herein supports crop planning and analysis through the modeling of the nutrient application placement and seed-type selection, in which the ML training model predicts crop responsiveness for seed-types planted in different ground type zones. Thus, the illustrative system and method describes an agriculture soil management system and the method that generates recommendations of seed-types by building a generic model to accurately predict crop responsiveness to differing soil characteristics with different seed-types planted in one or more plots of land.

The illustrative system and method described uses machine learning to build Explainable Artificial Intelligence (XAI) to generate soil health training models that provide a general classification of soil attributes into a limited set of soil zones called “ground types.” The soil health training model may be used to understand soil fertility, soil health and other soil characteristics associated with various spatial representations. Additionally, the illustrative system and method describes a precision agriculture management training model that integrates the results of a first machine learning (ML) training model into a second ML training model that is associated with a predictive agriculture management system.

In the illustrative embodiment, the first ML training model interpolates estimated soil chemistry covariates from an imported data-set of sparse soil sample and electro conductivity (EC) analytical results. The first ML training

model generates estimated values for soil chemistry covariates at a variety of different field locations.

The results of the interpolated values for the soil variables are then traversed through a second ML training model with an ensemble of the soil covariates and other covariates, e.g., measured crop-yield. The illustrative system and method apply a collection of observations that are based on the path of interpolated soil covariate values, which then traverse the second ML model. The path further leads to a first data set of associations between soil chemistry nutrients, which are treated as “clusters.” The clusters are refined into a classification of soil zones that are referred to as “ground types.” The observation of clusters and classification of ground types may then be used for predictive and recommended farm operations and farm management at large scale without requirements for small plot trial testing.

There are various use cases for the system and method presented herein. More specifically, the illustrative system and method can optimize crop productivity by large scale modeling of certain soil and environmental attributes normally not exposed to growers as first order crop limiters. Additionally, the illustrative system and method can assess soil health and fertility, chemical balances, and nutrient levels. Furthermore, the illustrative system and method can be used to form a large scale farm acreage platform, which can receive data sets that are used to train the explainable artificial intelligence model (XAI).

The illustrative system and method can also be used to provide optimal planting recommendations and yield predictions. The illustrative system and method can be used at a large scale for specific soil types that differ from field to field, farm to farm or region to region. The illustrative embodiment includes a system and method that accesses crop related information, including the collection of soil data from ground samples or from third party soil data.

Additionally, the system and method may be configured to collect additional historical external information, such as input data-sets that are used to train an artificial intelligence training model, from various sources like satellite (NDVI, stand-up, etc.), weather history, ground slope and elevation, future crop types to be planted, previous yield results, planted crop cover, applied fertilizers, organic matter applications, and other such historical external information.

Furthermore, the illustrative system and method cleans and normalizes each data-set prior to storage in digital format in at least one geospatial or other structured database. Further still, the illustrative system and method uses the XAI training model to identify clusters of soil variables with associated attributes. Further yet, the illustrative system and method classifies the results of one or more clusters into a set of ground types where each type has defined co-associated soil chemistry attributes. Further, the illustrative system and method may further recommend different crop-types, seed genetics and specific as-applied nutrients for optimal planting results in each resulting ground type classified.

The illustrative system and method find correlations of observations that co-associate similar ground types in geospatial areas. The identification of “clusters” of similar ground types allows farm managers and their suppliers to make decisions regarding where and what to apply for optimal seed selection and optimal crop-yield production across large-scale acreage.

In operation, the system and method apply Explainable Artificial Intelligence (XAI) and machine learning training models running in a cloud computing environment, on various farmland acreage, importing various data-sets from disparate sources, applying the data-sets to one or more

machine learning models to create farmland recommendations that are used to optimize agricultural production. The illustrative system and method apply machine learning models that traverse through the pre-computed grid of “voxels” containing interpolated soil chemistry and environment data-set features. Also, the illustrative system and method apply a machine learning training model based on the random forest (RF) training model. Additionally, the illustrative system and method observe the path each “voxel” takes as it traverses through the RF tree from root to leaf-node. Furthermore, the illustrative system and method count and vote on correlations and observations where “voxels” end-up in the same end leaf-node. Further still, the illustrative systems and method also co-associate like voxels into clusters (also referred to as ground types) into geospatial areas and compute the co-occurrence frequency of correlated voxels. Further yet, the systems and methods reduce the order complexity of the space by the creation of an NxN distribution Matrix that includes geo-spatially located voxels and apply reduction training models to find complementary clusters of soil chemistry characteristics.

Note, the system and method also support determining carbon and CO₂ extraction and interaction between plant and respective spatially related soil zones.

A system and method for interpolating soil chemistry variables to a plurality of locations within a plot of land is also described herein. The system and method include a trained machine learning training model with at least one set of measured soil chemistry variables, soil conductivity and possible environmental characteristics that are used as inputs to accurately predict soil composition estimations at arbitrary locations within each of a plurality of plots of land. The illustrative embodiment uses machine learning method in combination with distance weighting models to predict soil chemistry characteristics at any arbitrary location within one or more plots of land. The introduction of a hyper parameter tuning loop optimizes the coefficients used in the model to optimize the estimated soil characteristic predictions.

The computer platform (apparatus) describes the Machine Learning method applied to a system and method to estimate higher resolution soil chemical characteristics for any locations to produce an interpolated, higher resolution grid of soil chemistry estimates that can be used in any high precision agriculture practice. Thus, the system and method show a precision agriculture soil management practice using the system and method for building a generic model to accurately predict soil chemistry at any arbitrary location in different plots of land.

The systems and methods described herein use explainable artificial intelligence (AI) and machine learning (ML) in a cloud computing environment to enable a crop prediction engine. Additionally, the systems and methods import various data sets from disparate sources, apply AI and/or ML to the data sets for different farmland acreage. The systems and methods described herein are used to optimize agricultural production by improving crop yields and/or modeling crop yields for different seed types, using different nutrients in different locations.

Generally, geographic, weather, agronomic and environmental factors affect crop production. These factors that control crop production vary. Growers have various options to change planting strategies and control soil composition, however, there are many variables that growers are unable to control.

The systems and methods presented herein improve crop production. Additionally, the systems and methods presented herein may be used to determine carbon and CO₂

extraction and interaction between plant and respective spatially related soil zones. Furthermore, the systems and methods described herein applies at least one data set of soil chemistry, spatial boundaries, previous planted crop-type, cover crops and previously recorded crop-yield data sets as independent input variables used to train a crop prediction engine using machine learning.

Further still, the systems and methods described herein provide at least one learned model for the purpose of soil classification and soil fertility zonation, which is hereafter called “crop response zones,” “clusters,” or “ground types.” The terms crop response zones,” “clusters,” or “ground types” may be used interchangeably. The classification of crop response zones are crop specific and may allow crop planning and analysis through the precise placement of nutrient applications and specific seed-type selections. The machine learning used by the crop prediction engine predicts crop responsiveness for seed types planted in different ground type zones.

Thus, the systems and methods presented herein provide a more precise agriculture soil management and recommendation of seed-types to accurately predict crop responsiveness to differing soil characteristics with different seed-types planted in one or more plots of land.

Further yet, the systems and methods presented herein describe a new class of machine learning (ML) computing platform and software tools to aid farmers and their suppliers in predicting crop yield, managing risks, achieving sustainability objectives, and optimizing input costs. The illustrative systems and methods focus on the creation and use of ML to increase total factor productivity for commodity and biofuel crops while also improving water, nitrogen, and phosphorus use efficiency.

Additionally, the illustrative systems and methods leverage massively multi-scale, multi-modal data to gain insight into the driving forces behind ecosystem services. Current methods of remote sensing measure multiple aspects of cropping systems but fall short in providing meaningful, actionable intelligence for enhancing a farm or field’s agronomic health. Using geospatial technology, soil chemistry analysis, environmental DNA (eDNA) sequencing, and related technologies to discover and define soil regimes, the ML tools of the described systems and methods enable farmers to tailor crop inputs, accounting, and farm management operations for a high degree of soil heterogeneity; and this is a critical missing link, since above ground signatures (e.g., plant growth/activities) are known to be closely linked to below ground properties (e.g., soil moisture/texture/salinity, nutrient, microbial activities).

More specifically, there have been recent studies using UAV and geophysics which have quantified co-variability between above ground signatures and below ground signatures and further identified key controls and limiting factors in ecosystem functioning. In particular, simultaneously using UAV data and geophysical data have identified the soil-plant co-variability and the limiting factors for a soybean yield.

In general, the systems and methods provide a crop prediction engine that can be trained with ML to predict a crop yield in a particular location. Referring to FIG. 1 there is shown a management flowchart for farmers running a typical precision farming operation. The FIG. 1 flowchart shows the typical process used by farmers to grow crops on plots of land under their management. Most of the prior methods use manual communications between partners and non-computer-automated procedures to accomplish crop production and farm management.

Block **1000** presents an illustrative method that shows the typical interactions performed by farmers during a typical growing season. Suppliers may include seed suppliers and manufactures **1300**. Additionally, suppliers may include chemical and nutrient suppliers and manufacturers **1400** and equipment suppliers that may provide farm equipment for purchase, rental, or services for supplies and to apply various products to the fields.

Additionally, the prior precision farming applications typically require some form of client device for storage **1080**, which stores data sets used by the farmer or farmer consultants to improve farm management efficiency with additional precision. Data collection is necessary for farm management analysis, which typically includes yield analysis. Farm management analysis is used to determine what kinds of crops to grow, seed selections to be used and the nutrients required to meet desired crop production output. Furthermore, farmers may interface with banks and crop insurance companies **1200** for the finances and risk mitigation through crop-insurance as necessary for farm management operations.

FIG. 1 presents an illustrative method **1000** that a farmer may apply for seasonal activities during the management of at least one farm using precision farming techniques. The illustrative method **1000** is initiated at block **1010** where the farmer obtains recommendations from an agronomist that may be considered as the “specialist” or “consultant.” By way of example and not of limitation, the recommendations may include application prescriptions in preparation for field treatment and planting information prior, during and post planting seasons. In addition to prescriptions and recommendations, the agronomist may also play a part in advising budget and financial information needed by the farmer to meet management and financial goals.

At block **1020**, the consultant quotes the cost of services which may include seed-type, nutrient applications, and application rates across the farm by field or sub-field area. The method then proceeds to block **1030** where the agronomist or consultant may need to sample the soil in various areas of the farm or by field to get more precise information about particular soil characteristics prior to seasonal planting. Soil samples may be gathered with a spatially sparse resolution due to the high cost of lab analysis and sampling procedures. In other instances, agronomists may resort to third party databases such as Soil Survey Geographic (SSURGO) database or Natural Resources Conversation Service (NRCS) soil maps available from the United States Department of Agriculture (USDA) that contain historical soil chemistry and soil texture data to calculate recommendations.

At block **1040**, agronomists and crop consultants may import and manually process additional data to assist them with application prescriptions and recommendations. Some additional information may include actual farm boundaries, field boundaries, cover crop activities, previous seasonal harvest information along with short and long term weather forecasts. While most of the farmers, agronomists and consultants use university or third party formulas to look up recommendations-some will apply blanket applications that do not take in account the precision recommendations provided by well-known agricultural software applications.

After the farmer makes the farm management and planting decisions based on the decisions made at blocks **1010**, **1020**, **1030** and **1040**, the farmer proceeds to block **1050** where the farmer buys seeds and fertilized from a local farm supply retailer based on the consultant’s recommendation. More specifically, the purchases of recommended seed-types

and chemical nutrients are made from illustrative seed manufacturers **1300**, chemical manufacturers **1400** and other such suppliers. In some instances, farmers may select these supplies from a local retailer or have the supplies delivered to the farms or application locations. In other instances, the farmer may hire third party services or rent equipment for seed-planting or nutrient applications at the consultant’s or agronomist’s recommended quantities and application rates.

At block **1060**, the farmer applies nutrient and seed type products at recommended rates using precision GPS based farm delivery equipment. For precision farming, sub-field application precision is typically accomplished by farm equipment or farm implements that employ the Global Positioning System (GPS) allowing for 1 to 5 Hz application cycles from precision based equipment based on application delivery rates and speeds.

At block **1070**, the crop is ready to harvest and the farmer uses combine equipment equipped with GPS and yield reporting. As the growing season comes to an end, the farmer prepares for the harvest. The farmer may have secured harvesting equipment by purchase through bank loans **1200**, personal funds or through equipment rental **1500** to complete the harvesting of production crops. For harvesting, most modern combine equipment is also equipped with GPS and crop-yield monitoring sensor capability. The yield information obtained by the combine is a critical component of precision agriculture practice because previous seasonal yield results are critical to not only understanding profit and loss.

This yield information can also be used by the crop prediction engine systems and methods described in further detail below; more specifically, the yield information can be used to train the crop prediction engine. Precision application and harvesting equipment are often equipped with data collection apparatus as described in the illustrative embodiments presented below. Also, data collection information may be used by both the crop prediction engine described below and the previous well-known solutions.

Referring to FIG. 2, there is shown an illustrative flow-chart for a crop modeling software application used by growers to estimate farm management operations and predictive yield. At process block **2000**, a variety of different data sets are shown. In general, data set ingestion and processing of the different ingested data sets occurs at process block **2000**. Process block **2000** also indicates the transient and permanent data set types that may be input for processing. Data sets may be obtained from one or more of the farmer’s partners, associations, agriculture organizations, third party satellite companies, government agencies and the like.

Transient feature data for crop-yield **2060** represents land or field characteristics that vary from time to time. In the context of agricultural crop response zones, examples of transient feature data may include yield and/or soil characteristic data. Yields for example may vary within a field from one harvesting season to another. Yield data **2060** may include historical yield maps that represent spatial and temporal yield patterns for the sub-fields. Yield data **2060** may include information about yields of crops harvested from an agricultural field within one year or within several years. Yield data may also include additional geometry information such as a field boundary, a field size, and a location of each sub-field within the field. Yield data may be provided from different sources. Examples of the sources for yield data may include research partners, agricultural agencies, agricultural organizations, growers, governmental agencies, and other such sources.

Also, transient feature data for soil maps **2010** represents land or field characteristics that vary from time to time. Based on weather patterns, erosion, soil carbon emissions, crop-types, nutrient applications and other environmental conditions, soil chemistry can also be considered a transient feature. Soil characteristics for example may also vary within a field or sub-field and from one season to another. Soil characteristics may also include historical data maps representing spatial and temporal patterns but also may not vary as much as other environmental or crop yield transient conditions. Examples of the sources for soil characteristics may include governmental agencies, institutions, agricultural organizations, universities and sensor data from growers or sensor data directly from laboratory analysis of soil samples.

Data for soil characteristics **2010** of a field may be obtained based on soil samples collected from sensors in one of more plots of land or fields. In one illustrative embodiment, soil sampling may be performed using various sampling techniques. For example, for precision agriculture collecting soil samples may be performed at an approximate resolution of one sample per two acres or may vary with samples taken at higher or lower spatial resolutions. Lower resolution samples may be taken at one sample per 10 acres or at one sample per field leading to lower precision data sets for modeling. The samples may be collected at grid points within a field and the grid may roughly form a rectangle or may have no fixed geometry constraints. The original sample measurement data is typically gathered from samples analyzed by local or national laboratories and results are reported in the form of a textural report of the soil chemistry attributes surrounding the sample location. Soil data sets of this form may be stored in files on computer servers or may be only available in paper form.

When soil samples are provided from different sources, i.e., different laboratories, there might be some differences in soil sampling methods, terminology and units. The differences may be caused by nomenclature or different accuracy measures with which the samples were collected. In some instances, differences of chemical characteristics may be a result of different sampling depths at which the soil was sampled. To minimize the impact of such differences, the data sets may be preprocessed by one or more computing devices or manually manipulated to normalize the samples in some data sets. Thus, preprocessing of soil data sets may include removing duplicated samples, samples with no associated values, samples with no geographical coordinate information, and samples with incorrect coordinates and geographical information.

Although some satellite data sets may be considered as a permanent data set feature, some satellite characteristics for an agricultural field may also be considered transient. Satellite feature-sets are typically determined based on temporal satellite maps. Satellite image data may be provided at different spatial, spectral, and temporal resolutions. The satellite maps may provide information about agricultural crop assessment, crop health, change detection, environmental analysis, irrigated landscape mapping, yield determination and soil analysis. The images may be acquired at different times of the year and multiple times within a year.

Transient satellite data, aircraft fly-over and drone-collected data sets are typically also used for crop growth, pest control and nitrogen content analysis. Many of the previous learning platforms are dependent solely on normalized difference vegetation index (NDVI) readings to determine prior season crop production and yields used as input covariates for ML model training. Such transient data sets are important

for bulk understanding of trends but often times do not yield the necessary results for high precision yield prediction modeling.

Permanent feature data **2020**, **2030**, **2040**, are more commonly used and represent characteristics that remain relatively unchanged from one season to another. In the context of agricultural crop response zones, examples of permanent feature data for a field may include characteristics of soil texture, topology and terrain of the farm or field. This permanent feature data is considered permanent because such data usually does not change from one harvesting season to another. Permanent feature data may be obtained from soil survey maps, satellite maps, and bare-soil maps. Permanent feature sets may be provided as data sets from satellite imaging **2050** such as RapidEye, SSURGO soil image data sets, farm/field polygonal boundary data sets and National Elevation Dataset (NED) or USGS elevation map data sets.

Soil survey characteristics may also be provided in the form of soil survey maps as shown in process block **2020**. One source of the soil survey maps is available from the SSURGO database that contains soil survey data of most areas in the United States. Soil survey maps from SSURGO may also be considered a permanent data set. Although soil survey maps may represent a qualitative assessment and lab-analyzed sample data, soil survey maps are considered permanent because most soil survey maps available by government agencies are historical in nature representing overall soil characteristics. The SSURGO survey maps, for example, provide a low resolution of soil measurement data that may be used in the absence of or in combination with high resolution soil sample data. Although the soil chemistry and texture data available in the SSURGO survey maps may be sufficient for the purpose of supplemental data, they are not sufficient standing alone for crop response zone creation. In a particular implementation, the applicable soil texture data is at mukey (a map unit key) level **2**, which indicates that the value of soil texture properties is uniform over the entire spatial polygon and may serve as a substitute for other soil characteristic data set sources.

A typical soil survey dataset from SSURGO is organized as a set of individual map units, each of which covers a polygon area. The data associated with each polygon may include soil chemistry properties and soil texture data, and the data may be provided at different spatial resolutions. The data may or may not be associated with specific geographical point locations.

Historically, the SSURGO data for a set of fields of interest is provided as a set of spatial polygons. The set of polygons may be processed by determining whether the soil texture data was missing for an entire polygon, and if so, a k-Nearest Neighbor (kNN) set of data points may be used to interpolate one or more missing data points. Furthermore, the sand, silt and clay percentages may be normalized to add up to a 100%.

Since the SSURGO maps provide a high resolution of soil measurement data, the soil texture data available in the SSURGO maps may be sufficient for the purpose of a field-zone creation. In a particular implementation, the applicable soil texture data is at mukey (a map unit key) level **2**. That indicates that the value of soil texture properties is uniform over the entire spatial polygon.

Topology characteristics **2030** of a field may include geographical and elevation characteristics of the field. Topology characteristics may include elevation data for an agricultural field, and other topographical properties that may be derived from the elevation data such as the ability

wetness index by calculated slope information as a component of the digital elevation modeling. The wetness index may also be referred to as a Composite Topographic Index CTI, a Topographic Position Index (TPI) indicator, an aspect, a flow direction, and a slope. Digital elevation data sets may also be obtained from different sources, including the National Elevation Dataset (NED). The NED dataset usually provides a resolution of about a third of an arc-second.

Satellite images at this resolution may depict variations in organic matter and drainage patterns. Soils higher in organic matter can be differentiated from lighter sandier soil that has a lower organic matter content. This information may be used in conjunction with other types of maps to define management zones for a field.

Other examples of satellite imagery **2050** include bare-soil maps. Bare-soil map data sets **2040** may include bare-soil characteristics determined based on bare-soil satellite imaging. Examples of such maps may include satellite images from RapidEye images. A typical RapidEye image for a farm or field may contain per-pixel (5 meter by 5 meter) percentage reflectance values for five different bands: red, red edge, blue, green, and near-infra-red (NIR). Using the RapidEye images may provide soil characteristics that may not be available when other types of images are used. For example, the RapidEye data usually represents topsoil better and deeper than other types of images.

For example, a set of bare-soil maps **2040** may be pre-processed for each field. The images that contain cloud cover may contaminate image quality and may be discarded or use an ensemble of VIR and SAR satellite technology to compensate for cloud cover contamination.

In general, process block **2000** represents one or more of the raw input variables, features, feature-sets, data sets or covariates used to train the crop prediction engine. At process block **2100**, input data sets may be uploaded to permanent storage through one of many client devices as illustrated in further detail below in FIG. 3A. Uploads from program instructions in process block **2100** may also be data imports from other data-sources retrieved in the form of Application Programmable Interface (API) software, or by other such processes and systems for importing data. Besides uploading data features, pulling data, importing data, or downloading data from database systems or storage platforms may also be represented in process block **2100**. Some client devices are used to upload data sets dedicated to only the uploading or data retrieval process.

The upload process may be part of the crop management software running at least partially in one or more client devices. The upload process may also be enabled through client devices with network connections directly from third parties such as equipment manufacturers, agriculture consultants and farm co-ops or retailers.

By way of example and not of limitation, the process of uploading of data sets **2100** to the illustrative crop prediction engine **4000** (shown in FIG. 3A) may be performed by the various client devices **3010** through **3080** shown FIG. 3A, or the client devices **4400**, **4500** and **4600** shown in FIG. 4, the application computing cluster **4100** shown in FIG. 3A, or any other such device capable of receiving or generating a data set **2000**.

Referring back to FIG. 2, process block **2110**, process block **2120** and process block **2130** represent program instructions for pre-processing, normalization processing and smoothing of imported data sets **2000** after importing through data upload **2100** program instructions. Program instructions for process blocks **2110**, **2120** and **2130** may be

executed selectively, optionally, sequentially, or in parallel. The manner in which the tasks are performed may vary based on the implementation and the quality of received data set data. For example, some of the received data from various data sets may need preprocessing, but not smoothing. Other data may need only normalization processing, while other data may be normalized naturally as it traverses a random forest (RF) machine learning model and, thus, does not need pre-processing or normalization. Selecting one or more of process blocks **2110**, **2120** and **2130** may be based on a manual inspection or a machine based inspection of the received data and may be performed in process block **2000** prior to the data set upload by client application program instructions at process block **2100**.

Data sets **2000** may include sub-field observations that further include contaminated observations. Contaminations may be caused for a variety of reasons, for example, the way the crops are harvested, or by the way the data in each data set is collected or recorded. The contaminated observations may include outliers, invalid data, redundant data, missing data, and the like. In one embodiment contaminated data may be imputed to resolve areas where training data is required to enable crop-yield predictions by the ML model.

Program instructions for pre-processing data sets, indicated in process block **2110** program instructions, may include identifying data items that are outliers, invalid, redundant, missing or collected data from outside a field boundary. Such data may be removed, substituted, or imputed from an average from nearest neighbor data, supplied from various data set sources or substituted by various data sets from previous seasonal results. For example, missing yield data from one season may be substituted or averaged from another season. Pre-processing may also include identifying, and removing, the data set observations collected from one or more plots of land on which multiple crops were planted in the same season. Pre-processing may involve data set removal when data is considered to be redundant across one or more fields.

Pre-processing of data set data may also be performed to reduce or eliminate invalid data, which is also referred to as removal of noise observations or “de-noising” of data within particular data set inputs. Noise reduction is the process of removing noise from a raster image, vector, polygonal or textural based data set and may be necessary for feature preservation. De-noising of data within data sets may be based on a manual inspection to identify when noise is present or through machine-based inspection of the received data. De-noising may also be performed in the program instructions of process block **2000** prior to data set uploading. Pre-processing may include identifying the noise observations, and replacing the noise observations with approximated values or by other method(s) known to those knowledgeable in the art.

Process block **2120** represents the program instructions for normalization and imputation of yield data between different crop-yield data sets received from different sources. Imported data sets may be analyzed to determine that sufficient crop-yield data are available for ML model training or if crop-yield data are sparse or missing for at least one field within one or more farms. If less than two years of crop-yield data for any field are provided, then the yield maps for years not provided may be supplemented and sourced by other processes. Supplemental crop-yield information may include data sets from fly overs, drones or satellite images and may also include crop-yield data sets from organizations, government agencies, third party agronomist groups, agricultural co-op retailers or by other

organizations. Additional data set information may be used to supplement farm or field areas where crop-yield data is sparse, noisy, or missing. The additional crop-yield data sets may be averaged, added, subtracted, or substituted with other data sets to provide multiple years of crop-yield data sets for crop-response modeling.

Additional preprocessing and filtering of the crop-yield data sets **2120** may include adjusting the data values to account for grain moisture. By adjusting for grain moisture, crop-yield data records can be corrected for different seasonal values where the crop-yield data is different from the standard moisture level of 15%. Additional processing may also include correcting yield productivity data to account for data inaccuracies when experimental crop-yield data is provided. This may include correcting the crop-yield data if the data was pre-smoothed by a one or more of the crop-yield data providers. This type of additional processing is recommended to reduce the effect of improperly smoothed crop-yield data on the results of the crop response zone creation.

Pre-processing of data set input variables, performed by the program instructions of process block **2110** and the normalization of crop-yield data **2120**, may also require transformation from one geo-spatial coordinate system type to another. This transformation to a common coordinate system allows the unification of imported data sets into a common format. For example, transformation from latitude-longitude coordinate system to the Universal Transverse Mercator (UTM) coordinates results in mapping all data set coordinates onto a grid that has been previously defined for one or more locations. Transformation from other coordinate standards to a single standardization format allows data set records to be mapped to one common format for multiple different locations and/or the same location when different imported data set records have different reference coordinate systems. As a further example of coordinate transformation, if the received data was sampled in the system other than the UTM, then pre-processing of data may include adjusting the received data sets to some known grid or area UTM resolution. This may also include programmatically projecting the received data set data onto the UTM coordinates. Missing sample values may be interpolated at the UTM coordinates from the available data using a Gaussian model with a constant trend and parameters that are obtained using a maximum likelihood estimator.

Additionally, the pre-processing of input data **2110** may integrate topological information from digital elevation maps, Compound Topographic Index (CTI) or from farm/field equipment that directly monitors field elevation during planting, harvest, and nutrient applications. The elevation map services may be used for multi-resolution visualizations to explore hillsides, aspect, slope, as well as contour maps in raster image formats. Included in some governmental agencies, like USGS, is the ability to download through API point values for elevation based on GPS coordinates. The data sets originating from Digital Elevation Maps (DEM) may be used to calculate slope data and CTI wetness index. Pre-processing may also include conversion of DEM data sets into a normalized wetness index used as input covariates to train the crop prediction engine.

The ensemble of DEM data sets and digital elevation input from farm and/or field equipment may be normalized and used to determine elevation and slope information used to build a composite wetness index for the areas of interest.

In the illustrative embodiment an ensemble of covariates containing multiple elevation data sets may be used to build a Random Forest (RF) generalized DEM model. By way of

example and not of limitation, Inverse Distance Weighting (IDW) or other forms of interpolation, as known to those in the art, may be used to average the multi-dimensional DEM covariates within the farm and/or field location boundaries. Additional steps may be needed to normalize, apply weighting and transformation to the separate inputs formatting covariates prior to the interpolation. The interpolated output of the results may be used to build a comprehensive wetness index model for one or more farms or fields.

In the illustrative embodiment a combination of the multiple DEM sources require transformation between various data units to form a common unit format. Additionally, transformations between raster, point and polygon formats may be needed to unify the separately sourced DEM input data sets to form an ensemble of unified DEM data. For example, processing data set transformations to UTM coordinate representation and further performing transformations between point and raster coordinates from both farm equipment based DEM data and database DEM registries may be required. Interpolated DEM results may need further conversions and normalizations in order to form the ensemble of elevation averages used for wetness index determinations to train one or more crop prediction models.

Additionally, further pre-processing of the ensemble of DEM interpolated variables may include extracting cell values of the elevation raster where crop-yield or soil characteristics projected within a spatial point lies. If no cell raster, point, or polygonal area is found for the DEM projection within the farm and/or field boundaries of interest, then an indication of no DEM values may be returned.

In projecting the image data onto the UTM coordinate system, values of the image data at the location points of the various data sets **2000** may be obtained by rasterizing the data sets. The results may be transferred to data set raster cells. If one cell of one or more data sets is covered by multiple imagery bands of data points, then an arithmetic mean of the values may be used to associate the data points with the raster cells. For example, pre-processing SSURGO soil data coming from governmental agencies or third parties for analysis would require a projection of the coordinates of the spatial polygons into UTM coordinates. Once such transformation is completed, the SSURGO polygons may be overlapped onto the spatial locations of the crop-yield training data collected for the particular plots of land being modeled.

Process block **2120** further represents the program instructions for yield data set normalization of the received yield training vector. In the illustrative embodiment, the crop prediction engine uses the RF module to complete the data normalization as a function of the nature of the random forest, and thus process block **2120** may not be needed. Crop-yield normalization may also be required across different crops, farms, and fields.

Historically, data density processing includes using an Empirical Cumulative Distribution Function (ECDF) transformation. The ECDF transformation may be performed on the crop-yield data sets for each field and year so that the transformed crop-yield data is within a certain range across different crops and fields. For example, the ECDF may be applied to the received yield data to transform the data into transformed yield data in the normalized range of [0, 1]. Once the yield data is transformed, the transformed yield data may be compared across different years and across different crops, such as corn, soy, or wheat.

In the illustrative embodiment, a separate random forest module may be used for each different crop type to be

planted—by using the random forest module there is no need for an ECDF transformation between crop types.

Process block **2130** represents program instructions for smoothing received data. Data smoothing may include testing whether any crop-yield data records are missing, whether the crop-yield data records need to be further smoothed, or whether certain crop-yield data records need to be removed or interpolated. Spatial smoothing is a process of removing inaccurate observations from the collected crop-yield observations and defragmenting the obtained delineated zones. Historically, the spatial smoothing may be performed using a kernel-smoother or a stationary Gaussian process.

Depending on the quality of the received raw data, data smoothing may be performed on either raw data or pre-processed data. A kernel smoother is a statistical technique for estimating a function of observations when no parametric model for the function is known. The resulting estimated function is usually smooth and may be used to remove the noise observations from a set of yield data. Kernel smoothers are reliable and useful nonparametric estimators and may be selected to perform the spatial smoothing of the yield data. Examples of kernel smoothers that can be used to smooth the yield data include Gaussian kernel, inverse distance weighting kernel, rectangular kernel, triangular kernel, bi-square kernel, tri-cube kernel, and tri-weight kernel.

Process blocks **2140-2170** represent the programming instructions required for clustering farm and/or field areas into crop response zones. Clustering is a process of grouping data, i.e. crop response zones, into clusters and determining cluster labels for the clusters. Clustering the zones **2150** may be performed by merging small zones with larger zones. For example, zones with sizes smaller than a particular size may be merged with their most similar large neighboring zones. The particular size may be set manually or by automatically using database lookup, programmatically or by use of fixed configuration data.

Process block **2140** represents the program instructions for pre-processed data representing transient and permanent characteristics of an agricultural field used to delineate a set of crop response zones for the agricultural field. Historically, the set of delineated crop response zones may be represented using stored digital zone boundary data. Zone characteristics may be created by applying centroid-based approaches, such as the K-means approach, or a fuzzy C-means approach.

The process executed in process block **2140** may be repeated by program instructions of process block **2170** one or more times until the quality of the created response zones is satisfactory. The process may be repeated using different criteria, different parameters, or different parameter values.

A set of delineated crop response zones may be analyzed to determine whether some of the zones may be merged as represented by the program instructions in process block **2150**. For example, a set of delineated management zones may be analyzed to identify small zones and to determine whether the small zones may be merged with neighboring larger zones. Small zones may be identified automatically by a computer system, or manually by a user of the computer system. The computer system may display information about the set of first response zones to a crop grower in a graphical user interface that is programmed with widgets or controls to allow the grower to remove undesirable fragmented small zones, or to merge **2150** the fragmented small zones with larger zones. The merging of zones results in obtaining a set of merged response zones. There may be

components of clusters that have no resolved values or that may be out of scope due to noise, bad data, missing data, or other modeling reasons. These results may be undefined for many reasons and may be dropped from further prediction and analysis. In the case where prediction of zones cannot be accomplished, a simple median smoothing or strict substitution may be used to resolve the inconsistencies.

The program instructions executed in process block **2150** may be repeated one or more times until the program instructions of process block **2170** indicate that no small zones are identified in the set of crop response zones. The process may be repeated using different criteria, different parameters, or different parameter values. Small crop response zones that cannot be identified manually or by computer programming identification are passed to process block **2160** for further post-processing.

In process block **2160**, a set of response zones is post-processed. Post-processing of the crop response zones may include eliminating the zones that are fragmented or unusable. The post-processing executed in process block **2160** may be repeated one or more times when further processing is identified by process block **2170** until the quality of created response zones is satisfactory. The process may be repeated using different criteria, different parameters, or different parameter values.

Metadata about the created response zones is generated and stored. Furthermore, a test may be performed to determine whether the process of delineating response zones needs to be repeated. If the delineation process is to be repeated, then further delineating of the response zones is identified by program instructions of process block **2170** and repeated by the program instructions of process block **2140**.

The response zone delineation process is performed for different values of a response class. A response class refers to one or more areas in a farm and/or field that have relatively homogeneous crop-yield limiting factors. The areas are not restricted to a spatial continuity. For example, several response zones which are spatially separated from each other could belong to the same response class and could be operated or managed in the same manner.

Process block **2180** represents the repository for processed response zones that store the delineated pre and post processed data sets as identified and labeled as crop response zones. Crop response zones are categorized as clusters, for one or more of many geo-spatial boundary areas for each of the farms or fields. The characteristics of the stored crop response zones represented by the processed cluster characteristics and identified by the geographical zone boundaries may also be called ground types in the illustrative embodiment.

Referring to FIG. 3A there is shown an illustrative system having distributed client devices that interface with a crop prediction engine **4000**. The illustrative system includes industry standard client devices such as desktops, notebooks, laptops, Android, and iOS mobile or stationary client devices.

In the illustrative embodiment one or more users upload data, issue commands, and retrieve results using one or more client devices **4400**, **4500** and **4600** over Internet **3000** to interface with program instructions running on both the client devices and the apparatus platform of the illustrative embodiment shown in FIG. 4. The program instructions running on the application computing cluster **4100** and the data science computing cluster **4300** form the backbone of the crop prediction engine **4000** used to predict crop yield.

Process block **1080** in FIG. 3A provides a representation of a storage device with programming instructions to store,

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send and receive various data sets stored by the farmer or the farmer's agronomist, consultants, suppliers, or other representatives that are typically local to the farm under management. In another embodiment the farmer data sets **1080** may be stored in remote storage maintained by the farmer's equipment suppliers, software vendors and/or consulting partners.

Process block **2000** in FIG. 3A includes a representation of data sets provided by third-parties that are generally supplied for entire regions and may not be exclusive to a specific farm or field. These data sets may be from database servers or from permanent storage devices and may contain public or private data sets. Data sets from third-parties are typically accessed over the Internet but may also be acquired from direct storage devices on their respective host platforms.

In the illustrative embodiment, farmer data sets **1080** and external data sets **2000** are uploaded and/or queried remotely through at least one Internet **3000** network connection connected to the Application Interface **4110** by one or more computing apparatus running program instructions on one or more computing clusters **4100**, **4300** or in one embodiment on at least one computing cluster and the crop prediction engine. The external data set storage of process block **2000** stores and maintains non-farmer specific data. Non-farmer specific data may be data sets collected by sensors on farm equipment and/or sampling equipment local to the area where specific farm management is performed. Process block **2000** represents data sets that are typically stored in raster format from geographic regions where the feature set characteristics are typically remotely gathered for large portions of land. For example, external data sets **2000** may be from satellite, flight recordings, agriculture or government agencies and private companies that sell data sets and the like.

As used herein, a "portion of land" or "plot of land" refers to any amount of land in any shape or size. For instance, a "portion of land" can refer to a farmer's entire property, a field, a plot of land, a planting region, a zone or a crop response zone, and the like. Likewise, a portion of land can include one or more "sub-portions" of land, which refers to a subset of the portion of land of any shape or size. Various types and formats of data may be stored in both the sensor data sets from the farmer **1080** and external data set storage **2000** for access by the other components of the crop prediction system **4000** performing one or more machine learning operations in order to train the crop prediction engine.

Prediction information from the system **4000** is used to predict crop production for a portion of land, and to assist in identifying one or more sets of farming operations including recommended applications like the addition of recommended nutrients and seed types to optimize crop production. For each of the farmer data sets **1080** and external data sets individual files of data may be stored as flat files, in columnar storage, in binary format, or other such formats that may be accessed via one or more storage devices, relational databases, columnar databases, NoSQL storage, horizontally scaled databases, and other such databases.

As used herein, the term "crop prediction engine" refers to a hardware component, e.g., a cloud based server, that implements one or more "machine learning prediction models" or a "prediction model" that uses one or more machine learning operations to predict a measure of crop production based on information including field information, or that is trained on information including field information using one or more machine learning operations. In the illustrative

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embodiment, the crop prediction engine generates a predicted measure of crop production.

By way of example and not of limitation, the crop prediction engine **4000** can be trained by any machine learning operation, such as those described herein, or any combination of machine learning operations for predictions of crop production.

The term "crop prediction information" refers to any measure that relates to expected crop production, such as crop yield, crop quality, crop value, or any other such suitable measure of crop production. Crop prediction information can also refer to a set of farming operations expected to result in the measure of expected crop production when performed in a specified manner, at a specified time/location, and the like.

"Field information" can include one or more of past and present crop production information, past and present geographic information, past and present agricultural information, past and present agronomic information, past and present sensor data associated with crop production, any other information related to the planting, growing, and harvesting of a crop, and any other field parameters as described herein.

Referring to FIG. 3A there is shown an illustrative system having distributed client devices that interface with a crop prediction engine **4000**. In the illustrative system, the system includes at least one of the following client devices, namely, a farmer client device **3010**, an agronomist client device **3020**, a crop purchaser client device **3030**, a retail supplier client device **3040**, a seed manufacturer device **3050**, a chemical manufacturer client device **3060**, a banker client device **3070**, and a crop insurance agent client device **3080**.

Additionally, the system environment of FIG. 3A includes the crop prediction engine **4000** communicatively coupled to components that include application computing cluster **4100**, disk storage **4030** and data science computing cluster **4300**. Furthermore, the application computing cluster **4100** includes an application database **4150** and the data science computing cluster **4300** includes a data set database **4350**.

As illustrated in FIG. 3A, the multi-dimensional input data sets represent the "features" of the environment that may be stored locally in disk storage **4030** or remotely stored in a data set store **1080** (shown in FIG. 1) or the input data sets may be represented more generally as a data set **2000** (shown in FIG. 2). The feature data sets are input prior to any processing by the crop prediction engine **4000**. The feature data sets may represent the raw input that after possible pre-processing and de-noising may form the independent variables used to train the crop prediction engine **4000**.

The data sets generated from the various client devices and/or various data sets shown in FIG. 3A may be implemented within the same computing device or may be implemented in a mixed fashion between one or more different computing device at one or more physical locations. In another embodiment, the network communication between a client device and the crop prediction engine **4000** may not only include the Internet **3000** but may be embodied as one private, public or VPN networks where one or more of the client devices of FIG. 3A are communicatively coupled to the crop prediction engine **4000**.

In yet another illustrative embodiment, the crop prediction engine **4000** may be integrated with one or more client devices in a manner where the client devices of FIG. 3A communicate directly with each other and thus do not require a network connection to one or more remote crop prediction engines running on one or more remote computing clusters.

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Referring to FIG. 4, there is shown an illustrative high-level process block diagram of networked computing systems. The networked computing systems include a client device **4400**, which may be any device having computer functionality.

Client device types may include a mobile telephone, a smartphone, tablet computer, a desktop computer, a laptop computer, a workstation, a personal digital assistant and may include smart farming equipment such as a smart tractor, smart harvester, smart planter, and fertilizer as-applied implements, including smart soil sampling and irrigation devices and the like. Other client devices may be embedded into unmanned vehicles, drones, or other aircraft and may be remotely controlled or be autonomous.

In the illustrative embodiment, client devices are configured to communicate with at least one crop prediction engine **4000** (shown in FIG. 3A) of the illustrative embodiment via internet **3000** or through one or more other networks' architectures using one or more different network protocols for data transport and communication. In such an embodiment, the communication is by device client via a web-browser interface but may also be of another method like an application programming interface (API), wired interface such as Universal Serial Bus (USB) or on a private network where a client device is typically accessed by a native operating system resident on a client device.

Referring back to FIG. 3A, the farmer client device **3010** may communicate with the crop prediction engine **4000** via the Internet network **3000** to upload farm related data sets collected by various smart farm equipment equipped with at least one computing device and Global Positioning Systems (GPS). Some uploaded information may be input to a client device manually by the user such as tilling or no tilling, irrigated or not, previous cover crop information, application of nutrients or pesticides by air, flat applications of manure, applications that do not have GPS, quantity and rate monitoring and other information that is not autonomously generated by smart farming equipment. Additionally, the farmer client device **3010** may be used to request and receive prediction information and recommendations such as predictions of crop production and recommendations of soil nutrients and seed products to apply to achieve a predicted or desired crop-yield.

The interactive communication of data sets and visualization between the farmer client **3010** and the crop prediction engine **4000** allows the farmer to optimize crop yield at lower application costs, allowing the farmer to identify farming operations that can optimize crop production and efficiency to improve the overall seasonal return on investment.

In another illustrative embodiment, the farmer's client device **3010** may be used to communicate with other client devices attached to the network through the application interface **4110**. The communication between users of a client device and the crop prediction engine **4000** allows for acquisition of required supplies, negotiations for product costs and delivery's, planting, and harvest scheduling, and expected crop-yield output based on farm management and application practices.

Thus, the illustrative crop prediction engine **4000** is configured to circumvent the delays and added expense of the historical practice illustrated in FIG. 1 by circumventing the typical retailer supply chain. Circumvention of standard retail purchase and delivery using the illustrative embodiment opens up options for the chemical manufacture client device **3060** and seed manufacture client device **3050** to sell directly to local farm management personnel.

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For example, the crop prediction engine **4000** may prescribe nutrients, seeds or other farm management supplies and based on such predictions autonomously place purchase orders directly to manufacturers and suppliers. Additionally, a client device may be used to directly obtain bids from crop purchasers and/or crop brokers **3030** who can view estimated production volumes and prices directly. The prediction engine **4000** inputs data from multiple client devices to source information and derive the optimal soil and seed application costs for desired crop production in preparation for planting by allowing the farmer, agronomist, or crop consultant to rely on trained AI models to understand the multiplicity of soil and seed characteristics decisions. Thus, recommendations for optimal crop efficiency with precision application may be obtained through the crop prediction engine **4000** described herein.

In the illustrative embodiment, the agronomist client device **3020** may communicate with the crop prediction engine **4000** via the Internet network **3000** to access both farm information collected by smart farming equipment and to access crop prediction information generated by the crop prediction engine **4000**. When permission from the farmer is granted through the farmer client device **3010**, data set information stored by the disk storage device **4030** may be downloaded through the agronomist client device **3020** interface. Other data sets provided by the agronomist may be uploaded over the Internet network **3000** to the crop prediction engine **4000** through the application interface **4110**. The agronomist may use recommendations from the system, modify those recommendations, upload data sets such as laboratory soil sample reports, field, or farm boundaries, known environmental variables, fertilizer and/or seed recommendations including prescriptions. The agronomist may also obtain market pricing based on crop-yield predictions from the crop prediction engine.

Other information shared by the farmer with one or more agronomists and/or consultants may include information about farm management operations such as tilling or no tilling, irrigated or not irrigated, previous cover crop information and other information that is not autonomously generated by smart farming equipment. Additionally, the agronomist may request yield information stored in the crop prediction engine **4000** such as previous year's crop-yield results from harvest of specific farms or fields. The interactive communication of data sets and visualization between the agronomist client device **3020** and the crop prediction engine **4000** allows the agronomist to further recommend suggestions that may optimize crop yield at lower application costs, optimize crop production, and improve the farmers overall seasonal return on investment.

In another embodiment the agronomist's or consultant's client device may be used to communicate with other client devices attached to the network **3000** through the application interface **4110**. The communication between users of a client device through the system may be used for acquisition of required supplies, negotiations for product costs and deliveries, planting and harvest planning and expected crop-yield production and return on investment based on farm management and farm and/or field application practices.

For example, the user of the agronomist client device **3020** can change the type of fertilizer to be applied, based on supply and demand, and can change the harvest data by moving the date based on expected weather changes. Thus, the agronomist through the agronomist client interface associated with agronomist client device **3020** can modify suggested farming operations as a result of the predictions from the crop prediction engine. The agronomist client device

3020 may also be the same as the retail supplier client device **3040** or the crop purchaser client device **3030** or the same for any other client device shown in FIG. 3A.

In the illustrative embodiment, the crop purchaser's client device **3030** communicates with the crop prediction engine **4000** through the application interface **4110** via the Internet network **3000** to receive crop prediction information for future crop production in one or more portions of land. The purchaser, through the client device **3030** may send bids or purchase requests to secure purchase rights for estimated or actual crop harvests (for instance, sending bid contracts to the farmer client device **3010**). The purchaser's client device **3030** may be used by one or more crop-brokers or other crop recipients who has been granted permissions by the farmer to share or view farm projects that may include future or current harvest information predicted by the crop estimation engine **4000** on the application computing cluster **4100**. For example, the purchaser through the purchaser client device **3030** may enter agreements to obtain from the farmer a portion or all of an expected crop harvest. Thus, the purchaser through the purchaser client device **3030** accesses the crop prediction engine **4000** via an interface **4110** that allows the user of the purchaser client device **3030** to identify predicted crop production information from one or more farmers who use one or more embodiments to automate the generation of crop acquisition agreements with the one or more prospective crop purchasers. A crop purchaser (crop recipient) may receive a harvested crop directly from a farmer or from the farmer through a crop broker.

The crop purchaser client device **3030** communicates with the crop prediction engine **4000** via the Internet network **3000** to receive information via the application computing cluster **4100**. For example, corporate users may have one or more registered farmers, brokers or consultants registered in a corporate organizational account such that any corporate user of at least one of the crop purchaser client device **3030** may identify expected crop production of one or more farmers, including the crop type and expected crop production quantities of the crops produced by one or more farmers. In one embodiment the application computing cluster **4100** may include shared farmer accounts, agronomist accounts, consultant accounts or customized projects supported by the crop prediction engine **4000**. For example, the accounts can be from a group of farmers in a single location, multiple locations, from a geographical region or a corporate division. A corporate user of the crop purchaser client device **3030** can use this information to enter into crop acquisition agreements with one or more farmers or one or more brokers or one or more crop recipients.

In the illustrative embodiment, the retail supplier uses one or more client devices **3040** to communicate with the crop prediction engine **4000** through the application interface **4110** via the Internet network **3000** to send and receive information about farm management operations, supplies and services for historical, current, or future crop production relating to one or more portions of land. Typical use of the crop prediction engine, between farmers and retail suppliers, may be for acquisition and scheduling of supplies and services that may be based on the estimations and recommendations from the systems crop prediction engine **4000**.

For example, the user of retail supplier client device **3040** may be an agriculture co-op, farm supply company or seed and fertilizer retailer who has been granted permissions by the farmer or corporate vendor or partner to view crop prediction engine projects that may include farm chemical and seed application requirements and harvest information that are created or stored on the application computing

cluster **4100**. The retail supplier through the retail supplier client device **3040** may enter into agreements to obtain information and requests from one or more farmers or enter into various agreements with various requests from corporate suppliers through one or more other client devices illustrated by blocks **3050**, **3060** and **3070** of FIG. 3A. Information transfer between farmer client devices **3010** and retail supplier client devices **3040** with other client devices users may relate to client requests like farmers requests for retail services or agronomist services through client device **3020**, or requests for other supplies or services through any other client devices. The retail supplier client device may also be used to share logistics and order information with other manufacturers such as seed manufacturers through client device **3050** and chemical manufacturers through client device **3060**.

In another embodiment the retail supplier client device **3040** may be used for logistics for purchase and delivery of fertilizers, pesticides and seeds as required by the retail supplier's farming clients, agriculture consultants or agronomists. For example, corporate users may have one or more registered retail suppliers, registered in a corporate organizational account, such that any corporate user of at least one of the retail supplier client devices **3040** may identify expected farm management and crop production requirements of one or more retail suppliers. A corporate user of the retail supplier client device **3040** can use the crop prediction engine **4000** and associated crop prediction information to enter into crop acquisition agreements with one or more manufacture suppliers or one or more farmers or one or more agronomists. Thus, based on information in farming projects shared by one or more client devices with the retail supplier client device **3040**, the retail supplier may be contracted through the illustrative embodiment for supplies and services provided by the retail supplier.

The seed suppliers or manufacturers, chemical or fertilizer manufacturers, loan officers or bankers, and crop insurance agents, hereafter called the "corporate users," may be considered corporate suppliers of products and services sold, leased, or licensed that communicate through their respective client devices as represented in FIG. 3A by client devices **3050**, **3060**, **3070** and **3080**. The use of the crop prediction engine **4000** by corporate users is similar to the use and operation of the system by the crop purchaser through a client device **3030**. Corporate users may have corporate accounts and levels of access as illustrated in FIG. 3B and may communicate and share projects with retail, agronomist, consultant, and farm clients, hereafter called "local clients" through local client devices **3010**, **3020**, **3030** and **3040** connected to a network such as the Internet network **3000**.

The crop prediction engine **4000** includes an application interface **4110** which includes programming instructions for the network interface. The application interface **4110** is the connection to one or more networks used for the transfer of information between the client devices and the crop prediction engine **4000**. The application computing cluster **4100** includes the computing components to manage the client's computation and storage requirements. In one embodiment, the application computing cluster **4100** and the data science computing cluster **4300** may be the same components with a single computing cluster and database structure for both. In the illustrative embodiment, the data science computing cluster **4300** and related data set database **4350** are used to perform various operations described herein.

The crop prediction engine **4000** receives data from various sources including, but not limited to, a corporate

user, data set types **2000**, local sensor data sources **1080**, and other such data sources to perform machine learning operations on the received data produced by one or more crop prediction models. The crop prediction engine **4000** has the ability to predict and recommend precision application of chemicals, seed types including seed genetics and recommended amounts and rates of farm and/or field application. The data from these data set sources, which may also be referred to interchangeably as “features,” can be combined into a feature-set that may be extracted from the combined data and used as training data to support crop prediction models spanning different temporal systems, different spatial coordinate systems, and different measurement systems.

For example, sensor data streams can be a time series of scalar values linked to a specific latitude/longitude coordinate system. Similarly, LiDAR data can be an array of scalar elevation values on a 10 meter rectangular coordinate system, and satellite imagery can be spatial aggregates of bands of wavelengths within specific geographic boundaries. After aggregating and standardizing data from these data streams in a universal coordinate system, such as UTM reference system, feature sets can be extracted and combined. For example, feature sets that can be extracted and combined include a soil wetness index from raw elevation data, or cumulative growing degree days from crop types and planting dates.

“Crop quality” can refer to any aspect of a crop that adds value to a farmer, crop purchaser or crop recipient. The various possibilities for crop quality and crop quantity inferences can be large. In general, crop quality may refer to a physical or chemical attribute of any particular crop. For example, there are many different attributes that are related to a particular crop, namely, a genetic trait, a modification, an epigenetic signature, moisture, protein, carbohydrate, ash, fiber, fat, oil, color whiteness, weight, transparency, hardness, presence of foreign matter, absorption, milling degree, kernel size or volume distribution, grain length or breadth, density, broken kernels, stress cracks, immature grains, measurements and measurement expectations for gluten, sodium dodecyl sulfate, sedimentation, toxin levels, molds, insects, or other material damage and attributes related to crop quality.

Crop quality may also be inferred from the maturity or growth stage of a crop, the storage of a harvested crop (e.g., the type of storage: bin, bag, pile, in-field, box, tank, other containerization), the environmental conditions (e.g., temperature, light, moisture/relative humidity, presence of pests, CO₂ levels) to which the crop encountered during storage, preserving the crop (e.g., freezing, drying, chemically treating), or a function of the length of time of storage.

In another embodiment the food grade quality may be inferred from the variety of plant, damage levels, soil chemistry or one or more production practices used to grow the plant. In some embodiments, crop quality is a calculated, derived, inferred, or subjective classification based on one or more measured or observed physical or chemical attributes of a crop, or a farming operation used in its production. In other embodiments, one or more quality metric is collected, measured, or observed during harvest. For example, dry matter content of corn may be measured using near-infrared spectroscopy on a combine. In yet another embodiment, a quality metric is a grading or certification by an organization or agency, for example grading or certification by the USDA, organic or non-GMO certifications.

The system of FIG. 3A allows users to name at least one “project,” share at least one project having viewing rights and/or editing rights with other client device users. The

access may be provided from at least one client device having a client browser that is communicatively coupled to the Internet **3000**. In the illustrative embodiment, “projects” are set up and managed by users and contain information such as data sets, features, predictions, and ROI estimations for one or more plots of land. The projects may contain information for any access levels where the user has permission as described in FIG. 3B.

Referring to FIG. 3B there is shown an illustrative system that supports access levels within corporate management in a typical organizational hierarchy. The illustrative system in FIG. 3B includes eight possible levels of organization, in which the hierarchy assigned to each of the eight levels and the respective access privileges for each level is shown. In one embodiment the application administration interface is initially used to set-up one or more corporations to follow one or more of the active organizational hierarchy levels out of the eight possible levels. In another embodiment there may be more or less levels than eight set-up by the administration user interface software. In yet another embodiment the corporate names assigned to each level may be the actual usernames assigned to each account holder and names may be set by the organization name custom to the user account for each level. Names of the organizational hierarchy may be set by the administration interface or by the root user (or root users) assigned to the corporation during initial set-up of the main account. Set-up of the main account, the number of levels, the usernames per level and the organizational level name is typically performed by the application administrative interface of the crop prediction engine. In another embodiment where the root user is a single sole-proprietor of any company or entity, the term corporate user may not apply. For the purpose of this specification, a single root user may also be referred to as “a corporate user” in the context of a single level of the organizational hierarchy.

In FIG. 3B the top level of access of the organizational hierarchy is called the “Corporate HQ Executive” level **3110** which has level-1 (highest) access privileges and may be considered the highest level of the organizational hierarchy as described in the present hierarchy template. The users with level-1 privileges can view and access not only information from their own account but in addition any user account, including all the information within the account below this user level. For example, from block **3110**, the indicated EVP “Bret” can view all the projects, see crop prediction reports, establish return on investments, determine correspondences or any other information for system access levels 2 through 8, i.e., any access level below Bret’s access level. In another embodiment any number of additional corporate HQ executives **3160** may be added to the organization by the system administrator or by the root user assigned to that access level of the illustrative system.

The organizational hierarchy and assigned names continue for the levels selected by the administration users under advisement from the corporate root user for the remaining assigned levels of the organization. For example, Level-2 access users **3120**, **3121** and **3122** may be assigned level-2 access privileges in the overall organizational hierarchy. In the example diagram of FIG. 3B, block **3120** illustrates a possible configuration of the division level access that has been granted to level-2 root user Daren. For example, in addition to assigning Daren **3120** as the root user for this division, an individual division name that fits the nomenclature of the corporation’s organization names be assigned in the organizational hierarchy template under “division name 1” as illustrated in FIG. 3B. Blocks **3131** and **3122** and other level-2 division managers (Vice presidents

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Scotty and Gene, for purpose of this example) may also be assigned to the corporate organizational hierarchy.

The assignment of users in the organizational hierarchy of FIG. 3B continues to the assignment of regional directors as indicated in blocks 3130-3133. In one embodiment regional directors are granted level-3 access and manage one or more locations 3140-3141. In the illustrative embodiment, the levels of assigned hierarchy may continue as needed by the corporation filling out the organizational hierarchy as needed for one or more particular corporation levels.

For example, regional director Alice manages at least four locations run by location office managers Bo 3140, K 3141, Foe 3142, and other additional location office managers 3143. The location managers may not see reports, predictions or crop production results for each other without explicitly granting permissions to share projects that contain information from lower levels (i.e. levels 4-8) with each other. Although because Alice is the regional director, she has the privileged access to see projects and information for all the locations and corresponding location office managers under her management from the agronomist/consultant level-5 3150-3152 down through the client (Farmer) level-6 3161, the client (Farm) level-7 3170, and on to the client (Field) level-8 3180. Alice may run reports, do ROI analysis, and use any of the application system features on one or all of the locations she manages.

In one embodiment, levels of the corporation such as the agronomist level-5 may not exist in the corporate organizational hierarchy. In this case level-5 may be pulled out and the organizational hierarchy would apply the next level (level-6), the client level, as the next access level in the organization. As a further example of the organizational levels may be collapsed, again referring to FIG. 3B, if Sue 3150 is the agronomist assigned to Bo 3140 for the office location "name 1" she may be responsible for recommendations and application prescriptions to client-1 3160.

In the illustrative embodiment the client with level-6 access is typically the farmer or landowner who has at least one farm under management. When the hierarchy has more than one farm under one client, the organizational hierarchy may assign at least one farm, with access level-7, to the organizational hierarchy 3170. In the illustrative embodiment, there is no fixed indication that all levels in the organizational hierarchy work for the level above them. For example, not all levels are always corporate employees working for the same corporation and may be independent organizations in separate businesses as needed for supply and demand by contract, agreement, or license between parties at different hierarchical levels and may be set up as independent businesses from other levels within the hierarchy.

In FIG. 3B, the organizational hierarchy may have multiple clients assigned per location or agronomist working in a location indicated in blocks 3160, 3162 and 3163. In the illustrative embodiment, each client of the organizational structure has at least one farm under management as indicated in block 3170, in which each farm has at least one field 3180 within its boundaries and within each field are sub-fields that make up the different crop response zones 3185 that may provide one or more feature data set used to train the crop prediction models as described in the illustrative embodiment.

In the illustrative embodiment the crop prediction engine 4000 provides recommendations, predictions, and analysis data 3165, 3175 and 3185 to each of the respective levels 3161, 3170 and 3180 and may provide recommendations, predictions, and analysis to other levels 3110, 3120, 2130,

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3140 and 3150 to improve farm production effectiveness, and corporate ROI while reducing operational risks for the entire hierarchy of organizations.

For example, Jay is the farmer indicated in block 3160 with multiple farms under management 3161 having level-6 access to each of the farms 3170 of level-7. Jay may contract with a retail supplier through the retail supplier client device 3040 for the application of fertilizer to one or more farms 3170 under Jay's management. Jay may grant access to a single retail supplier by sharing a project that contains only the details of a single farm with one or more fields where the fertilizer application is to take place.

In the illustrative embodiment, separate projects may be built, at any level of the organizational hierarchy tree. Further, each project may contain only specific information to be shared with any other level of the organizational hierarchy by giving specific access to Jay's farm/field project. Any users in the system can see what projects are shared with them and what projects have been shared to them by other users. Projects may be set with view only or edit/view authority (use rights) but only by the project owner that shares the projects. In the privacy and ownership terms and conditions any one level when sharing is giving view and use rights to the other level but, no ownership rights to the actual data shared.

In one embodiment, sharing project information with third parties within or outside the organizational hierarchy may have a billing and commerce component and client device interface, in which account credits may be purchased and subsequently used to gain access to shared information and reports produced by the crop prediction engine and any other associated platform components or functionality. Access levels based on shared information and account credits may be set by each level or a single level of the hierarchy organization or by the owners or its affiliates who sell or license the platform or a version of the platform for its commercial use.

Referring to FIG. 4 there is shown an illustrative high-level block diagram of networked computing systems. The physical components may be separated into one or more client devices 4400, 4500 and 4600 and include the associated computing components to receive and transfer data through local gateway 4050, the Internet network 3000, DNS domain controller 4010 and application interface 4110. The client devices interface with clusters of computing components, cloud storage device 4030, application computing cluster 4100, and data science computer cluster 4300.

The physical components of a typical computing client device 4400 are used for data set transfer, input/output, and visualization through Internet 3000 to one or more storage devices 4030 or one or more computing clusters to view outputs from the crop prediction engine 4000. The outputs are viewed on at least one of a display device, monitor, CRT, or the like, through one or more VGA, DVI or the like, output 4045 interfaces. Computing client device 4400 includes at least one main memory subsystem 4450 including Random Access Memory (RAM) a volatile memory or Non-Volatile Random Access Memory (NVRAM) for storing information and instructions to be executed by processor 4440 through at least one bus or other communication mechanism for communicating information between the physical components on the client device 4400.

The client device 4400 may include at least one interface connecting input/output (I/O) peripheral device 4040 like a keyboard, mouse, or touch screen to input user actions and requests into the client device for further processing by the system. The I/O hub may be connected to one or more buses

to control at least one Universal Serial Bus (USB) used for connection to external devices and other peripheral devices. The client device **4400** has at least one hardware processor (CPU) **4440** coupled with one or more buses for processing information. The hardware processor may be a dedicated custom CPU or a general purpose CPU. The computer client device of block **4400** may also include core logic **4420** having a hardware I/O hub used to communicate between local storage device **4430**, input peripherals **4400**, network adapter **4410** and at least one memory controller hub **4070** connected between at least one processor and at least one memory device **4450** and at least one graphics adapter **4460** for graphical output, connected to at least one graphical display device for display of information and data visualization. The network adapter **4410** may be connected to an external Local gateway **4050** used to interface to one or more local and/or wide area networks like the Internet Network **3000** as needed for network communication between at least one client device, other client devices and to other physical components such as the application interface **4110**, one or more DNS controller devices or remote storage such as a content delivery network (CDN), cloud attached storage or dedicated storage subsystems **4030** as illustrated in FIG. 4.

Application interface **4110** is used to communicate with one or more computing clusters with external networks like the Internet network **3000**, dedicated storage **4030** and the DNS domain controller **4010**. The application interface **4110** of the crop prediction engine **4000** interfaces through one or more busses to a load balancer **4020** that is dedicated to balance traffic between separate computing instances located within one or more computing clusters. Communications between the load balancer **4020** and the computing clusters may be through one or more elastic IPs **4025** allowing virtual dynamic reconfiguration of data routing between physical devices within one or more computing clusters.

Application computing cluster **4100** may be dynamically scalable for configurations that required the addition or removal of one or more computing instances and storage resources within the cluster **4100**. In the illustrative embodiment, the application computing cluster **4100** includes at least one web-server associated with instance **4120** that may act as a network proxy for access and cluster communication with client devices running web-browser interface software. The application computing cluster **4100** may also include at least one front-end computing instance **4130** for processing main-line application program instructions originating from one or more external client devices **4400**, **4500** and **4600**. Additionally, the application computing cluster may include at least one back-end computing instance **4140** for background computation of subroutines used to assist special operations where in the illustrative embodiment, the back-end computing instance **4140** may be one type of GPU or Math computing engine to assist and offload the computation requirements of the crop prediction engine. Furthermore, the application computing cluster may include having each of the computing instances **4120**, **4130** and **4140** connected through one or more client device busses to at least one primary application database illustrated by block **4150**.

In the illustrative embodiment of the crop prediction engine **4000**, the database **4150** is configured as a geo-spatial database such as a PostGIS database that includes a Postgres database structure with geo-spatial extensions. In the illustrative embodiment, the geo-spatial database enables faster geographical based search, query and join operations that are

typical in the geography referenced precision agriculture industry. Data sets of independent input variables and resulting dependent output variables are indexed and reference quickly in a geo-spatial database architecture based on unified GPS coordinates stored within the database. Other types of databases such as centralized, distributed, relational, columnar, hierarchical, NoSQL and the like may also be used for the primary application database for data storage, search, retrieval, and data association purposes.

The primary application database **4150** may be used to hold previously processed results computed from the pre-processing, scaling and normalization program instructions. Additionally, database **4150** may also store the predicted results from one or more prediction engine models. Furthermore, the database of **4150** may or may not be used for the database's geo-spatial extensions but may be used to store user account application information, account access levels and privileges, as well as stored references and default settings from universities, government agencies and the agricultural community.

In one embodiment the database **4150** of the application computing cluster **4100** may store the calculated predictive yield and application recommendations of the crop prediction engine used for map data visualization, return on investment calculations and recommendations for the amounts and rates of various seeds and nutrient products applied. Additionally, the primary application database **4150** may store and retrieve user account information to and from the primary application database. In one embodiment the primary application database **4150** may store and retrieve default information input through the browser based device client by application or administration users. In other embodiments, the application database **4150** may store and retrieve information from an external source such as an application programming interface (API) running on both a device client **4400** and one or more of the front end **4130** or back end **4140** computing cluster instances through the application interface **4110**. Thus, in the illustrative embodiment, the system of FIG. 4 may store and retrieve data to and from at least one device client **4400** and at least one computing cluster **4100**.

In another embodiment, additional computing clusters such as the data-science computing cluster **4300** may be used as the physical components for computing to train and build the ML application for the crop prediction engine **4000**. Cluster **4300** may contain the same physical components as that described above for the application computing cluster **4100** but run various software that is more specific to the task of ML research and development. In the illustrative embodiment, the data science cluster **4300** may be used by third parties to perform customizations as required by the systems users. Thus, the embodiment of the illustrative system uses the one or more of the physical components shown in FIG. 4 to run and compute the system application as further defined below.

Artificial Intelligence has been used for agricultural prediction through modeling. Popular statistical and machine-learning methods for detecting interactions may include decision trees and their ensembles, random forests (RFs), node harvest, forest garrote, rulefit3, as well as other methods more specific to soil and environmental characteristics with categorical input features such as logic regression and multifactor dimensionality reduction. Historical tree-based procedures grow shallow trees to prevent overfitting of the data. Shallow trees exclude the possibilities of high-order interaction detection without predictive accuracy.

In the crop prediction engine **4000**, random forests are used to circumvent these historical limitations. The use of random forests leverages high-order interactions to obtain high prediction precision and accuracy, with the exception that RFs alone do not support the interpretation of feature interactions in the resulting tree ensembles.

The illustrative embodiment of the crop prediction engine **4000** uses an RF training model to search for stable, high-order interactions by introducing an iterative process that sequentially grows feature-weighted RFs to perform soft dimension reduction of the input feature space to stabilize decision paths.

The crop prediction engine **4000** support for higher-order input feature combinations that are the most prevalent as they traverse through the RF decision paths of the tree from root to leaf. In addition to the prevalence identification, the higher predictive accuracy of the RF decision trees may capture more of the underlying soil and environmental attributes that lead to insights for chemical combinational interactions that may limit or pronounce crop-yield production for farmers. The crop prediction engine may determine a small subset or a single main feature input that contributes most to crop productivity. In the illustrative embodiment, the invariance of decision trees to monotone transformations mitigates some pre-processing requirements as known with a main component of achieving normalization of the feature data sets.

Due to the size, variability and lack of good signal-to-noise ratios found in most raw agricultural data sets, data normalization issues are a major concern in analysis and in the ability to achieve a stable agricultural prediction. The illustrative embodiment achieves improved predictive accuracy and extracts both known and compelling soil and environmental candidate interactions that derive one or more "Prevalence Differentials" that enable a predictive yield response.

In the illustrative embodiment, the crop prediction engine **4000** includes at least three basic steps to search for high-order feature interactions. Firstly, use of input data set feature re-weighting to adaptively regularize the random forest fitting. Secondly, the extraction of decision rules out of the re-weighted random forest, mapping from continuous or categorical into binary features. Thirdly, an ML bagging step that recovers the stable interactions with respect to boot-strap perturbation of the binary features.

In the illustrative embodiment, the crop prediction engine **4000** builds on a generalization formula of the Random Intersection Tree (RIT) training model. RIT performs a randomized search for high-order interactions among binary data set features in a deterministic setting. In the illustrative embodiment, searches for co-occurring collections of *s* binary features or order *S* interactions that appear with greater frequency in a given class "C." The illustrative RIT training model allows for precise recovery of such interactions with a high probability with substantially lower computational effort.

Referring to FIG. 5 there is shown an illustrative flowchart of the operations performed by a crop prediction engine **4000**, which uses machine learning processes. FIG. 5 summarizes the program instructions by grouping the program instructions into process blocks that are used by the crop prediction engine **4000** to predict a crop yield. More specifically, the process blocks **5000-5800** represent program instructions for receiving a multiplicity of data sets and the computational processing of those data sets in the illustrative method of the present embodiment to predict a crop yield from at least one trained ML model. Firstly, data

sets are loaded to the permanent storage **4030** associated with data set process blocks **5000**, **5100** and **5200** for pre-processing to identify covariates that are used to train the crop prediction engine **4000**.

In the illustrative embodiment, data sets are ingested by the crop prediction engine through application client devices **4400**, **4500** and **4600** (shown in FIG. 4) interfaces from client devices **3010** to **3080** (shown in FIG. 3A) by the program instructions associated with the crop prediction engine application software. Data sets are typically uploaded from the local storage, attached or removable storage at the local client devices. Client devices are typically connected through the Internet gateway **3000** to the illustrative cloud storage **4030** as described above.

The process blocks **5300**, **5400**, **5500**, **5600**, **5700** and **5800** are associated with operation performed by the crop prediction engine **4000**. In the illustrative embodiment, the data-science computing cluster **4300** (shown in FIG. 3A and FIG. 4) may also be used for processing the program instructions within each process block.

In another embodiment one or both of the application computing cluster **4100** (shown in FIG. 3A and FIG. 4) and data science computing cluster **4300** may be used to process program instructions within the process blocks shown in FIG. 5.

Referring again to FIG. 5, process block **5000** represents the ingested files from field or equipment sensors that contain a multiplicity of information in multi-dimensional data sets, farm feature sets or farm data features and the like hereafter referred to as "sensor data." Process block **5000** includes imported sensor data such as lab results **5010** from soil samples that are site specific. The soil sample results may be in the form of text files, like PDF's or CSV format and typically contain values for the soil chemistry at each sample location. Soil sample locations may be identified by GPS coordinates or by field name embedded within the file or by manual identification and may depend on the sample sites location and sampling resolution.

Sensor data from electro-conductivity **5020** measurements may be uploaded as additional data set information may be used during computations as an indication of the Organic Matter (OM) of the soil located in at least one site. In one embodiment, soil texture data may also be uploaded, again as supplemental information generated as an additional farm feature set. Electro-conductivity data files may be continuous, point or area sampled and will most likely contain GPS positioning for each value within the sensor data set. File formats for electro-conductivity may be in text, CSV, PDF, or shape file format as well as other forms including manufacturer specific or custom formats. Additionally, previous yield or harvest uploaded data sets **5030** play an important part to train the model of the crop prediction engine **4000**.

In the illustrative embodiment, more than a single harvest of yield data is uploaded and stored for further computational analysis. Yield sensor data is typically collected automatically by harvester equipment using GPS position monitors, moisture sensors and elevation sensors and is typically collected in crop rows at the rate from one (1) to five (5) Hertz. Yield harvest upload files are typically noisy and dense data sets and may be in a shape file, CSV, or custom formats. In one embodiment, yield information may be a polygonal section where the yield is averaged and smoothed within the polygonal area as a single sample. Typically, the polygonal area is represented in one or more file formats as one sample per polygonal area of the harvest. In another embodiment the polygonal area per yield sample

may be computed by the product of the harvester arm width, the speed of the harvester divided by the harvester sample rate in Hertz.

With respect to process block **5100**, the ingested files from third parties such as government agencies, commercial companies, and agricultural groups, hereafter called “third-party” data sets, typically have been pre-processed to remove missing data and noise. However, the third party data sets may be in raw format and be unprocessed. Third party data sets in process block **5100** may vary in data format including raster, point, polygon, and multi-polygon and may need to be adjusted to a common coordinate system to align physically with other data feature sets. While sensor data sets **5000** are site specific and typically collected by sensors from farming equipment, the third-party data sets indicated in process block **5100** may span vast geometries covered by large data sets collected over time or from satellites that update weekly or even daily. The illustrative embodiment may use APIs to pull data into storage **4030**, into the application computing cluster **4100** or data science computing cluster **4300** for processing and database categorization and use. Data sets represented by process blocks **5000** and **5100** may be in a variety of file formats and may be converted to other file formats prior to computational purposes of the crop prediction engine.

At process block **5200** the data sets from data sensors originated by farm equipment and used for the application of fertilizers and seeds are uploaded. Planting data **5200** may be recorded by the programmatic interfaces. The planting data may include GPS tracking, seed-type or product selection, planting depth and planting rate in seeds per second or seeds per acre. Planting records are dynamically recorded and saved for uploading as a feature set for processing by the crop prediction engine **4000**. The crop prediction engine may use seed variety, like genomic, hybrid or non-genomic or custom seeds with special coatings or other traits to determine the seed types that are most cost-effective for crop production and have the best yield response in different soil chemistries or environmentally based clusters within one or more plots of land as identified.

Applied nutrient data sets **5210**, typically fertilizers, may be applied from a programmatic interface that selects fertilizer product and application rate according to precise location information provided by GPS coordinates. In one embodiment, the programmatic interface relates to the application of such products records GPS, fertilizer product type and application rate. The sensor data is typically saved into one or more data-files as the products are being applied to one or more plots of land like fields. The saved data file data sets **5210** are then uploaded and saved as a feature set for processing by the crop prediction engine **4000**. Applied nutrient data sets **5210** may also be manually input. Illustrative data sets that would be manually input include fly over nutrient applications, blanket manure applications and other nutrient applications that are manually input and typically have positional relationships to fields and possible sub-fields. Seed types and application rate data sets **5220** may also be communicated to the crop prediction engine model.

Process block **5000**, process block **5100**, and process block **5200** represent program instructions to upload, process and store input data sets, which are multi-dimensional feature sets, to determine the overall dimension and/or order of computing required by the crop prediction model **4000**. Note, the feature sets used to train the ML model may require pre-processing for noise reduction, averaging and coordinate system unification.

At process block **5300**, program instructions representing an illustrative training model are used by the crop prediction engine **4000**. In the illustrative embodiment, a random forest (RF) training model is implemented as a computational model. In operation, each of the RF trees is built into a set of tree estimators and calibrated using first out-of-bag training data, which assesses initial conditions, parameter settings, and first pass model quality using techniques like R-squared error minimization prior to application of actual training data sets. Tuning of the RF training model may be iterative and may use different feature weights based on the desired properties inherent to the set of training data as known to one of skill in the art. Once the RF training model has been tuned, a data set of multi-dimensional set of features is applied to each tree in the random forest.

Note the distinction between an “algorithm” and a “training model” for machine learning purposes, which are described in this patent. For the purposes of this patent, an algorithm is derived by statisticians and mathematicians for a particular task. Thus, in machine learning, algorithms have already been derived. When an algorithm is implemented as machine code in a computer, the algorithm’s utility increases because the machine (computer) can handle more computations. Every algorithm has some mathematical form, which when implemented in machine code is developed to form a machine learning algorithm. A “training model” is an equation that is formed by applying the data sets to the equation, which is associated with the algorithm. Thus, the RF training model in process block **5300** is created using data sets from process blocks **5000**, **5100** and **5200** and applying those data sets to the RF training model, which springs from an RF algorithm.

Since the data sets being analyzed at process block **5300** are relatively large, then the number of possible observations from tree, branch, and leaf-nodes for each tree of the random forest will also be too large to effectively resolve. Thus, the method in FIG. 5 continues with process block **5400**, in which the number of features in the data sets are reduced.

At process block **5400**, the high order level and number of feature sets is reduced from the feature sets at process block **5300**, block **5200**, block **5100**, and block **5000**. In the illustrative embodiment, the Random Intersection Trees (RIT) **5400** training model is used to effectively reduce the high-order levels and number of feature sets from the RF model. RIT reduces the complexity (order) by limiting the observations of features as they traverse the RF to only the features that show main-effect or co-interactivity, i.e., covariate importance.

The method then proceeds to process block **5500** where the covariates from the RIT process block **5400** are ranked. The RIT reduction technique of process block **5400** ranks the most important covariates at process block **5500** by re-interjecting the results back into the RF model (in process block **5300** for further observations).

The method then proceeds to process block **5600** where such observations are then used to form a stump model which further resolves to one or more response surfaces indicating an estimated output pertaining to a specific spatial geometry, which is also referred to as a “voxel.” By way of example and not of limitation, the voxel has a specific spatial geometry with defined soil, environmental and yield characteristics. Each voxel then has at least one response surface representing the most important chemical-nutrient or seed-product contributors that generate the models estimated output response.

Additionally, the stump model response surfaces enables the crop prediction engine **4000** to compute a Reduced

Order Surrogate Model (RoSM) represented by the program instructions of process block 5700 to build a Generalized Additive Model (GAM) that further builds a functional equation in the form of a linear function composed of non-linear terms that can be used to easily compute seed and nutrient application recommendations as illustrated by the program instructions of process block 5800 resulting in the form of a downloadable or written prescription file.

The illustrative embodiment of the crop prediction engine 4000 uses a generalized Random Forest (RF) Machine Learning model to understand how input covariates associate and interact with each other through observations of how these covariates traverse decision trees in the random forest. By understanding which covariates are coupled to one-another and which are important contributors for crop yield production, users of the system can take actions that explain and recommend where to plant certain seed-types and what nutrients to apply to achieve the estimated crop predictions.

Thus, the input variables, when applied to the crop prediction model generated with crop prediction engine 4000 can accurately predict an estimated output given a multidimensional set of input features.

The Random Forest (RF) learning model is composed of decision trees. One advantage of decision trees is they may look at data one dimension at a time and use simple rules like less-than, greater-than, yes or no and the like to send operations on a simple binary direction (left or right) for each branch within a single tree. The direction may be based on pre-weighted feature set data where weights may be set for thresholds at each branch resulting in new nodes per branch based on simple binary logic decisions as the tree is built. This process continues recursively from root to branch to leaf until the tree has a fairly homogeneous set of end points called leaf-nodes that can make one simple prediction for the conditional expectation given a single dimensional set, e.g., single feature, of input data.

As an example, a one-dimensional single feature data set may traverse through an RF model. Given a single dimensional decision tree with one dataset "xi," in which (i, i+1, . . . n) where "n" is the total count of numerical input values in "xi" such that all of the splits within the tree are based on one input variable, then each conditional estimated output would resemble a histogram with each leaf node corresponding to one histogram bin. The height of each histogram bin represents the estimated output "y" for each node in the tree.

In the illustrative embodiment of the crop prediction engine 4000, each additional feature set, which is a data set representing an input feature, of input covariates adds another dimension creating another histogram that changes the conditional estimated output to a higher order function of the input data sets

$$\hat{y} = \hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

where; (x) is the multi-dimensional set of input features $[x_i, x_j, x_k, \dots, x_p]$, p is the total number of input features, B is the total number of trees in the random forest and T_b represents an ensemble of tree estimators. For each additional input feature additional trees are built to effectively model the higher dimensional predictive output "ŷ" as a function of the set $[x_i, x_j, x_k, \dots, x_p]$ of "p" features. In the illustrative embodiment, input features may represent a multiplicity of

histograms resulting from a multiplicity of trees that when averaged together effectively generate a smooth function representing the estimated response or predictive output in the higher dimensional space. Thus, in the illustrative embodiment the random forest is used to fit an estimation function that may results in a smooth curve based on a sufficient number of tree estimators used to build the prediction model.

An important differentiator is to observe where the splits are made within the tree. In higher dimensional data, the dimension "p" of input data set matrix [X] is the number of input features and the estimated number of observations in "p" dimensions are estimated to be $\sim 2P$ resulting in an extremely large number of possible observations to compute and analyze. For example, the soil, environment, and seed variety variables may alone have over 10,000 features or $\sim 210,000$ observations which may require more computing resources and compute time than economically feasible.

One benefit of using the generalized random forest (RF) model in higher dimensional space is the shallow nature of each tree. In one embodiment, each split in the multi-dimensional forest reduces the number of observations by approximately half since at each split approximately half of the data may be sent to the right of the branch and the other data to the left resulting in an exponential fall-off of feature data from root to final leaf node. This natural reduction in tree depth from the RF model results in the average number of levels of depth to only be approximately $\sim \log 2N$ where N is the total number of observations represented by the input data sets.

In the illustrative embodiment of the crop prediction engine 4000, the lower number of levels of tree depth for any input data set represented in input covariates matrix [X] for the possible tree estimators may be exploited to extract the internal data representation from the forest. The exploitation of the data is accomplished by looking for subsets of features that occur together as they transition through the tree from root to leaf. In the observation of the input data as it traverses through the trees the goal is to determine which of input dependent variables occur together as they traverse the tree from root to leaf. Sets of variables that follow the same path through the tree increase the probability of importance for the feature and may contribute or be more important to the estimations made when computing a predictive output. For example, the crop prediction engine takes advantage of the fact that in the simplest form of tree traversal, one observation looks for subsets of variables that pass through the same path from root to leaf. This implies a simple form of a linear equation $\vec{y} = \alpha x_i + \beta x_j + \gamma x_k \dots$. Additionally, if y represents a function of a complex non-linear equation, the same principle follows when the covariates from different features pass through the same path. The observations assist in determining which subsets of covariates are the most important predictors as computed by the crop prediction engine 4000.

As a result of determining the same path for multiple data sets of inputs, the crop prediction engine 4000 can determine how to map a function for estimated response. In the illustrative embodiment, there is an understanding of the crop responsiveness to a chemical application or an environmental event when that variance is applied in a specific location is extremely important information to the agriculture industry. For example, in the simple case of a linear equation of estimated yield, the predictive dependent variable may be written as $\hat{y} = \alpha x_i + \beta x_j + \gamma x_k$ where y is the estimated yield output, x_i is one data set which may be organic matter (OM), x_j is another data set which may be the

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Cation Exchange Capacity (CEC), x_k is yet another data set which may be Magnesium (Mg), and where a , B and y may be specific scaling coefficients for each of the input variable data sets OM, CEC and Mg. In the simple case of a linear equation, one can estimate the predictive change in estimated yield by holding the CEC and Mg input variables constant (assuming β and γ are also constants), while varying the a coefficient such that the value of a determines the estimated yield changes for each unit change in OM. The majority of big data modeling seldom follows this simple example. A simple form of a linear equation to estimate yield as the dependent variable is seldom a regular set represented by a linear equation and most likely represents a much higher dimensionality response.

Therefore, it is desirable to understand not only the main-effect but also the coupling of different features as one feature may have an effect on another. In the illustrative embodiment on FIG. 5, a method is shown that determines which of the subsets have co-dependencies on estimated yield ($\hat{y}$) and the marginal impact on y as they vary in order to determine chemical nutrients and seed-type applications and recommendations needed to achieve the desired production output performance.

Based on the well-established binomial theorem, when given a multi-dimensional input set of feature data, the method proceeds to calculate the number of possible observational subsets by 2^p where p is the dimensionality of the input data or number of features. In one embodiment, as applied to soil and environmental input characteristics, the number of observational subsets may be beyond the ability to effectively compute the estimated output or to predict the dependent variable output. To reduce the set of observations into something that is manageable, forward regression may be used. The forward regression reduction process follows the law of marginality, which in the illustrative embodiment may not satisfy the accuracy requirements required for several reasons.

First, the forward regression training model assumes that the predicted output dependent variable is an additive and multiplicative function. In the field of soil science, the output function may rarely be of this form and is often more complex. Second, an interaction between input characteristics may be important even when the "main-effect" of a single additive term of the function has no or little importance at all.

For example, the forward regression may determine in the estimations that organic matter is not important according to the main-effect of the forward regression training model, but in fact the observation of the interactions of the model for the crop prediction engine may prove it to be one of the most important soil characteristics in the estimation of crop-yield performance. Thus, to reduce the number of observations and keep the accuracy high, for example when observation levels are above order 7 and where use of the forward regression training model falls short of the desired accuracy and quality requirements, the present embodiment uses a novel method to sample into the observational universe of 2^p possible observations at substantially reduced computational costs.

In machine learning applications counting of higher order observational subsets based on multi-dimensional sets of input features is difficult and may not be achievable due to the inability to count observations in high dimensional space and having to count first to observe which feature subsets are more frequent in the multi-dimensional feature analysis. For example, given a typical soil and environmental set of input covariates, where possible observations may be above order

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7 and feature data sets may be above 3,000, the goal of the illustrative embodiment is to determine from the input characteristic features where two or more features occur together and are thus considered to have a high probability of coupling or importance.

The illustrative method implemented by the crop prediction engine 4000 uses the Random Intersection Trees (RIT) training model to reduce the order of observations to just the observations that are considered important to the predictive output response. The illustrative training model utilized by the crop prediction engine 4000 addresses this problem by the use of RIT to compute intersections between binary vectors that represent observations of coupling of covariates as they pass from limb to limb of each tree. For example, RIT is based on the idea that given a product of binary vectors where a "1" in the product vector indicates the position within each binary input vector where there is commonality between two or more observations represented by one or more output binary vectors.

In an alternative embodiment, the RF method of achieving predictive output may be replaced by Neural Networks where the RIT training model to reduce high order multi-dimensional data sets may still be applied thus reducing the non-responsive observations of the input feature sets.

In the illustrative embodiment, the use of RIT requires that the crop prediction engine 4000 encodes each traversal path through each RF tree from root to leaf into at least one binary vector. For example, for each $[x_i, x_j, x_k, \dots, x_p]$ input feature on a given path from root to leaf for each tree in the random forest a one is marked "1" if a particular variable is included and marked a "0" if absent in the path through any particular tree. This process may be repeated for all paths in all trees. The resulting binary vectors may be sparse because the terms in a path are typically much shorter than the number of input features in the multi-dimensional data sets represented by $[x_i, x_j, x_k, \dots, x_p]$. As a first order reduction, once the paths for each tree are encoded into binary vectors a simple bitwise binary product may result in a substantially reduced number of variable sets of observations where the observations are more likely to be coupled and may be more important for the estimated predictions computed by the crop prediction engine 4000. The process as described above may be repeated recursively with randomly chosen binary vectors by intersecting them as $\text{Path}_n, \text{Path}_m$ to build random intersection trees. The resulting binary vector is then used to intersect with one or more randomly chosen paths to form the next branch of the RIT. This process may again repeat recursively as additional levels of the RIT are built. The surviving paths at the bottom of the tree that have not been eliminated by calculating the random intersections of other paths (from the process of multiplying binary vector products) will be the paths that are important with strong interactions between coupled covariates from the different input data sets. In the illustrative embodiment, paths represented by "1" at the bottom may be infrequently found as they have survived many binary product intersections based on the number of levels in the RIT. Thus, the output of the RIT may establish the most frequent sets, hereafter called "item-sets", of interactions of which the value for each interaction may be assigned.

Once the solution to the item-set counting problem is solved as described above in the illustrative embodiment, and the most frequent solution sets are found, these results may be subsequently used to extract the functions that the RF has previously learned. Based on the illustrative method in FIG. 5, the method has identified and established the multiplicity of dimensional input variables that map together

and have a strong non-linear dependence to one another. Additionally, the method of FIG. 5 has established which of the collections of features are also mapped together that also may have a strong non-linear dependence to one another.

The method continues by ranking the most frequent observations and sets of observations by the introduction of weighted prevalence. Weighted prevalence may be determined by the product of the measure of node purity and the frequency of the determined item-set found. Wherein, node purity of a particular node in any RF tree is determined by a low variance between the coupled variables present at the leaf-node under purity observation. For example, a leaf-node may be considered to be pure if the corresponding voxel has the same yield. Wherein a voxel may represent the geo-spatial location of the smallest cubic area of soil that contains multidimensional soil chemistry and other environmental covariates used for model prediction. A voxel may be a polygonal area representing a two dimensional surface or may be represented by individual points that fall inside a voxels polygonal surface boundary. The points within a voxel may be random or non-randomly located and may represent a multidimensional set of input features.

Thus, in the illustrative embodiment the quality of calculation may be ranked by how accurate are the predictions multiplied by how frequent are the predictions that may result in how important are specific features to the expected response of the output variable.

Additionally, the ranking method as described in the illustrative embodiment puts everything on the same scale. The outcome may be the result of the main-event or a complex interaction between multiple data-set inputs giving interactions in a normalized order between different input item-sets.

The solution of the illustrative method continues by ranking the most frequent observations and sets of observations by the introduction of weighted prevalence. Weighted prevalence may be determined by the product of the measure of node purity and the frequency of the determined item-set found. Node purity of a particular node in any RF tree is determined by a low variance between the coupled variables present at the leaf-node under purity observation. For example, a leaf-node may be considered to be pure if the corresponding voxel has the same yield.

The voxel may represent the geo-spatial location of the smallest cubic area of soil that contains multidimensional soil chemistry and other environmental covariates used for model prediction. The voxel may be a polygonal area representing a two dimensional surface or may be represented by individual points that fall inside the voxel's polygonal surface boundary. The points within the voxel may be randomly or non-randomly located and may represent a multidimensional set of input features.

Thus, in the illustrative embodiment the quality of calculation may be ranked by how accurate the predictions are multiplied by how frequent are the predictions that may result in how important specific features are to the expected response of the output variable. Furthermore, the ranking method as described in the illustrative embodiment puts everything on the same scale. The outcome may be the result of the main-effect or a complex interaction between multiple data set inputs giving interactions in a normalized order between different input item-sets.

Once the quality of the data sets and particular variables within the data sets are ranked for importance, i.e., the main-effect and co-dependencies, the crop prediction engine 4000 uses additional different training models to calculate the optimal response surfaces. Once the important features

are resolved then the method applies at least one of a simple classical regression, plane-fitting by linear regression, generalized linear modeling, multi-variate regression spline modeling, a lattice network modeling, and the like to build a predictive response surface. Thus, by using the crop prediction engine 4000 to see through the eyes of the random forest and pull the response surface directly out of the random forest, the method of FIG. 5 accurately and quickly predicts the response surface to any variation found in the most important and frequent input variables.

The resulting response surface estimator is called a stump model. The stump model may use a similar method of traversing the RF model as that defined previously but is re-applied with an awareness of ranked importance for both the main-effect and co-dependencies of the input variables. The simplest form of the stump model used for estimated predictions is to use the same one-dimensional method as previously described herein by tracking the splits through the RF model but in a single dimension.

Once the most important coupled covariates are identified, the crop prediction engine 4000 uses at least one computing engine to look at every tree and every path where the coupled covariates occur together, determining the splits for each tree and determining where a split corresponds to a box or stump. This method of stump model usage is similar to the histogram bin as previously described. For example, if the two most significant contributors happen to be CEC and Mg in a single path in a single tree a simple graph may show the CEC (x_i) and Mg (x_j) contributions where the height of the stump represents the estimated yield for the single path in that single tree. The actual total response is the subsequent average of all stumps generated across all significant paths in all trees for the most important contributing variables. The stumps will be of all different shapes, sizes, and values as each is dependent on the paths through the tree estimators used to model the final output prediction. A similar approach may be taken when more than two significant contributors are recognized by the regression through the stump model such that the number of significant contributor variable sets is greater than two.

Thus, the output of the stump model results in the derived function for crop yield modeling resulting in a smoothed surface that represents a non-linear multi-dimensional model for predicting yield. The illustrative embodiment presents at least one method for generating at least one response surface, using the form of a stump model, in the number of dimensions that are important while building a crop prediction engine capable of making accurate crop-yield estimates. The crop-yield response is then based on variances of the important dimensions.

Note, the same method associated with the crop prediction engine may be used for the prediction of other dependent variables such as amounts of carbon sequestration, biofuel generative modeling, life cycle carbon emission models and the like.

The results of the response model, for one or more input dimensions, may be stitched together into the form of a Reduced Order Surrogate Model (RoSM). The RoSM training model can be used to build a Generalized Additive Model (GAM) which represents a linear function of non-linear terms. The resulting GAM may be used as a predictor for any one of the input variables, represented as a high-order curve used for a generalized predictive model for the output prediction of yield as a function of any single important input dimension. The GAM response may also be used to predict other dependent output variables such as

carbon sequestration, biofuel generative modeling, life cycle carbon emission models and the like.

Once the GAM has been built making generalized recommendations for crop management and production is streamlined. The RoSM has a dramatically simplified output equation as compared to the complexity of building the predictive model, as understood by the “reduced” order nomenclature in the name Reduced Order Surrogate Model. From the RoSM, the compactness of the model allows for fast and simple predictions. The “Surrogate” component of the nomenclature represents that it is a “Surrogate” of the entire RF model to simplify the productivity of the crop prediction model.

The simplification supports selecting the peaks of the response curves from the predictive RoSM to achieve the application amounts for a predictive yield (output) response. Once the GAM for one or more specific plots of land has been created, it may be used as a standard equation for the crop prediction as a function of the soil, environmental and seed characteristics without needing the entire predictive model apparatus and method.

The normalized yield values and assigned relative maturity values are used as predictor variables for machine learning models. Also, additional hybrid seed properties such as, crop rotations, tillage, weather observations, soil composition, may also be used as additional predictor variables for the machine learning models.

In the illustrative embodiment, machine learning techniques based on random forests are implemented. In another embodiment neural network modeling may be used instead of RF modeling to determine probability of success scores for hybrid seeds planted at the geo-locations associated with target fields. The dependent variable of the ML model and subsequent modeling is a probabilistic value ranging from 0 to 1, where 0 equals a 0% probability of a successful yield and 1 equals a 100% probability of a successful yield.

In other embodiments, the target variable may be a probabilistic value that may be scaled to fit one or more particular output response types. For example, the dependent output variable for yield may be scaled to bushels per acre with additional scaling for wet or dry measurements. In an embodiment having alternative dependent variable projections, the output may be scaled to represent the return on investment (ROI) based on the delta of hybrid seed costs vs. yield and harvest market prices. A successful yield is described as the likelihood that the yield of a specific hybrid seed is a certain value above the mean yield for similarly classified hybrid seeds. For example, a successful yield may be defined as a yield that is 10 bushels per acre above the mean yield of hybrid seeds that have the same assigned relative maturity value.

In the embodiment where a random forest training model is implemented as the machine learning technique for determining probability of success scores for each of the hybrid seeds for the target fields, the random forest training model may represent an ensemble machine learning method that operates by constructing multiple decision trees during a training period and then outputs the class that is the mean regression of the individual trees. For example, a hybrid seed standup classification may use ML to determine the best seeds based on previous years’ yield results. Random forest is used to determine the best seeds based on final yield but also may be used for stand-up output projections based on seed-type, seeding rates, seed depths and other chemical and environmental characteristics. In the method of stand-up projection, the ML model is trained with one or more years of stand-up covariate data from previous years or seasons

with hybrid seed, soil, and environmental classifications as additional imported data sets.

The illustrative method of FIG. 5 may also be used to cluster and train the model to predict which hybrids work best for production output based on standup count when target fields have characterized soil and environmental characteristics integrated into the data sets for modeling. Thus, the system and method specified herein may also apply to seed-type varieties such as genetic hybrids. The splits within the trees for every branch would then include moving one direction for one hybrid and/or seed-type and moving in the other direction for a different hybrid and/or seed-type.

With respect to RF modeling, the process is repeated recursively in the same manner over multiple dimensions of input data sets through all trees in the forest. The process also includes the use of the RIT to reduce the number of observations and further reduction to a linear set of non-linear equations to create a generalized additive model (GAM). Thus, the illustrative embodiment describes how to build an accurate and reliable prediction model for the determination of either optimal stand-up count or optimal yield production based on 1) modeling output from clusters of the soil and environmental characteristics and 2) hybrid and/or seed-type selections within one or more target fields on at least one plot of land.

In one embodiment RF as described herein is used to build at least one crop-yield model to predict crop yield and follows the general steps of the method as illustrated in FIG. 5. First, build an RF model as a function of soil chemistry and other available environmental covariates. Second, define a distance metric between pairs of feature vectors. Third, serve as input to Random Intersection Trees (RIT) that identifies common predictors and interactions within the feature space defined.

The basic approach in the illustrative embodiment for building a crop prediction engine used as a model to estimate as defined in the specification of the illustrative embodiment is based on Random Forest and can be written as an ensemble of tree estimators $T_b(x)$ according to function

$$\hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

Where x is a feature vector including; (1) soil chemistry variables such as B, Ca, Cu, Fe, K, Mg, Mn, Na, NO_3 —N, P, S, Zn, OM, pH, Buffer pH, CEC, and the four percent nutrient saturations % K, % Ca, % Mg, and % Na, from analysis of sparse soil samples; and (2) other point-wise environmental features such as soil conductivity, soil texture, elevation or wetness index and the like. And, where B is the total number of trees fitted with $b=1, 2, \dots, B$.

The process steps of FIG. 6 represent the program instructions for the process steps used to fit the Random Forest yield model according to the illustrative embodiment. Process block 6100 of FIG. 6 represents the program instructions used by the crop prediction engine 4000 in a mathematical representation to prepare one or more sparse soil training matrices [Xtrain], based on the available sparse soil sample analysis of the chemical properties, into subsets as defined herein. Building the multi-dimensional soil input matrix may require a first processing step to make them spatially-independent or sparse from one-another wherein, the sample data sets to achieve independence from possible samples that may be in close proximity of other samples. The spatially-independent covariates are then used to build

at least one soil training matrix which is the first step **6100** in preparation to train the crop prediction model **4000**. The soil training matrix may be represented mathematically by $X_{train}^{soil}(N \times p^{soil})$ where N is the total number of voxels contained within the bounds of the plot of land being modeled and p^{soil} is the number of soil chemistry variables (soil features) used for modeling the RF observations. Care must be taken in preparing the training observations to ensure that training data are de-noised and spatially-independent. Spatial data encountered in the art of precision agriculture are typically noisy and also may exhibit strong spatial autocorrelation issues. The illustrative embodiment may use Median Smoothing (MS) of one or more of the covariates in the final training matrix $[X_{train}]$ and input response vector $[Y_{train}]$ in the neighborhood of each soil sample to address certain noise issues.

Process block **6200** shows a mathematical representation of the programming instructions used to prepare the environmental training covariate matrix X_{train}^{env} . In one embodiment the environmental covariate data may be from one or more data sets such as sampled soil conductivity measurements, elevation values from the neighborhood of $[X_{train}^{soil}]$, or soil texture information from national databases or localized sensor measurements. Preparation of the said environmental covariates may include one or more processes to de-noise and smooth noisy environmental covariates by one of many illustrative methods such as Median Smoothing applied each covariate within a neighborhood of each soil sample in data set $[X_{train}^{soil}]$ to form the environmental training matrix. For example, in the illustrative embodiment the spatial geometry in the neighborhood of $[X_{train}^{soil}]$ may be represented by g_i (the geometry of the neighborhood where the environmental variable is located). To de-noise and smooth the environmental training data one embodiment may apply Median Smoothing for each of the environmental covariates at location g_i to form the smoothed environmental covariate training matrix $X_{train}^{env}(N \times p^{env})$.

Process block **6300** of FIG. 6 represents the program instructions used by the crop prediction engine **4000** in a mathematical representation used to combine the spatially-independent soil covariate training matrix $[X_{train}^{soil}]$ with the de-noised and smoothed environmental covariate training matrix $[X_{train}^{env}]$ into the full training matrix $[X_{train}]$ in the form $X_{train} = [X_{train}^{soil} \ X_{train}^{env}]$ where the full training matrix is applied to the entire set of voxels in one or more plots of land by applying product matrix $N \times p'$ where $p' = (p^{soil} + p^{env})$ and N represents the total number of voxels within the same one or more plots of land.

Process block **6400** shows a mathematical representation of the programming instructions used to prepare the yield response training vector y_{train} . In one embodiment the harvest data from yield monitoring equipment may be densely sampled at a typical sample rate of 1 to 5 Hz. In various embodiments the sample rate of the yield monitoring equipment may be different. In one embodiment the yield data set from the yield monitoring equipment at harvest time may be noisy and may contain dense yield data such that pre-processing to de-noise is required and performed by at least one type of Smoothing training model. In the illustrative embodiment Median Smoothing prior to being used as the dependent variable training vector may be performed in the neighborhood of each soil sample in X_{train} to form y_{train} , the response noisy response training vector may include one training vector. Preparation of the said y_{train}^{noisy} response training vector may include one or more processes to de-noise and smooth by one of many illustrative methods such as Median Smoothing the response vector within a

neighborhood of each soil sample data set in $[X_{train}^{soil}]$ to form the smoothed response training vector matrix. For example, in the illustrative embodiment the spatial geometry in the neighborhood of $[X_{train}^{soil}]$ may be represented by g_i (the geometry of the neighborhood where the response vector is located). To de-noise and smooth the training response vector one embodiment may apply Median Smoothing to the noisy response vector y_{train}^{noisy} at location g_i to form the smoothed yield training matrix $y_{train}(N \times 1)$.

Process block **6500** represents the program instructions used by the crop prediction engine **4000** of the illustrative embodiment, in a mathematical representation used to fit the Random Forest regression model. Fitting may be accomplished by training the RF model with the prepared X_{train} and y_{train} matrices. This procedure **6500** results in a fitted model and may be represented mathematically by;

$$\hat{f}_{RF}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

where $\hat{f}_{RF}(x)$ is the fitted model representing the output response from the crop prediction engine with spatial geometry for one or more plots of land as defined the area of at least one $N \times N$ voxel, and (x) represents the combined soil and environmental training vectors $(\vec{x}_{train}, \vec{y}_{train})$ and

$$\frac{1}{B} \sum_{b=1}^B T_b$$

represents the Random Forest of B tree estimators.

Process block **6600** of FIG. 6 represents the programming instructions used in one embodiment to build a spatial cross-validation framework to facilitate evaluation of the fitted Random Forest. The cross validation may be used to iterate input variable weights and set initial model conditions to reduce r^2 error and improve the quality of the crop prediction engine **4000**. Cross-Validation (CV) folds can be one or more of: compact circular clusters, horizontal strips, vertical strips, diagonal (south-west to north-east) strips or anti-diagonal (north-west to south-east) strips. FIG. 7 illustrates one example of a set of fields with anti-diagonal cross-validation where the strips represent training data, test data and buffer data. Using the five different spatial CV fold designs helps combat anisotropy in the soil and environmental feature sets that can occur due to the direction in which the field was planted or harvested. The CV method of the illustrative embodiment may permit improved estimates of the model's generalization error. Typically, a buffer of a given size around each CV-fold is set to prevent leakage due to spatial autocorrelation. In one embodiment where data is spatially autocorrelated, samples may be present in both the training and test data sets that may force the model's generalization error to be significantly underestimated.

Referring to FIG. 7 there is shown an illustrative spatial image associated with a cross validation (CV) framework. More specifically, FIG. 7 provides an illustrative example of an anti-diagonal cross-validation strip **7100** within the cross-validation framework. Additionally, FIG. 7 shows multiple plots of land with defined geometric polygonal boundaries **7140** used for CV testing. The CV framework in the process block **7100** has a cross-validation strip composed of training data **7110**, test data **7120** and buffer data **7130** that is used for validation and tuning of the RF model.

Referring to FIG. 8 there is shown a flowchart of an illustrative process for training covariates for estimation of soil characteristics and environmental characteristics at various geo-spatial resolutions. In the illustrative flowchart of FIG. 8, there are two separate, but complementary, interpolation models that are applied to each soil chemistry variable sampled. The first interpolation model combines a “distance field” with a RF model to generate a Random Forest (RF) interpolation model. The second interpolation model is a specialized model of the Inverse Distance Weighting (IDW) model that is used to estimate the value of each soil chemistry variable at arbitrary locations within one of more plots of land.

In the illustrative embodiment presented in FIG. 8, the ability to estimate soil characteristics and environmental characteristics at different locations within one or more plots of land is used to build training data sets when an actual data set from farm sensors is either not available or when sensors are located too sparsely to obtain high spatial-resolution training data for accurate estimations by ML modeling. The illustrative method shown in FIG. 8 can be used for precision accuracy in the delivery of recommendations for additive nutrients and farm management operations. In various embodiments, the method of FIG. 8 may be used for other feature sets and to train the ML models such as yield, weather, elevation, and the like.

In the illustrative embodiment, interpolated soil characteristics, and environmental characteristics, hereafter referred to as “soil chemistry variables” or “soil variables,” are used to estimate one or more different clusters of soil variables that are co-dependent or form the main contributors and/or limiters to crop production. In the illustrative embodiment, soil chemistry variables (xtrain) may be defined as having at least one or more of the following illustrative soil chemistry variables: B, Ca, Cu, Fe, K, Mg, Mn, Na, NO₃—N, P, S, Zn, OM, pH, buffer pH, CEC, and the four percent nutrient saturations % K, % Ca, % Mg, % Na, and other such soil chemistry variables. In various other embodiments, the soil chemistry variables may include additional soil chemistry variables that are not described.

The method for interpolating soil chemistry variables described provides a high resolution interpolation method, which is generated from the sparse input training data. Additionally, a clustering analysis and ranking of the co-dependent soil variables is performed, which allows the ML models to provide recommendations for managing the soil chemistry variables. By way of example and not of limitation, the recommendations may identify the need to add, reduce or balance one or more soil variable quantities to optimize yield production efficiency and improve soil health. Thus, the system and method for interpolating soil chemistry variables permits the estimation of soil chemistry variables at any location within one or more plots of land at any location with a limited data set of soil chemistry variables.

At process block 8100, the first interpolation model combines a “distance field” with a RF model to generate a Random Forest (RF) interpolation model. In the illustrative embodiment, the RF interpolation model looks at each soil chemistry variable to build a forest of tree estimators fitted with at least one set of “ancillary covariates” such as soil chemistry, soil conductivity or other environmental covariates. Other ancillary covariates include, but are not limited to, growing days, planting dates, moisture indexing, local weather conditions such as rain accumulation, wind, temperature, humidity, average sunlight based on cloud cover or

sunlight hours, Soil organic matter, texture, top-soil depth, fertility indexes, and soil electro conductivity . . .

In general, “distance fields,” which are also known as a “distance map” or a “distance transform” are used to generate spatial predictors, which are then supplied to the RF model. For purposes of this patent, the term “distance fields” is defined as a representation of the distance from any point or variable value to the closest point or variable value within a domain of points or variable values in one or more data-sets. The domain may represent a set of similar dependent or independent variables consisting of geospatial, chemical, environmental, or other variable types.

More specifically, the illustrative embodiment uses Euclidean Distance Fields (EDF) to supply the random forest (RF) model with several spatial predictors. The fitted RF model is denoted as fRF (a), where a are the ancillary variables. The fitted RF model is then used to predict the values of the dependent variables at arbitrary locations within one of more plots of land.

In alternative embodiments, other interpolation models such as ordinary kriging (OK), regression kriging (RK), geographically weighted regression (GWR), contextual spatial modeling (CSM), and the like may be used as various spatial predictors for EDF. In yet another alternative embodiment, bagged multivariate adaptive regression splines (MARS), support vector machines (SVM), Cubist, Neural Network (NN) and the like may be used as machine learning alternatives to the machine learning RF method of the illustrative embodiment.

At process block 8200, the second interpolation model is a specialized model of the Inverse Distance Weighting (IDW) model that is used to estimate the value of each soil chemistry variable at arbitrary locations within one of more plots of land. The IDW interpolation method is used to obtain a smart smoothing effect based on the assumption that in the field of geography everything is related to everything else, but near things are more related than distant things. The assumption ensures that soil samples closer to a point of interpolation are weighted more heavily than soil samples that are more distant, as known to those skilled in the art and shown by the mathematical representation;

$$\hat{f}_{IDW}(s) = z(s) = \begin{cases} \frac{\sum_i \omega_i(s) z_i}{\sum_i \omega_i(s)}, & \text{if } s \neq s_i \text{ for all } i, \\ z_i, & \text{if } s = s_i \text{ for some } i, \end{cases}$$

where:

$$\omega_i(s) = \begin{cases} \frac{1}{\|s - s_i\|^p}, & \text{if } \|s - s_i\| < r, \\ 0, & \text{otherwise,} \end{cases}$$

and:

$$\{(s_1, z_1), (s_2, z_2), \dots, (s_N, z_N)\}$$

represents the set of N known locations ($\vec{s}_i$), and values ($\vec{z}_i$) for each soil chemistry variable, and where r and p are search radius and power parameters, respectively. The weighting coefficient $\omega_i(s)$ decreases as the distance increases from the interpolated point such that the resulting geo-spatial regions defined by radius r surrounding the interpolated points $\hat{f}_{IDW}(s)$ resolves into a set of mosaic tiles, hereinafter referred to as the “IDW-

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voxels” or “voxels” that are used to define an area of a plot of land representing a smoothed nearly constant multivariate surface for larger values of power variable p and within the area of the radius r .

In the illustrative embodiment, where the soil chemistry variables approach M dimensions, and soil sample locations (St) are far away the power p is set closer to the value of the total number of dimensions M and the radius r is increased to encompass a larger interpolation area. Thus, for computation using the IDW interpolation method, the dynamic parameterizations of r and p may be varied and optimized based on the calculation of the density estimation of points within the area of circumference of distances r_0 (center) to R (circumference). Other forms of distance weighting may also be used such as the Lukaszuk-Karmowski metric or modified Shepard’s method.

FIG. 8 presents a flowchart that includes feature-set storage and programing instruction flow, which is used to estimate soil variables at any location within one or more plots of land. As previously described, feature set data is collected and stored as either farmer data-sets **1080** (typically from sensors local to one or more plots of land) or from remote sensors, government agencies, agricultural groups and the like stored as external or third party data-sets **2000**. The stored data of the illustrative embodiment undergoes several stages of data pre-processing in preparation for ML model training. In the illustrative embodiment, pre-processed soil variables along with their known sampling locations are used to build at least one soil training matrix $[X_{soil}]$ as shown in process block **6100**, which is associated with FIG. 6.

The soil training data at block **8100** is used to build an interpolated data set at loop **8110** of random forest models that are used to estimate at least one soil variable using at least one environmental and at least one distance field vector to train the multiplicity of RF tree estimators used for each soil variable. The estimated RF interpolation outputs are the dependent output variables of the RF models, in which each output includes a single feature variable from the multi-dimensional soil variable matrix. In operation, the RF interpolation models may be run multiple times at loop **8110**, typically in a program loop, for each independent soil variable that is to undergo interpolation.

Continuing to process block **8200**, the method implements a second interpolation model. By way of example and not of limitation, the second interpolation model includes an Inverse Distance Weighting (IDW) model for each soil chemistry variable. In the illustrative embodiment, the IDW interpolation model may run at loop **8210**, which calculates each soil chemistry variable (z_i) at N known locations that are represented by one or more location vectors (s_i). Note, for each soil chemistry variable having IDW parameters r (radius) and p (power), the IDW parameters are used to define the interpolation bounds and smoothing factors, respectively.

At process block **8300**, the method continues by forming an ensemble of the RF and IDW interpolation models for each soil chemistry output variable that is “blended.” The combined “complementary interpolator” at block **8300** may be represented as a form of the RF+IDW ensemble interpolator represented by:

$$\hat{f}_{RF+IDW}(a, s)$$

where vis one of the soil chemistry variables to be interpolated from the $[X_{soil}]$ input covariate matrix and $\vec{a}$ represents the vector of ancillary input variables from the RF interpolator model, which is used to

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predict the values of the variable at arbitrary locations within one or more plots of land. Additionally, v represents the individual soil chemistry variable under interpolation and $\vec{s}$ represents the IDW point geometry location vector for the IDW input covariates y_{train}^v and x_{train}^{env} to be used in at least one of the IDW interpolation models.

The ensemble of the RF and IDW models **8300** is then used to calculate a non-optimized interpolated soil chemistry variable v as a function of the blending term a , which may be represented by the ensemble blending equation;

$$\hat{f}_{RF+IDW}(a, s) = \alpha \hat{f}_{RF}(a) + (1 - \alpha) \hat{f}_{IDW}(s)$$

By using a linear combination of these two complementary interpolation models, the combination of the strengths of both methods such that the random forest model compromises smoothness of the solution for accuracy, while the IDW model does the opposite by compromising accuracy for smoothness.

The method then proceeds to process block **8400** where a hyper-parameter tuning loop **8410** is used for optimization of the blending term a and the IDW model parameters r (radius) and p (power) are used to optimize each interpolated soil chemistry variable from the entire feature set of soil chemistry variables. Discontinuities and spatial variations of soil characteristics found in different plots of land are resolved by constructing error surfaces and adaptively partitioning the interpolated surfaces.

The hyper-parameter tuning loop **8410** uses the multi-fold cross validation method for parameter tuning as illustrated in FIG. 7. To assess the interpolation accuracy, error terms are calculated for each fold of the multifold cross validation surface. In the illustrative embodiment, a ten-fold cross validation method is implemented by splitting the data samples for each soil chemistry into folded cross validation strips using a portion of the input samples for interpolation and a portion for the multi-fold cross validation.

In one embodiment the error terms may be calculated by comparing different methods of error calculation and using at least one of the following methods: mean error (ME), mean absolute error (MAE), mean relative error (MRE), root mean square error (RMSE) and other such methods of error calculation. In the illustrative embodiment, the coefficient of determination or R^2 (R squared) error provides a measure of how well observed outcomes are replicated by the model. The R^2 error may be used to effectively normalize the error term where the accuracy is sufficient for hyper-parameterization optimizations.

More generally, hyper-parameter optimization at block **8400** uses programmatic loops **8410** with a fixed set of estimated a , r and p parameters for loop initialization for at least each soil variable and each fold in the CV framework as illustrated in part by FIG. 7. Samples from the input feature sets of training data are composed of training data **7110**, test data **7120**, and buffer data **7130** for tuning.

The error terms are calculated and compared to at least one tuning loop with each of the tuning parameters adjusted for least error in each hyper-tuning-loop of the RF+IDW ensemble interpolator. In various embodiments, where the discontinuities and spatial variations of soil characteristics are relatively homogeneous across one or more plots of land, the hyper-parameter tuning loop may not be needed or used to obtain reasonable adaptive partitioning of the interpolated soil variable surfaces. In an alternative embodiment, the number of soil chemistry variables to be interpolated may be reduced by a first pass calculation by the crop prediction engine **4000**. The first pass response may determine the

important soil variables that contribute a main-event or, within a group of coupled soil chemistry variables the important contributors to one or more crop-yield or seed-type yield response surfaces. Thus, the set of soil chemistry variables applied to the soil interpolation method may be reduced in number by first observations from ML modeling of the important soil chemistry variables in one or more plots of land.

The method then continues to process block 8500 and stores the resulting optimized interpolated soil chemistry values (z_i) for each of the estimated locations (s_i) into one or more storage devices, which are connected to the application computing cluster 4100 application database 4150 or the data science computing cluster 4300 data set database 4350 or in disk storage 4030 for use by the crop prediction engine 4000.

Referring to FIG. 9 there is shown a flowchart of a method for clustering ground types from interpolated covariates and environmental covariates. Additionally, FIG. 9 shows a flowchart for optimizing spatial resolution soil chemistry and environmental variables using voxels. Furthermore, the illustrative system and method of FIG. 9 uses a geo-spatially positioned grids of interpolated soil chemistry variables, i.e., "voxels," by making observations as the voxels traverse through the systems random forest yield response model. Each of the interpolated voxels describes a polygonal boundary that includes one or more interpolated soil chemistry and environmental variables located at one or more geo-spatial locations in one or more plots of land.

In the illustrative method of FIG. 9, voxel results are generated from the first RF based variable interpolation training model, which includes multidimensional point data. The voxel results are then applied to a second RF yield response model to identify clusters, i.e., "ground types," of homogeneous soil regimes. The resulting output from observations of the second RF model are used for subsequent statistical comparison to build further training models that uncover the drivers that may promote or limit crop production performance.

The illustrative system and method also utilizes clustering training models to assess effects of seed coatings, seed genetics, fertilizers and the like when careful experimental designs or small plot trials are not available.

By way of example and not of limitation, quarter acre voxels are used to represent the smallest polygonal area for each data set of interpolated or measured soil chemistry variables. Also, the size, shape, or components of a voxel may be specified differently, and voxels may represent other sets of independent variables other than those assigned to soil chemistry characteristics, environmental characteristics, or the combination thereof. Additionally, voxels may represent polygonal areas of crop yield, planting or other crop related feature-sets used for model training, predictive yield estimations or aggregated area visualization. Furthermore, voxels may contain one or more geographically located points that together or singly represent one or more dimensions of feature-set data and typically may represent a small area within one or more plots of land for training the ML model or represent predictive data output from the model.

In the illustrative embodiment, the system and method as specified herein determines where clusters of similar soil and environmental characteristics are located within one or more plots of land. The system and method use the random forest yield training model, as previously described, and simply traverses each voxel through an RF yield model for observation of the path of traversal through the forest of tree estimators. The traversal of a voxel through the random

forest starts at the root node and propagates through the tree ending up in one and only one leaf node. The method then repeats for one or more tree estimators in the forest. Thus, each independent variable associated with each voxel is evaluated at each branch through each tree such that each of the applied voxels eventually end up at one and only one leaf node for each tree, which results in each leaf node in each tree having no voxels, one voxel or multiple voxels present after the traversal of all voxels through one or more tree estimators has finished. When multiple voxels end up in the same end-point, i.e., a leaf node, such voxels may be considered homogeneous with each other, and they are considered clustered or co-associated with one another.

Voxel co-association (to generate clusters) has no geographic components and, as a result, clusters of voxels with like characteristics naturally group together as voxels that fall into the same leaf-nodes of an RF tree. The clustering of voxels having like features is the result of the RF training model's ability to observe sets of input covariates that co-associate. The only reference to geometry, during the clustering observation, is to track assigned geometry locations previously given to each voxel for geographical location referencing across one or more plots of land.

Thus, the illustrative system and method of FIG. 9 can predict clusters of homogeneous voxels leading to a training model that can predict which clusters contain the soil and environmental similarities that co-associate and that produce the best crop production response given one or more differing sets of input variables. Additionally, the illustrative system and method describes a process for "learning" a model topology and how to use the model topology to extract observations of feature sets performed by a random forest. The system and method begins with the supervised yield prediction model to define the RF distances between each pair of interpolated samples.

A voting mechanism may be used to add up voxels and to understand voxels that are not coupled across the entire set of trees estimators and their corresponding leaf-nodes. The illustrative method tallies up the votes from all leaf-nodes of all tree estimators to determine how often the same voxels end up in the same leaf-node. The process of summation of like covariates may be known as "simple" clustering. Next, the method calculates the co-occurrence frequency S_{ij} which is a number that represents how many times within [T] trees both V_i and V_j end up in the same leaf-node.

$$S_{ij} = \frac{((V_i + V_j) | \text{when } V_i \text{ and } V_j \text{ are in the same node})}{T}$$

Where V_i and V_j are the observations of voxels that traverse through the trees. When V_i and V_j end up in the same leaf-node, they are considered to be similar. If they end up in different leaf-nodes or in different trees they are considered to be different and not similar. For example, even when two voxels have the same value of estimated output $\hat{y}$ they may not be considered similar because the feature data represented by the voxels themselves may be different.

The illustrative system and method may also cluster voxels based on a similarity metric rather than a distance metric as described in the illustrative embodiment described herein. Thus, the illustrative embodiment may use the calculated co-occurrence frequency to gain an understanding about the magnitude of the estimations of similarities between the input variables.

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Calculation of the co-occurrence frequency enables the model to then determine a random forest distance metric D_{ij} which is simply the distance metric between V_i and V_j as represented by:

$$D_{ij} = \sqrt{1 - S_{ij}^2}$$

Wherein the distance metric D_{ij} may be used to build at least one distance matrix of $N \times N$ dimension including N voxel observations where the N voxels may represent the area of one or more plots of land. The $N \times N$ distance matrix would have a zero value diagonal since the distance between a point and itself is always zero and the remaining entries of the distance matrix are the respective distance metric values of D_{ij} where $i=1, 2, \dots, N$ rows and $j=1, 2, \dots, N$ columns are the span of the matrix.

The random forest distance metric may be considered the estimated distance between the feature components within each voxel with the physical location of each voxel assigned a geographical location typically within one or more plots of land. The building of a cluster solely on the representation of the distance matrix may not be an accurate representation since nothing in the distance matrix is informative about the space at which the observations exist and dimensionality may be unknown with only a distance metric between points or between observations of voxels. For example, distances may be arbitrary in high-dimensional space and the distance matrix as computed from the random forest estimator, which knows nothing about the space the points or voxels actually reside in. Thus, the use of the raw random forest distances may not pull the signal from the noise and may result in non-contiguous clustering. For example, given a distance matrix of N voxels on one or more plots of land with $(1-N)$ voxels on each side and given a raw distance matrix D_{ij} , silhouette analysis may be used to determine the effects of high-dimensionality on the properties of the clusters that show poor separation.

The illustrative embodiment may use a generalization of Uniform Manifold Approximation and Projection (UMAP) training model to “clean” the distance matrix by reduction of the order of the high-dimensional space resulting in lower dimensional data which improves the clustering results of voxels into a more contiguous set of ground types.

Some embodiments may use dimensionality reduction training models such as K-means, t-distributed stochastic neighbor embedding (t-SNE) and others to clean distance matrix data sets. The reduction training model t-SNE for example, only looks at the entropy of distances between points, where in contrast, UMAP adds a gap term to also look at gaps between points that are near and far away from each other. The use of K-means and t-SNE limits the ability of the model to understand gap distances between high order points along a non-linear manifold as represented by the distance matrix.

The ability to accurately clean the distance matrix may require the use of a manifold learning training model with the ability to learn the space surrounding the models manifold. The UMAP training model is typically known as a data visualization technique for taking high dimensional data and visualizing it in a lower dimensional space. For example, assuming the distance matrix data-set can be described to live on or around some arbitrary manifold in high dimensional space and in a highly non-uniform manor along the manifold, the UMAP data visualization training model uses a number of steps to learn the complexity of the manifold to reduce the order by learning what points along the manifold are near each other and what points are far from each other. Additionally, the UMAP data visualization training model

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learns the complexity of the manifold to reduce the order by comparing all points with a fuzzy method that wraps each fuzzy bit with a simplicial covering using triangles for the points that are near to each other so that two fuzzy points form a line, three fuzzy points are two dimensions and many fuzzy points within the simplicial covering may be of higher-dimensions. Since the dimension of the simplex covering at a point along the manifold is an indicator of the overall dimensionality of the whole manifold, the higher order points within the simplicial covering may be reduced to a single point along the manifold. Thus, the UMAP training model uses a fuzziness factor and forms fuzzy groups made up of triangles where the simplicial covering is a rough estimate of the dimension of the manifold at that location along the manifold and the dimension of the simplex covering at a point on the manifold is an indicator of the overall dimension of the entire manifold.

After application of UMAP, high-order reduction to a lower dimensional space is obtained and clusters have a much better clustering response in contrast to clusters in a high-dimensional space. Thus, for the distance matrix, by reducing the order of the data the signal rises well above the noise. In the illustrative embodiment, hierarchical density-based spatial clustering of applications with noise (HDBSCAN) may be used to cluster the cleaned data after UMAP processing. The HDBSCAN training model is a hierarchical clustering training model designed to group points with distance proximity of a certain radius.

In the illustrative embodiment, the UMAP and HDBSCAN process is wrapped in a hyper-parameter optimization tuning loop to choose the optimal values for the UMAP and HDBSCAN hyper-parameters. For cluster validity assessment the use of silhouette analysis or Density-Based Clustering Validation (DBCV) may be used to run gap statistics and optimize the number of clusters for the best separation in each step of the hyper-parameter tuning loop.

In an alternative embodiment, a vector quantization training model such as K-means clustering may be applied for cluster analysis before or after the UMAP dimensionality reduction step. In yet another alternative embodiment, UMAP and K-means may also be run iteratively to improve tuning parameters and achieve cluster optimization. K-means clustering or one of its many variants may be used to further group points into clusters of arbitrarily sized area in multidimensional space.

In the illustrative embodiment, the resulting clusters may then be classified into ground types for the purpose of representation on maps, in spread-sheets and for predictive analysis and determination of best estimation for crop planning, supply purchasing and return on investment in agricultural management. To represent the clusters on maps and in training models representing crop response zones, the illustrative embodiment uses the Voronoi Tessellation training model to move data-representation from points to polygonal geometries. Voronoi cells are also known as Thiessen Polygons and may be thought of as the partition of a plane into regions close to each of given set of objects. In the illustrative embodiment, the input objects to be tessellated represent the predictive clusters including point vectors describing soil chemistry and environmental characteristics as described above. The output of the Voronoi Tessellation is a set of 2D polygonal boundaries with embedded objects that may represent a given finite set of points in a Euclidean Plane. The Corresponding Voronoi cells include every point in the plane whose distance is less than or equal to the distance to any other point. The Voronoi cell walls represent all the points in the plane that are equidistant to the two

nearest sites and Voronoi vertexes represent all the points equidistant to three or more sites.

FIG. 9 shows a detailed flowchart of a method for clustering ground types from interpolated covariates and environmental covariates. Additionally, FIG. 9 describes the process steps performed by the crop prediction engine 4000 to cluster the interpolated soil chemistry variables into predicted soil chemistry characteristics located in one or more voxel areas in one or more plots of land.

The input data-sets, data-set delivery apparatus and network programming instructions 2000 of the illustrative embodiment are used to build the random forest tree estimators, train the RF model and be the input covariates used to predict which sets of the feature data-set inputs are co-associated with each other.

As previously described in FIG. 2, input feature-set data is preprocessed at block 2110 for cleaning and normalization to form data-sets used to train the ML model. The same input data-sets are then used in a first RF model to compute interpolated values in process blocks 8100, 8200, 8300, 8400, and 8500 (described in FIG. 8) for soil chemistry and environmental characteristic within the feature-sets of data as previously described.

The programming instructions for the clustering process as illustrated in FIG. 9 begins with the building of a supervised RF yield response and prediction model 5300 used to define the distances between each pair of interpolated samples.

As previously described, process block 5300 operations are performed by crop prediction engine 4000. The operations performed at process block 5300 include building a random forest (RF) training model and using first out-of-bag training data, which assesses initial conditions, parameter settings, and first pass model quality using techniques like R-squared error minimization prior to application of actual training data sets.

Process block 5300 receives the interpolated values from process blocks 8100, 8200, 8300, 8400 and 8500. The data fed into process blocks 8100-8500 are received from the data set ingestion and processing at process block 2000, which is then preprocessed at process block 2110.

Process block 2000 receives transient and permanent data set types. Data sets may be obtained from one or more of the farmer's partners, associations, agriculture organizations, third party satellite companies, government agencies and the like.

At process block 2110, the pre-processing of data sets from block 2000 is performed. More specifically, the process block 2110 operations include identifying data items that are outliers, invalid, redundant, missing or collected data from outside a field boundary; and such data may be removed, substituted, or imputed from an average from nearest neighbor data, supplied from various data set sources or substituted by various data sets from previous seasonal results.

The random forest training model built at process block 5300 includes building RF training models of tree estimators at process block 5330. The data set that is received by process block 5330 includes a bootstrapped data-set 5310. The bootstrapped data-set 5310 is also used to select an out-of-bag 5320 data-set from a subset of unused training vectors from the bootstrapped data-set 5310. The bootstrap data-set 5310 is also used to select the out-of-bag 5320 data-set from a subset of unused training vectors from the bootstrap data-set 5310 used for quality and accuracy determination of the tree estimators in the RF yield regression model 5360 of the random forest.

Note that the bootstrapped data set includes a statistical resampling technique that involves random sampling of a dataset with replacement, which is often used as a means of quantifying the uncertainty associated with a machine learning model.

The quality of the RF training model at process block 5330 may then be determined by applying the out-of-bag data-set 5320 to each tree in the forest, comparing the correctly predicted outcomes to the incorrectly predicted outcomes and determining the out-of-bag error 5340, which is simply the ratio of incorrect to correct predictions based on the application of the out-of-bag data to all trees in the forest.

If the out-of-bag error is not within an acceptable range 5355 the process starts over again 5310 with alternate covariate or covariate weighting selections used to tune output results. In another embodiment, the RF method may be changed to incorporate additional comparison terms in each branch of each tree such that the modification of the RF method used to build the forest has less out-of-bag error. After one or more iterations through the RF quality loop 5355 and when the prediction quality of the RF yield regression model is in an acceptable range 5356, the random forest yield regression model, at process block 5360, is considered to be trained, has acceptable predictive quality and is ready to have the interpolated soil and environmental feature-sets applied 5357 from the stored values 8500 computed by the variable interpolator.

Thus, the clustering method of the illustrative embodiment uses a second supervised yield prediction model 5360 to define estimated RF distances between all pairs of interpolated sample inputs from the first model using the previously defined Random Forest soil interpolation prediction model 8100-8500.

The clustering computing process as illustrated in FIG. 9 continues by applying Uniform Manifold Approximation and Projection (UMAP) 9100 to create a lower-dimensional embedding of the interpolated samples as previously described.

The UMAP training model cleans the distance matrix by reduction of the order of the high-dimensional space resulting in lower dimensional data, which improves the clustering results of voxels into a more contiguous set of ground types. UMAP adds a gap term to also look at gaps between points that are near and far away from each other. The ability to accurately clean the distance matrix may require the use of a manifold learning training model with the ability to learn the space surrounding the models manifold. The UMAP training model is typically known as a data visualization technique for taking high dimensional data and visualizing it in a lower dimensional space. The UMAP data visualization training model uses a number of steps to learn the complexity of the manifold to reduce the order by learning what points along the manifold are near each other and what points are far from each other. Additionally, the UMAP data visualization training model learns the complexity of the manifold to reduce the order by comparing all points with a fuzzy method that wraps each fuzzy bit with a simplicial covering using triangles for the points that are near to each other so that two fuzzy points form a line, three fuzzy points are two dimensions and many points within the simplicial covering may be of higher-dimensions. The UMAP training model uses a fuzziness factor and forms fuzzy groups made up of triangles where the simplicial covering is a rough estimate of the dimension of the manifold at that location along the manifold and the dimension of

the simplex covering at a point on the manifold is an indicator of the overall dimension of the entire manifold.

In the illustrative embodiment the lower-dimensional embedding may be followed with a generalization of the Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) at process block **9200** to cluster the interpolated samples into sets of geographically located points. After application of UMAP high-order reduction to a lower dimensional space is obtained and clusters have a much better clustering response. In the illustrative embodiment, HDBSCAN may be used to cluster the cleaned data after UMAP processing. The HDBSCAN training model is a hierarchical clustering training model designed to group points with distance proximity of certain radius. Also, the UMAP and HDBSCAN process is wrapped in a hyper-parameter optimization tuning loop to choose the optimal values for the UMAP and HDBSCAN hyper-parameters.

The resulting sets of geographically located samples undergo a hyper-parameter tuning loop **9300, 9400** to assess the cluster validity. Cluster validity may be determined using Silhouette Analysis (SA) and/or Density-Based Clustering Validation (DBCV) to choose the optimal values for the UMAP and HDBSCAN hyper-parameters **9500**.

In the illustrative embodiment, the DBCV **9300** method allows the construction of at least one “relative validation index” for optimization of density-based, arbitrarily shaped clusters. Each of these indices are further used to modify the hyper-parameters **9500** when the acceptable accuracy **9410** is not achieved by one or more hyper-parameter tuning loops and additional tuning is needed. The hyper-parameter tuning loop completes when the acceptable accuracy **9420** meets one or more pre-determined accuracy levels or has iterated some maximum number times.

At process block **9600**, the final step to cluster soil variables moves the clustered data from the point geometries to smaller subset of polygonal geometries also represented in a collection of voxels, that when combined, represent at least one tessellated polygonal cluster in at least one voxel in at least one or more plots of land. To move from point data to polygonal surfaces the illustrative embodiment uses the 2D Voronoi Tessellation training model **9600**.

The clusters are classified into ground types for the purpose of representations on maps, in spread-sheets and for predictive analysis and determination of best estimation for crop planning, supply purchasing and return on investment in agricultural management. The illustrative embodiment uses the Voronoi Tessellation training model to move data-representation from points to polygonal geometries. The input objects to be “tessellated” represent the predictive clusters including point vectors describing soil chemistry and environmental characteristics as described above. The output of the Voronoi Tessellation is a set of 2D polygonal boundaries with embedded objects that represent a finite set of points in a Euclidean Plane. The corresponding Voronoi cells includes every point in the plane whose distance is less than or equal to the distance to any other point.

In the illustrative embodiment, the desired output in the illustrative embodiment is to associate $\frac{1}{4}$ acre grids to the voxel area where each voxel contains the desired output of co-associated soil and environmental characteristics that are clustered into polygon geometries, herein after called ground types or more commonly called crop response zones in the agriculture industry.

FIG. **10** provides an illustrative image of the final output results from the method presented in FIG. **9**. More specifically, FIG. **10** shows the resulting predictive multidimensional data embedded into ground types made up of voxels

assembled into one or more plots of land. Each ground type may have at least one voxel representing the soil chemistry and environmental characteristics that may be used as a predictive model to understand enhanced or limited crop production and management practices.

A dated but widely deployed tool for creating management zones within individual fields is the USDA’s Management Zone Analyst which uses a Fuzzy C-means clustering training model. While most existing clustering techniques in precision agriculture define continuous regimes the clustering approach employed by the illustrative embodiment is a more agile method of crop production management across one or more plots of land following where the data lead rather than imposing traditional constraints of broadly contiguous management zones. In the illustrative embodiment, crop response zones need not be contiguous nor confined to a single plot of land but can be modeled at large scale providing improved understanding of the cost advantages for crop production and management efficiency.

The systems and methods described in FIGS. **11** through **14** relate to the use of Random Forest (RF) training models as shown in FIGS. **5** and **6**, which are further developed for the determination of one or more Response Surfaces (RS). The RS is a multidimensional window into the Random Forest yield prediction model. The RS training modeling is used in statistical analysis to explore the relationships between several independent input variables and one or more output “response variables.” RS analysis is applied to a sequence of calculations and observations to obtain one or more optimal responses per one or more main-effect input features.

Response Surfaces are typically simplified by reduction from multidimensional space to at least one second-degree polynomial model. The RS training model is used to approximate one or more estimated optimum responses between the important input features and may be used for simplification of graphical display. For example, FIG. **15** illustrates an RS training model generating an estimated yield output as a function of two feature set variables, namely, sodium (Na) and magnesium (Mg). In the illustrative embodiment, optimal RS are based on predictive ML modeling using Random Forest and subsequent order reduction using RIT training models.

The corresponding order reduction techniques of the illustrative embodiment are sufficient enough to determine which independent variables, e.g., soil chemistry variables, affect a response variable, e.g., crop yield, of interest. The multidimensional input variables may be from any input feature-set of independent variables that affect the estimated output dependent variables.

For example, in one embodiment the input features may be used as “control parameters” and other environmental covariates may be used as “conditioning parameters” that are learned while building the yield prediction model. The estimations of response surfaces may be ranked in order of importance and interpreted for estimating how specific input features contribute to the main effect and also to understand which feature-set items couple to affect crop yield performance. In one illustrative embodiment, one or more additional feature-sets such as land elevation, wetness index, weather and crop management practices may be input as additional feature-sets and may be used as input features by the illustrative system and method presented in FIGS. **11** through **15**.

The illustrative embodiment presented below describes how to reduce the p-dimensional feature space to a subset of just a few important feature items that define the estimated

main-effect and the corresponding prevalence of feature interactions. By reducing the order of the feature space, the domain of the resulting RS becomes the Cartesian product of the ranges of each of the features contained in the interaction. In the illustrative embodiment all features not present in the interaction are integrated out simplifying the resulting response.

The illustrative embodiment builds estimated response surfaces from extraction of decision rules from each tree in the Random Forest. The decision rule extractions on paths where input covariates travel through the same path of the tree estimators forms a single stump. In the illustrative embodiment, each decision stump includes a decision rule that may result in two or more dependent variable values. For example, a left value which may represent the average yield over the samples where the decision rule is true, and a right value which may represent the average yield over the samples where the decision rule is false. Thus, a decision stump forms a tree with a height of one, and each stump of height one has one root node and two child nodes. In one embodiment a decision stump may have more than two branches such that each branch results in a fractional decision resulting in more than two values. The collection of decisions from each stump may be considered multivariate in general because the decision rule is a function of all the features in the interaction.

Referring to FIG. 11 there is shown a flowchart of the process steps used to generate an estimated Response Surface (RS) training model. Beginning at process block 11100, program instructions select a voxel from one or more voxels associated with one or more plots of land. Note, voxels are associated with each feature set of independent variables for the selected voxel location. Also, input feature sets must retain a location index for geo-spatial coordinate location to maintain an association between the location of input features and output responses using geo-location systems and methods.

The voxel location may be represented by a variety of geography point values that lie within at least one defined voxel area, and this point geometry may be used for the association between data-sets that build and traverse the ML model. In another embodiment a collection of point values may be used in a polygonal format to represent the association between data-sets used to build and traverse the ML model.

After selecting the voxel area for model estimation 11100, the illustrative embodiment continues to block 11200 where the computing instructions build, train, and optimize the Random Forest (RF) yield response training model. The Random Forest (RF) training model uses a variety of tree estimators built from at least one data set of input features. The RF may be "built" by using randomly sampled data records from the input data that includes independent and dependent variables. In the illustrative embodiment, the minimum set of input variables includes soil and environmental independent variables and dependent variables like crop yield.

In another embodiment, additional feature sets may be used to further augment the results. In still another embodiment, each RF model is built for a specific crop-type with the purpose of generating crop yield predictions based on the input variables supplied within that specific crop-type. In yet another embodiment, the RF training models may be built generally to accommodate one or more different crop types. The RF training model 11200 is built using a first set of boot strap data from input data-set records that are randomly selected. The RF training model is then tuned by running the

out-of-bag input data records for each tree through the forest of tree estimators and comparing the predicted dependent output variable to the expected output results. Thus, the out of bag error is used to tune the tree estimators by re-building and optimizing selections of the independent variables that have more significant results to minimize the out-of-bag error from the variety of tree estimators.

The method continues to process block 11300 where the programming instructions determine which interactions between the variables in each set of independent input variables have a high co-occurrence frequency. This process is accomplished by observation of the traversal of each input variable from each feature set through each tree in the forest of tree estimators. Independent input variables that follow the same path and end up in the same leaf-node of a tree are considered to be co-occurring and may be considered to have more importance for estimations of the output model's yield response. The observations of the co-occurring independent variables as they traverse through any tree from root to leaf-node determines the decision path. Determination of co-occurring input variables may operate as a voting process over the entire forest of trees by first observing the variables associated with the feature sets that end up in the same leaf-node for each tree and then accumulating the results across the entire forest. As a result of the voting process, the independent variables that co-associate with higher frequency determine the interactions that are estimated to have the most importance to the model's response.

At block 11400, the most significant interactions are identified to reduce the number of observations from the space of possible outcomes. The most significant interactions are determined by performing the illustrative process step to integrate out the non-interactions to reduce the number of interactions. The most significant interactions are used to further extract the predictive response of the model.

In the illustrative embodiment, Random Intersection Trees (RIT) are used to reduce the order even further by building simple binary path vectors for each decision path where a true decision is represented by a "one/true" value and a "zero/false" value that are represented as individual elements in at least one binary vector. The dot product of each binary vector for each independent variable (that has been previously identified as important) from each path are used to further reduce the number of observations and further integrate out non-co-occurring covariates.

The method then proceeds to process block 11500, in which the interactions are ranked. More specifically, prior to determining the magnitude of the effective response, each interaction may be ranked for importance in order to further reduce the response to a much smaller set of the most important contributors. In the illustrative embodiment, the ranking of the features from the identified feature interactions is accomplished by "prevalence" which is based on the product of the node quality and the relative frequency of co-occurrence within the leaf-nodes of the trees in the Random Forest indicated by where the important interactions have settled. The node quality may be estimated by looking for homogeneous yield in each node between the coupled covariates that end up in the same node. Selecting feature interactions with high prevalence permits identification of subsets of simple decision rules that frequently co-occur on decision paths of the Random Forest. These simple decision rules are combined to form the decision rules that build the decision stump model.

At process block 11600, a variety of decision stumps are formed by observation of each interaction S of the decision paths for each tree within the Random Forest. The trees may

be traversed and the decision rules for each “s” (decision block) in the interaction “S” (important co-occurring covariate set) are extracted wherein “s” represents at least one component of the interaction set “S.”

At process block **11700**, the Response Surfaces (RS) for each resulting interaction is formed. In the illustrative embodiment, the Response Surfaces (RS) for each resulting interaction are formed by the crop prediction engine **4000**.

Referring now to FIG. **12** there is shown an example of the traversal of one decision path **12000** through one tree from root **12100** to final leaf-node **12600** where an interaction “S” shows three co-occurring variables P, K and OM that resolve to the at least one or more dependent variable nodes **12600**, **12700**. The decision path may include other independent variables **12300** that are not considered important under the example of the interaction in FIG. **12**. For example, the decision rules for each “s” (decision block) **12200**, **12400** and **12500** are along the decision path **12000** for $P < 40$, $K < 89$ and $OM < 1.4$, respectively, are illustrated. In the illustrative embodiment, the decision rules for each “s” in “S” are extracted starting with the co-occurring variable **12200** which is closest to the root node **12100**. In the decision path example of FIG. **12**, the rule $P < 40$ **12200** may be considered as the starting point **12150** (the decision rule closest to the root) for the extraction.

The predicted or trained yield values **12600**, **12700** are associated with one or more voxels for one or more “true” results **12050** or one or more “false” results **12160** along the variety of possible paths. Thus, the extracted rules for each path are combined to form at least one decision stump for the interaction, resulting in an ensemble of stumps across all decision paths containing the interaction.

FIG. **13** shows an illustrative third-order stump from the decision path example presented in FIG. **12**. In the illustrative embodiment, each decision stump in the entire ensemble of stumps always has a tree height of one (1), as represented by referenced number **13200**. The tree height of one (1) represents a direct binary relationship, (True or False) on the final path to the dependent variables of the tree. The decision logic **13100** for the stump example shown in FIG. **13** is a logical AND gate for each decision block along the path where the interaction variables are under observation. When the outcome of the decision **13100** is determined to be “True,” the estimated yield for a path may be found in the last node **13300** of the path. The method iterates for all paths through all trees building an average yield response for the ensemble of trees where all decision paths satisfy all of the interaction’s decision rules. Similarly, when one or more of the outcomes in the decision path of the interactions is “False,” the estimated yield for a path may be found in the last node **13400** of the path. More generally, the estimated yield is repeatedly calculated and averaged for the ensemble of tree stumps to determine interactions that do not meet the decision stump conditions to achieve a predicted average yield.

After forming the ensemble of decision stumps for a given interaction “S,” the stumps may be evaluated at representative points within the range of each “s” in “S” and wherein each stump in the ensemble of stumps contributes to the final crop yield response. Denoting the response surface for the interaction “S” as $z(s)$ the response surface may be represented by the following equation:

$$z(s) = \frac{1}{B} \sum_b \sum_k \frac{w_{k,b}}{W_b} z_{k,b}(s)$$

Where b is one of the B trees in the Random Forest, k is one of the k_p stumps formed from tree b , s is a point for the interaction S at which the response surface is evaluated, $Z_{k,b}$ is the k th stump from the b tree, and $W_p = 2k W_{k,b}$ with $W_{k,b}$ is any weighting metric for the stump response (e.g., the size of the node).

For the illustrative embodiment, the determination of which of the large set of interventions to keep and which to cull may be used to estimate the main independent variables and most important variable combinations. These important interventions are regressed to form the leading drivers and leading limiters for estimated dependent variable output, e.g., crop yield. In order to determine the most important drivers and limiters from the set of important Response Surfaces $z(s)$ the order must be again reduced to something manageable, or in the case of the illustrative embodiment something displayable in three dimensions. To accomplish further simplification a Reduced Order Surrogate Model (RoSM) may be used to stitch together the most important modeled Response Surfaces. For example, the final RoSM is a low order Random Forest surrogate model that can be used for crop yield driver estimation simplification forms. By way of example and not of limitation, it would be preferable to reduce the list of interactions to under 30 for evaluation purposes.

Referring now to FIG. **14** there is shown a flowchart for building a Reduced Order Surrogate Model (RoSM) training model. The method is initiated at block **14000** where a list of Response Surfaces (RS) is created for each interaction. The method then proceeds to process block **14100**, in which each RS from the list of RSs is prepared for further analysis.

The method then proceeds to process block **14200** where a regression process such as a least squares estimations are used to form the initial hypothesis that determines the most important interactions from at least one culled list from process block **14100**, process block **14750** and process block **14850** of ranked interactions.

The illustrative method continues with relative accuracy testing using cross validation at decision diamond **14250** and tuning of the RS weighting coefficients at process block **14260** and model refitting at process block **14270** to achieve the desired accuracy and model estimation expectations for each of the interactions in the list. The illustrative regression process examines the estimated model output for the main-effect and the main co-occurring features from one or more set of interventions that have exhibited the highest co-occurrence frequency from the RF model. Once a pruned list of low order interactions is built by ranking and reduction analysis, the method iterates through at least one list of Response Surfaces starting with the surface that trends to be the closest to the main-effect, moving through the list adding and removing certain interventions from the list.

In the illustrative embodiment, the iteration process for each tuned loop is termed a “progression” and is used to find the independent variables that have the most contribution to the dependent variables in the model, e.g., crop yield drivers, including those that hinder the model estimated performance or limit, e.g., crop yield limiters, the model’s response.

At process block **14400**, the analysis for relative accuracy determines how important each progression is to the training model accuracy and has the ability to optimize the interaction set by addition or removal of RS sets during the first set of iteration loops shown at decision diamond **14600**. The determination of whether or not the progression has a major contribution to each of the training model estimation accuracy is used to either remove the interaction from the list or save the progression for further analysis and estimations.

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At decision diamond **14500** the determination about model degradation is performed. At block **14550**, the process may find one or more progressions that when removed from the model may improve relative accuracy of the estimated response and in this case the progression may be labeled and saved for further iterative analysis as a possible “crop-limiting” interaction.

At decision diamond **14450** the determination to improve the model is performed. At block **14460**, the analysis may find a progression that when added to the model improves the training model relative estimation accuracy and may also be saved for further analysis as a possible “crop-driver” interaction. Additionally, the training model may set iteration limits and bounds when testing for accuracy such as limits to the number of inner tuning iteration loops for each possible crop-driver and crop-limiter progression.

At decision diamond **14700**, the top drivers are identified, and at decision diamond **14800** the top saved crop limiters are identified. At process block **14750**, the response surfaces from the saved top crop drivers and, at process block **14850**, the top saved crop limiters from the previous RoSM may be iterated in a second tuning loop, which is represented by block **14900**, before being selected as the training models top candidates for covariate contributions to the estimated crop yield response model.

Additional iterations and tuning loops may be used to further optimize the model for estimation accuracy. Thus, the process of observations, additions and removal of terms and accuracy testing with least squares and cross validation may be used to improve and optimize the Response Surface Methodology (RSM).

At process block **14950**, the tuned coefficients are then solved for and written into a finalized set of Generalized Additive Model (GAM) equations which may represent the one or more covariates that contribute to both the main-effect and/or the most important combined interactions from the entire array of feature data-sets. The relative importance of the GAM is that it forms a linear equation of non-linear terms with one or more “linear” additive terms for each interaction. The form of the training model then allows for simplification by adding or removing the additive terms for both simplification and visualization through order reduction of the top interactions. Thus, the GAM presents a training model where each independent variable in the interaction of additive terms forms a linear equation with respect to each independent variable in the estimation model. The final optimized GAM provides a training model where the independent variables are linear additive terms while the additive terms’ coefficients may be complex non-linear representations that are now independent from the input variables.

In another embodiment, it may be beneficial to tune the RS evaluation list by calculating the R-Squared error or computing T-test for covariates to find the best and worst T-test scores. In yet another embodiment, the alternative of L1-L2 regularization (Lasso) may be used to evaluate more than one RS at a time. In still another illustrative embodiment, retractable convex optimization may be used adding a penalty in an objective function resulting in a form of Lambda times the coefficients from ordinary least squares approximation.

The final step of the illustrative embodiment is to integrate the interactions into at least one Generalized Additive Model (GAM) and resolve to one or more nutrient and seed-type prescriptions and/or recommendation used for “as applied” or “as planted” application for one or more plots of land.

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In the illustrative embodiment, the GAM training model may be represented as a linear equation of non-linear terms and may be represented by:

$$\hat{y} = [\alpha(\text{Interaction}_1) + \beta(\text{Interaction}_2) + \gamma(\text{Interaction}_3) + \dots]$$

where $\hat{y}$ is the model’s estimated crop yield, α , β , γ represent complex coefficients and $\text{Interaction}_{1,2,3}$ are the most important interactions that include combinations of co-occurring input feature variables. The final simplified model of the illustrative embodiment may then be used for large scale modeling of expected crop production operations having the ability to predict crop yield, Return on Investment (ROI) and overall soil health/value prior to or during at least one planting season.

By using a GAM training model to estimate the most important contributors or limiters that contribute to crop yield response, it is possible for growers to take actions based on model recommendations prior to planting. An additive model with quarter-acre resolution accuracy allows growers to understand what nutrients to apply, make changes to applications for environment or soil moisture recommendations and select hybrid seed products that work best in different soil regimes at large scale across one or more plots of land.

FIG. **15** shows an order-2 response surface **15000** showing the independent and combined effect of the % Mg and % Na nutrient saturations on corn yield. The order-2 response surface **15000** shows that the greatest increase in corn yield resulting from % Mg occurs at 20%-30% Mg for all concentrations of Na. Additionally, the order-2 response surface **15000** shows that the greatest increase in corn yield resulting from % Na occurs at 1%-2% Na for all concentrations of Mg.

FIG. **16** represents one exemplary embodiment, that is not limiting, of a top level system architecture **16000** of the present invention. FIG. **16** illustrates five segmented blocks **4000-4004**, each of which contains processing steps, software elements, and hardware elements that, in combination, comprise the components of the complete system **16000**.

Segment block **4004** represents the Internet of Things (IoT) input data sent from a plethora of sensors, devices, data storage and public or private data bases. As described above and illustrated in FIGS. **1** and **2**, blocks **1080** and **2000**, input data-sets **4004a-i** are typically from measured crop field operations **4004e**, **4004f**, **4004h**, satellite imagery **4004a**, and/or large data repositories **4004b-d**. The input data of block **4004** is used to both train and build the predictive ML models as well as provide input stimulus data used to exercise the pre-built ML models providing various predictive responses.

Segment block **4002** illustrates transactional data transport services provided by one or more computing cloud providers, data centers, Internet Service Providers or other computer and software platform providers. These providers supply computing resources and customer support for public or private computing service users. In the illustrative embodiment, Amazon Web Services (AWS) provides such computing resources in the form of an AWS DNS domain controller **4010**, an AWS S3 disk storage **4030**, and an AWS load balancer and gateway interface **4110**, which are communicatively coupled to one another through the Internet **3000**. AWS services in combination with the Internet (World Wide Web) **3000** distributes computing, storage and networking devices to provide access to, and transport, IoT input data **4004** between one or more physical devices, such as remote servers or one or more local computers that when

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enabled run the systems software under the methods and processes of the present invention. Input data-sets **4004** containing multiple forms of Agronomic and image data use Internet and transport services **4002** to transport and load the data sets **4004a-i** between one or more computing clusters **4100** and **4300**, and a plethora of browser based client devices **4003**.

Input data **4004** originating as sensor measurements from one or more farm implements (IoT devices) may be transported and stored to at least one client device **4400** via at least one input interface device **4040**. Input data may require subsequent processing by program instructions embodied as client device software **4401** running on at least one Hardware Apparatus **4400** which may be one of many client devices operating a client browser **4003** wherein a client device is not limited to a web-browser by illustrative example. The client device software **4401** may include one or more operating system kernel process, memory, scripts and services, and one or more web browser applications. A client device may also contain program instructions to enable a local web-browser, embedded firmware, custom software or other applications running as used for user instructions, graphical display and various application features. One such feature may run program instructions to allow the client device to transport stored input data through one or more local gateways **4050** to upload data via the Internet Network **3000** for subsequent disk storage, such as on the AWS S3 disk storage **4030** or for direct transport through one or more application gateways **4110** provided by one or more Transactional Data Service providers, e.g. AWS.

Raw or clean input data-sets **4004** are uploaded to the Application Back-End computing cluster **4001** by programming instructions under control of at least one network file system (NFS) software layer **4101** running program instructions on at least one Application Computing Cluster **4100**. In the illustrative embodiment, the NFS software layer **4101** is implemented as a Linux NFS distributed file system protocol. The purpose of the NFS Software Layer **4101** is to abstract the operations of the underlying computing cluster in order to service the application software. Specifically, in this embodiment, to assist in the Loading and Transformation process step **2100**, the AG/Satellite input data-sets **4004** are prepared for further processing by additional computing resources such as the Data Science Computing Clusters **4300**.

The Data Science Computing Cluster **4300** runs programming instructions preferably under the direction of another NFS Layer of system software **4301** to assist with the present invention's method of crop optimization. Crop optimization may be defined herein as running a specified sequence of programming instructions on at least one hardware apparatus, such as one or more Data Science Computing Cluster **4300** to train at least one predictive model. This predictive model may subsequently be used to estimate the best Grower operations for increased crop productivity and Farm efficiency. This optimization is used to minimize the plethora of variable environmental and soil conditions and optimize the use of limited applied crop nutrients.

The computing cluster uses certain programming instructions to direct the computing cluster **4300** to read, process and store Localized and Regional Agronomic historical data **4030** in preparation for pre-processing described above and illustrated in blocks **2110** through **2140**.

The computing cluster also treats the input data by pre-processing, normalizing, smoothing, delineating, and indexing to specific geospatial data formats at blocks **6100-**

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6600. After these various treatments, the input data may be further treated through additional novel data processing and ML training methods using interpolation to provide geospatially independent arrays of soil chemistry characteristics.

The interpolated soil characteristics resulting from the programming instructions of blocks **6100-6600** are used as input covariates to train at least one Random Forest yield estimation model at block **5300**. As described above, an ML training method is used to predict crop yield response. The Yield prediction model is further refined by programming block **6600** that is used to perform K-fold cross validation in order to reduce error and further enhance the signal to noise ratio.

Soil and environmental characteristic point data along with other point data such as predicted crop yield are then stored in one or more memory subsystems including but not limited to network attached storage, local storage, or memory, geospatial or other types of databases **4350** for future processing by additional programming instructions. Although typically accomplished by processing in parallel, and for the purpose of process flow illustration, the process extraction of ground type zone (AGT) clustering is the next step to developing one or more predictive crop-field optimizations and/or chemical/seed applications.

The predictive and agronomic input data of block **4350** is further processed by programming instructions running, e.g., in the data science computing cluster **4300**, for the purpose of clustering groups of similar soil and environmental characteristics into crop zones. This process is again based on ML trained models that model crop-yield responsiveness to a plethora of AGT zone characteristics.

At programming step **8400** the crop prediction engine **4000** refines the AGT clusters into at least one Voxel wherein each Voxel represents a defined geographical area representing the multidimensional data-sets are assigned. In the illustrative embodiment, the square area of a Voxel is between 0.1 and 0.25 acres. In some embodiments, a Voxel is the square area having a depth, wherein each Voxel depth layer may be a separate data-set dimensional data. In another embodiment, the data science computing cluster **4300** operates on point data geographically indexed by each transformed geo-coordinate representing a Voxel containing multiple levels of agronomic data-sets.

The crop prediction engine **4000** combines Voxels into Clusters of ground types that represent crop-yield areas that obey similar crop-production responses that are subsequently used for modeling to gain insights on crop-yield optimization and operational efficiency. The number of AGT clusters may be reduced by using a hyper-parameter tuning loop **8110-8410**.

The process continues block **8500** by storing and sending the calculated AGT cluster numbers as assigned to per-Voxel indices to the Application Computing Cluster **4001** for subsequent format conversion from point data to polygonal data representation in preparation for geo-reference graphical 2D and 3D display **9420**.

The Application Back-End **4001** software programming enables the cloud computing hardware **4100** along with the System Software NFS Layers **4101** to send the geographical representation of each Voxel. In the illustrative embodiment, the geographical representation is sent through one or more load balancers **4110**, one or more Internet Networks **3000** and terminating at one or more local gateways **4050** prior to output display on one or more client browsers **4003**. The output displayed may include: input variables, output dependent variables and other predicted crop or farm operational responses **4430**. These output responses **4430** may be dis-

played visually, by text lists or be embedded into downloadable reports or supplier order forms. Calculated and predicted outputs sent to the application computing cluster from block **8500** for mapping and display may include: AGT data-set clusters **4430i**, crop-yield predictions **4430h**, soil carbon predictions **4430f**, ROI on operations and harvest analysis **4430e**, digital elevation and moisture indexes **4430g**, various interactive maps **4430a**, nutrient recommendations **4430b**, seed recommendations **4430c**, digital prescriptions **4430d**, and ETL results for big data repository and data-completeness verification purposes.

Additional programming instructions **11300-11400** may be run by the Data Science Computing Cluster **4300** that operate interactively to observe how various covariate inputs behave as they are applied to the plethora of Random Forest Tree Estimators. The purpose of programming blocks **11300-11400** is to reduce the number of possible observations of yield response based on soil, seed genetics and environmental input characteristics. Without such reduction, the number of observations is too massive and expensive for computing clusters to resolve in a timely manner. This programming code uses interactions between variables with high co-concurrency to integrate-out the non-important variable observations.

The process continues in programming code blocks **11500** through **11600** by the combination of clustered AGT information **8400**, and the observations of co-associated input covariates **11400** to observe and understand the interactions between soil, seed and environmental variables within clustered object boundaries. The analysis further allows interaction observations to determine a series of response surfaces **14000** and subsequently tune by coefficient weighting **14260** to achieve the desired R-squared error and cross check validation **14250** responses. The analysis allows further observation, understanding and projections of crop-performance limiters and crop-performance enhancers as related to soil, seed and environmental characteristics.

The final step performed by the programming software running on the Data Science Computing Cluster **4300** with support from the NFS software **4301** is to use all observations to formulate Reduced Order Surrogate model (RoSM) and subsequently a GAM model to determine the optimum seed and chemical nutrients to apply to achieve the maximum crop performance **14000**. This "Prescription" is then sent to the Application Back-End **4001** for further processing and subsequent downloading via the Transactional Data Transport Services **4002** to the client browser **4003** for final output display and/or printed reports or manufacturer supply orders for seed and nutrient recommendations and purchases.

In one embodiment the output response may be in the form of programmable data-files used to run farm equipment or implements to deliver seeds or nutrients at Voxel level precision to at least one crop-field based on GPS coordinates.

With reference now to FIG. 17, there is shown an exemplary application of the system **16000** as a client browser user interface **4003** displaying a response surface with possible features that result from the combination of applying the present invention to existing hardware computing resources, NFS software and application tools. The client browser user interface **4003** may be a web-browser viewed on a client device. This embodiment illustrates hardware/software programming methods and processes as presented to end users. A subset of the possible product features **4050** achieved using the present inventions novel programming and hardware application base are displayed to the user

through client browser user interface **4003**. This example implementation has the ability to ingest and process Agri-economic input data to achieve insights, predictions and seed/nutrient prescriptions for improved crop-performance and improved crop operation efficiency. Illustrated in the client browser **4003** is a map view of a predicted crop-yield **4430h** response surface that uses land elevation for large scale farming operations. Programming instructions **3100** enable an organizational hierarchy of users to manage warehousing projections, delivery logistics, sales orders and customer recommendations that substantially improve operations, reduce chemical and fertilizer runoff pollution and predict the value of Carbon Credits via direct sequestration that pulls CO₂ from the atmosphere directly into the soil.

The systems and methods presented above optimize crop productivity by accessing certain soil attributes normally not exposed to growers as first and second order crop limiters. By assessing soil health and fertility, chemical balances, seed and hybrid seed choices, biologicals and nutrient levels from large scale farm acreage and applying those data-sets to train the XAI. This enables optimal planting recommendations and yield predictions, which are generated at large scale for specific soil types, and differ from field to field, farm to farm, or even region to region.

The systems and methods presented above are configured to use cost analysis (gathered from manufacturers) in different locations with differently priced products to understand the costs of possible amendment for applications of different combinations of products. For example, the determination of the amounts and rates of possible amendments may be accomplished by comparing the ROI objective function and the predicted yield output from the training model for the quantity and rate of application of amendments recommended by the ML-GAM model. The method may use nutrient models to convert the units from the GAM to actual parts per million (ppm) per acre as required for application by the precision farming equipment. The illustrative embodiment may optimize the amendment application ROI by using a simple gradient descent, which easily converges because of the non-complexity of the GAM in the low order model. For example, to be able to translate between desirable levels of soil nutrients or base saturation and how much fertilizer is suggested to apply, the units of the RoSM model must be mapped to parts per million. Thus, to optimize the estimated ROI, the present method may include creating a GAM derived from ML modeling, optimizing the GAM model for ROI with at least one objective function, and performing gradient descent optimization.

Referring now to FIG. 18 there is shown a high-level flowchart **1800** for optimal sampling with soil stratification. The method is initiated at block **1802**, where soil organic carbon (SOC) is modelled from public sources. More specifically, at block **1802** the crop prediction engine extracts, assesses, or queries variables from publicly available data. Public data may be used to conduct a general survey of the region, specifically known as the project region. By way of example and not of limitation, publicly available data sources may include Soil Grids, gSSURGO, and Polaris. Additionally, sources for landform and/or remote sensing data include Sentinel, USGS, NCEP and USDA and other such sources of public data well known to those in the art.

At block **1804**, the method proceeds to output a first project map with sample points. In the illustrative embodiment, sparse soil samples may be ordered to better understand SOC potentials and model the project region. The additional data of soil samples that contain soil chemistry, soil bulk density, and organic matter along with public and

imagery data may be used to create a rough stratification map used to define project boundaries.

At block **1806**, the method defines project boundaries. More specifically, block **1806** defines one process for the determination of project boundaries. By way of example and not of limitation, the first step in defining project boundaries is to exclude land that is not able to store carbon and thus not suitable for soil sampling (e.g., roads, oil wells, ponds, and buildings). Also, block **1806** may use an iterative weak learning method to improve classification of non-suitable SOC regions.

The method proceeds to block **1808** where estimates of a potential Return on Investment (ROI) for the project region in areas outside the excluded boundaries are provided by the crop prediction engine. By way of example and not of limitation, at block **1808** the crop prediction engine uses binary partitioning to stratify organic matter (OM) and maximally reduce variance in sub-regions. Additionally, block **1808** determines one or more stopping criteria for estimation algorithm recursive partitioning. The estimation is typically based on practicalities concerning land management practices of the region. One major consideration for project design ROI potential incorporates the identification of fields or land parcels that are potentially cost prohibitive for soil sampling. One embodiment may determine the ROI potential by the cost curve of pulling and processing soil (y-axis) versus the OM potential and variability of OM over time and space (x-axis). Block **1808** may further fine-tune the project boundaries based on the ROI analysis and practicalities of land management capabilities. Thus, this analysis may help determine and identify polygons for excluded regions and may set a baseline for Soil Organic Carbon (SOC) potential within one or more SOC projects.

Additionally, block **1808** may include an optional “pre-stratification” step that allows pre-soil-sampling to be done in an “informed” manner. Pre-sampling options augment the ROI determination and may be needed if there is no access to measured data for the region. This optional process step in block **1808** starts by use of the available public data and then applying that data to determine at least one set of stratification parameters as illustrated in FIG. **21** below.

The pre-stratification process step is used to fit the hyper-parameters in order to optimize the predicted ROI using a form of cross validation analysis. The resulting output from this optimization step sets the base areas to define the project boundaries and possible soil sampling regime. Pre-sampling helps determine and inform the machine learning (ML) models of the project boundaries and/or optimal stratification design to achieve a chosen degree of confidence.

Furthermore, block **1808** may also include an optional “pre-sampling design” step, which is conducted by following the sampling simulation method illustrated in FIG. **22** below. The pre-sampling design assists the project developer in deciding the number of samples to take to maximize the ROI potential. The illustrative pre-sampling design step uses measured data from the pre-sampling process to best determine a possible stratification regime.

The method continues with block **1810** where the stratification parameters are determined by the crop prediction engine. More specifically, block **1810** ranks which covariates are important in a particular field or parcel of land by presenting a variety of data parameters, associated with the project area, to one or more computational platforms. The covariates, their interactions, and the varying numbers of strata are tested to determine how best to stratify the project area into sub-areas.

With respect to the optional “pre-stratification” step in block **1808** impacting the stratification parameters of block **1810**, the stratification parameters in block **1810** are determined with a feature selection that uses the interactive Random Forest (iRF) Machine Learning (ML) algorithm to identify which covariates to use based on local feature importance. Alternatively, other machine classification algorithms such as decision trees (DT), neural networks (NN), support vector machines (SVM), naïve Bayes (NB), linear regressions (LR), K-nearest neighbors (K-NN) or other supervised, unsupervised, and reinforcement algorithms known in the art may be used.

More generally, the process step at block **1810**, determines the most important covariates for the project area with the available data including public and/or measured data along with the iRF modeling. The most important variable may be defined as the variable that most impacts the prediction outcome when changed during simulation runs. Stratification will occur around these most important covariates. Examples of covariate stratification parameters include cation-exchange capacity (CEC), calcium, nitrogen, zinc, and pH. Also, common local associations exist where one input variable is highly correlated to another input variable. For example, if CEC is found to be highly correlated with SOC, then CEC may be considered an important covariate in the project area.

Process block **1810** also ranks the covariates in order of importance to fit an individual regression tree, which is also referred to as a “decision tree,” which is described above in further detail. The illustrative process block **1810** applies a stratification regime that follows the same hierarchy of a tree structure. Regression trees for recursive binary partitioning use the covariates to create decision boundaries based on the location and/or distribution of the input covariates that most likely reduces model fitting error. In the illustrative embodiment, for each individual covariate and combination of covariates (the predictors), the method used fits a regression tree against OM (the response). In other embodiments, other covariates may be used against other response variables using the same steps as described above. Numeric split points from each regression tree are used as “learned decision boundaries” to define the strata boundaries.

Block **1810** generates an outcome that includes a list of important covariates and their interactions ranked in order of importance and the defined strata. However, the outcome of block **1810** is dependent on the stratification parameters derived from block **1808**. Thus, the outcome of block **1810** can be determined based on the outcomes of block **1808** that included project sampling (required), pre-sampling (optional) and project design (optional). Furthermore, the different outcomes of block **1808** can be repeatedly applied to block **1810**.

The process step of block **1810** continues by running soil sampling simulations with various input data by deciding the number of desired samples to simulate, in which the number ranges from the minimum number of one sample to a maximum number of samples as defined by the project management and budget. In the illustrative embodiment, the default maximum number of samples is the number that corresponds with 2.5 acres per sample. Note that decreasing the soil sample count will increase the acres per single sample. For example, if N represents the number of soil samples for a fixed number of acres, the area that the number of samples covers is normalized by measuring the number of acres and dividing by the number of soil samples, i.e., number of acres/number of soil samples.

The stratification design illustrated in block **1810** is generated from the feature selections input and the stratification analysis method as outlined above. The outcome may yield a single region, namely, a non-stratified design, in which case sampling will correspond identically to simple random sampling.

The method **1800** then proceeds to block **1812**, in which a Monte Carlo (MC) simulation is used by the crop prediction engine to optimize sampling efficiency with a two-stage design. The first stage selects the strata to sample. The second stage gathers a random sample from the selected strata. During the MC simulation sampling efficiency is continually increased to reduce overall uncertainty. In the illustrative embodiment, the two-stage sampling design at block **1812** is used to reduce error and uncertainty. Reduction of uncertainty is accomplished by use of the two-stage design, which simulates a full sampling experiment for a given stratification design and a sample count.

During the MC simulation, the first stage selects which strata to sample, and the second stage takes a random sample within the selected strata; this step is performed for all prescribed samples. This simulated sampling process is repeated many times in order to estimate the variation of the sampling design. For the illustrative embodiment, a Neyman allocation is used to define the optimal sampling strategy. In other embodiments similar methods may be used to determine the optimal sampling strategy. Optimal allocation is achieved by the application of Neyman allocation, which sets the sampling weights for any particular strata as a function of the size of the strata and strata variability. The Neyman allocation maximizes survey precision by minimizing variability of the overall variation of the mean, specifically by minimizing the amount of variance within each stratum and between each stratum.

The Monte Carlo simulation extracts the simulated sample mean from each of the preferably 1,000's of iterations and uses each sampled mean to naturally form an empirical sampling distribution. For example, running the simulation 5,000 times provides 5,000 sample means that are pooled together to form a single distribution of values.

By contrast, other soil protocols allow for the calculation of uncertainty from a single run. For example, a standard deviation is taken from a limited number of samples, that are estimated to have a Gaussian distribution in standard unit space. With other soil protocols, the project mean is conformed to a normal distribution and/or parametric distribution, which is not an appropriate fit since soil does not always follow homoscedasticity. Homoscedasticity, or homogeneity of variances, is an assumption of equal or similar variances in different groups being compared, which is an important assumption of parametric statistical tests because they are sensitive to any dissimilarities; uneven variances in samples result in biased and skewed test results.

Block **1812** also determines the uncertainty quantification and deduction. The uncertainty quantification defines the margin of error to query the uncertainty bounds. The additional steps performed at block **1812** include calculating the probability that any given run of the simulation will lie outside the error bounds. The portion of these passes and failures generates the second layer of probability, which for illustrative purposes includes nested probabilities and/or conditional probabilities.

By way of example and not of limitation, to calculate the first layer of probability, the project developer chooses the margin of error, which may also be referred to as the "accepted error tolerance." The error tolerance may be defined as the percent deviation of the predictive analysis

from the actual measured soil chemistry. The default standard margin of error is preferably 5% and often stays fixed by definition. In comparison, the certainty of the sampling varies. The uncertainty quantification simulates the number or percent of samples projected to lie outside of the 5% error threshold, which refers to 2.5% on either side of the distribution.

The operations of block **1812** continue by calculating the second layer of probability based on the relationship between margin of error and the probability of the sampling regime achieving the margin of error. In an illustrative embodiment, the SOC project developer chooses the desired degree of confidence and sets the threshold for the probability of exceedance. The probability of exceedance represents how likely it is that the true mean will be within the error tolerance, given the simulation, if sampling were repeated. For example, by choosing a 97% confidence level with a 5% margin of error, there will be a 97% chance that the project mean, e.g., for average SOC content, is within the 5% margin of error-if sampling was repeated. For example, if sampling was repeated 1,000 times, then 3% of the means, e.g., 30 samples, would be outside of the margin of error.

Block **1812** continues by evaluating the MC simulations to determine the optimal sampling design. The optimal sampling design is the point of the regime that meets the degree of desired confidence and also has the lowest sampling density, which provided the lowest cost sampling plan. The optimal outcome having least uncertainty with the smallest number of sample points finds the intersection of the degree of confidence for probability of exceedance and sampling density; this may be represented by spatially graphing the most bottom right intersect to determine the "least-dense" regime that first crosses the chosen error threshold (e.g., 1%, 3%, 5%).

In one embodiment, specific design rules may be used to reduce complexity. The process requires a decision to stratify or not stratify the land parcels or fields depending on what yields the best sampling results. There is the possibility that stratification could emulate simple random sampling. In this scenario, simple random sampling would be sufficient and preferable, as it is the more simplified process. As a rule of thumb for most practical applications, if it is not conclusive which sampling regime to choose, a "default" sampling regime may be utilized, e.g., 1 sample every 5 acres. For example, if two or more stratifications yield the same results, e.g., ROI within one standard error, then block **1812** proceeds with the stratification composed of the fewest strata.

The result of block **1812** is the determination of the optimal number and location of sampling coordinates.

At block **1814**, the sampling coordinates are exported from the crop prediction engine to one or more mobile applications. In the illustrative embodiment, the mobile application is designed for field use by soil collector personnel that are responsible for both locating the modeled optimal sampling points and bagging soil from those sampling locations. The illustrative embodiment exports the sampling location coordinates to a mobile device based on the optional sampling regime from the outcome indicated above. Thus, soil sampling coordinates are uploaded into the mobile application and soil sampling contractors use the mobile application to locate the sampling locations. Additionally, the mobile application assists with tracking the pulling and bagging of soil within each field or parcel of project area.

The method then continues to block **1816** where the soil collection process includes geo-tagging the collected soil samples. In operation, the soil collection contractors are sent

QR code labels that describe all aspects of the sampling activity. Soil collectors are instructed to pull the soil sample and place the soil samples into a soil bag, and then attach the QR code label for the sample at each specified location. The soil collectors then mail the soil sample to a soil lab indicated in the mobile application. Verification of the date, time, and location of the sample is performed by taking a picture of the QR code and uploading the QR code for each collected soil sample. Also, the illustrative mobile application tags images with the date, time, and location coordinates, e.g., GPS coordinates. Additional verification information associated with the tagged image is stored and/or uploaded from the mobile application to a back-end application platform such as the crop prediction engine. By way of example and not of limitation, the verification information may include the collector's username, sampling company, soil lab information, and may include other relevant information used for verification purposes. In the illustrative embodiment, the mobile application alerts the user to take a photo when the mobile device is within proximity of the sample location coordinates. Furthermore, the illustrative mobile application of block 1816 stores images and related information locally, i.e., on the mobile device, and uploads the information to the crop prediction engine described above when the mobile device has an appropriate network connection.

If a proximity-based location alert is not received, or if the soil collector cannot reach the location due to unknown obstruction, the mobile application allows the user to attach a note within the mobile application with an explanation that includes the reasons for the failed collection process. In the illustrative embodiment, the soil collection process works with in-field contractors and soil testing laboratories to seamlessly integrate with all parties via one or more API's that connect all parties.

At block 1818, the soil collection verification information is uploaded to the illustrative crop prediction engine described above. In operation, the mobile application may be used to validate the date and time of collection, and the location using the mobile device GPS functionality. In the illustrative embodiment, the mobile device's camera imager application embeds GPS, date, and time stamp on the image of the QR code attached to the tested soil sample. When an image is captured within proximity of the sample location, the image is uploaded and verified by the illustrative crop prediction engine. The verification process may include validating the information from the uploaded sample image and the QR code with the stored information that generated the location(s) for the initial soil sample and lab testing process. The crop prediction engine may include a back-end verification process that monitors the completion of this verification step 1818 and also may display the status of one or more users accessing the soil system application. Also, the soil collection agency, hired to gather the soil samples may use information embedded on the QR code label or the displayed information in the mobile or desktop application to send the soil samples to one or more labs for chemical analysis. The lab may also test the soil for organic and/or non-organic carbon quantity. After the soil chemistry and SOC are determined the soil sample lab results are uploaded via an API handshake to the illustrative crop prediction engine that is running on one or more back-end system server devices.

With respect to the SOC registries that represent the buyers of carbon credits, the SOC registries require a more rigorous verification process, such as that described in block 1818. More specifically, proof of the soil data location point,

date, and time for aggregated carbon measurements are required to meet their qualification standards.

The method then proceeds to block 1820 where the back-end servers, associated with the crop prediction engine, store the soil results for further analysis. An illustrative crop prediction engine is the CROPFORCE™ cloud-based platform developed and supported by Arva Intelligence, Inc. In the illustrative embodiment, the CROPFORCE™ cloud-based platform is used to analyze the total SOC and accumulated SOC. The illustrative crop prediction engine, i.e., CROPFORCE™, is an application platform service that is delivered as Software-as-a-Service (SaaS) to third-parties that license the application for large-scale SOC measurement, SOC tracking and SOC verification. Additionally, the CROPFORCE™ cloud-based platform provides other environmental agronomic insights for soil health and soil sustainability practices. Furthermore, the illustrative crop prediction engine, i.e., CROPFORCE™, is used to view and/or download lab results or the status for the analyzed soil samples.

At block 1822, the illustrative CROPFORCE™ platform provides further analysis and SOC predictions at scale for follow-on years of potential aggregated carbon in the soil. The illustrative CROPFORCE™ platform includes one or more soil prediction models that can estimate the year over year aggregated carbon. Soil prediction model estimations use many independent variables and may be based on the prior history of fertilizer, climate, crop, land practice and many other such input parameters. The soil prediction models are designed to predict outcomes for farmland, pastureland and/or timberland, termed SOC predictions.

In the illustrative embodiment, the CROPFORCE™ platform uses the "DayCent" model for SOC predictions, although other models such as MEMS, GREET, DNDC and the like may be used in other embodiments. In addition to SOC estimations, the DayCent model outputs other greenhouse gas (GHG) predictions that can be used to estimate the GHG emission reductions for methane and nitrous oxide that are based on certain regenerative practices. The illustrative CROPFORCE™ platform measures and predicts emissions reduction on a year-per-year basis for enterprise clients that manage large agricultural assets such as farms, ranches, and fields across large geographical regions and more granular single farms, ranches, and fields.

The method then proceeds to block 1824 where project sustainability credits are determined by one of the crop prediction engine and the mobile application. More specifically, a project determination for SOC and GHG reduction potentials is completed at block 1824. The project outcomes support payment "for certain regenerative practices" to those who manage the land with soil and environmental soil health in mind. Those involved in the management or ownership of one or more parcels of land can be compensated for changes made to how and what is applied to the soil and how the soil and associated land is managed. The payment mechanism includes the generation and verification of soil carbon accumulation, or carbon equivalents from regenerative and sustainability practices in the form of carbon credits.

At block 1826, carbon credits can be traded on one or more carbon exchanges. Carbon credits are typically measured in tons of CO₂ or CO₂e (carbon equivalents). The practice of CO₂ and other GHG reductions from the air, along with improving soil health and minimizing soil erosion (soil sustainability), may be provided through changes in land management operations and land management practices. Once measured (or approximated by one or more application platform models) and approved by the buyers of

carbon credits, payments can be made to land managers because their changes in traditional land management practices can go a long way to improving soil health, food supply and air quality.

The method **1800** describes a process that analyzes soil strata in one or more soil organic carbon (SOC) projects. In the illustrative embodiment, soil stratification and the associated sampling regime takes a data driven, iterative approach to optimization and identification of input variables most correlated to SOC.

The illustrative crop prediction engine is the CROP-FORCE™ cloud-based platform, which incorporates such input variable most correlated to SOC to directly measure and apply soil covariates, along with public soil survey data, climate data, topography data, satellite imagery, and other relevant information, such as land feature selection, to define and optimize stratification parameters used to optimize the predictive SOC potentials. The use of feature importance is derived as a by-product of iterative Random Forest (iRF) modeling and includes the ability to rank features as possible potential influencers for the soil stratification criteria. The method **1800** takes an empirical approach to test the different stratification regimes such that soil stratification results in a more targeted regime of sample optimization than well-known grid or zone sampling.

The method **1800** utilizes a Neyman allocation to determine soil sampling plans that are based on true data instead of relying primarily on model assumptions. Also, the method **1800** sets and changes the sampling size and tests different regimes against each other to determine the best fit to meet the project developer goals. Furthermore, empirical estimations of uncertainty (in contrast to parametric estimations) are possible.

Previously, a project developer would perform one sampling simulation then fit the number of samples to a Gaussian curve, then multiply the standard deviation of the curve by a constant to create a post hoc uncertainty quantification. The method **1800** provides an empirical sampling distribution from repeating the sampling simulation thousands of times, which reduces uncertainty before the cost and complexity of sampling is conducted. Additionally, the method **1800** generates empirical estimations of uncertainty. Furthermore, the soil stratification and sampling approach of method **1800** is an adaptable procedure that can enhance decision making at each stage of project development including at the initial project design, prior to soil sampling, or after soil sampling. Further still, method **1800** uses pre-stratification methods as outlined herein to reduce uncertainty by augmenting simulation analysis with measured data and enabling the ability to evaluate uncertainty for the overall project output.

Referring now to FIG. **19** there is shown a method **1900** for collecting measured soil data from public and private sources, generally termed “soil data sources.” The method **1900** is initiated at block **1902** where at least one SOC region is defined by one of the crop prediction engine and one or more mobile application; and this process step determines the extent of the SOC project. The SOC region may be determined by contracts from farmers, corporations, or landowners, which are interested in selling SOC credits while keeping soil sustainable for soil health reasons. These “supply sellers” can participate in helping measure and permanently retire CO₂ from the atmosphere into carbon sequestered into the ground. The sequestered carbon from one or more project areas generates carbon supply credits that can be sold, by the supply sellers, on one or more carbon market exchanges.

The method then proceeds to block **1904** where public data is gathered for the project region by one of the crop prediction engine and the one or more mobile application. More specifically, to determine the SOC project region the method **1900** uses publicly available data sources such as SoilGrids, gSSURGo, Polaris, and USGS.

At block **1906**, soil samples that were previously gathered are collected. Note, these collected soil samples are privately measured. In one embodiment, some projects have previously collected soil samples that are used to determine the amount of SOC on one or more plots of land. Some soil samples may also include other measurements such as Organic Matter (OM), Cation Exchange Capacity (CEC) and Soil Bulk Density (BD) to approximate or model the estimated SOC.

At decision diamond **1908**, the crop prediction engine makes a determination regarding the number of soil samples. For illustrative purposes, decision diamond **1908** is directed to determining if the soil samples are “sparse,” which would be less than optimal but sufficient to determine an optimal sampling plan with the soil stratification process described herein.

If there is no soil sample data, the method **1900** proceeds to decision diamond **1910** where SOC calculations are performed by the crop prediction engine. If there are none or insufficient samples to perform the SOC calculation at decision diamond **1910**, the method proceeds to block **1912** where soil samples are ordered for the particular project.

At block **1914**, soil data returned from the collection of soil samples is measured for chemical content, i.e., soil chemistry, by one or more soil labs. Soil chemistry is returned to the application via electronic transmission or paper report form. Soil chemistry data is stored in one or more private database tables and is tagged as “measured” data.

After measuring the collected soil samples, the method returns to decision diamond **1908**, where the crop prediction engine determines if the soil samples are adequate or sparse. If the samples are determined to be “sparse,” additional samples are needed and the method then continues to decision diamond **1910** where the illustrative SOC calculations are performed. The method then proceeds to block **1916** where the locations for the SOC samples are updated and the method then continues to FIG. **20**. Note, privately gathered soil chemistry data is stored in one or more private databases and is tagged as “measured” data.

If soil samples have been collected but are not available or have not been specified in the initial project plan, the method may pass on measured data at decision diamond **1908** and proceed to using public or private data for SOC analysis. In such instances, the machine learning (ML) models may use publicly available data to estimate the project environment variables, e.g., input parameters, to estimate the optimal SOC sampling density and sampling locations.

If sparse samples are to be ordered at block **1912**, the project manager may desire an estimate for both the minimum number of samples to be collected and the sample locations that best represent the mean soil chemistry of the entire plot of land. One benefit of the optimal sampling plan with the soil stratification process described herein is that optimal soil sample collection density and soil sample collection locations can be determined using only public data and without requiring actual measured samples or privately gathered soil chemistry data.

In the event that an initial sampling campaign was developed with the method **1800** described above, the crop

prediction engine makes a determination of whether or not the initial analysis is sufficient to control the uncertainty requirements. This determination can be performed by completing the sampling design with soil sample information and simulating the sampling design using a bootstrap training method. Next the variance is tested with progressively larger “m-out-of-n” bootstrap procedures to ensure that the project is in a soil regime with “n” sufficiently large enough to control the variance and therefore the risk. In one illustrative embodiment, the goal is to ensure that confidence intervals have at least a 97.5% chance of covering the true value.

Referring to FIG. 20, there is shown a flowchart **2000** for determining the valid SOC boundaries and excluded areas that have no SOC potential, which are also referred to also as “project exclusions.” Project exclusions are performed within the project boundaries where there may exist areas with little or no CO₂ absorption potential. These project exclusion areas must be identified and excluded from the SOC potential. Typically, these excluded areas include storage buildings, parking lots, homes, barns, ponds, streams, and the like. Boundary files of excluded areas from within the project must be analyzed and removed for the non-carbon-producing areas. The method **2000** excludes such areas from the total project area by using the summed area of a plethora of individual polygons.

In operation, the method **2000** for determining project exclusions include obtaining land management data at block **2002**. For example, the land under management for the region of interest may be cropland, rangeland, grazeland, timberland or other land with potential SOC generating feedstock.

The method then proceeds to block **2004** where the input variables are used to establish the potential SOC-ROI as a stopping criterion. Block **2004** provides the boundary conditions for the expected minimum and maximum potentials for each project. In one illustrative embodiment, the stopping criteria sets the min/max tons of SOC potential per acre of land or other unit area, such as one or more polygons.

At block **2006**, a stratification boundary is associated with an illustrative SOC area. The stratification boundary is a dynamic boundary until the analysis is completed, at which point, the stratification boundary is fixed. The stratification boundary includes the min/max boundary conditions and an iterative trial and error step that provides an iterative stratification analysis. By way of example and not of limitation, the iterative stratification analysis estimates the potential amount of CO₂ sequestration and CO₂ equivalents (CO₂e) and quantifies GHG emission avoidance for one or more polygons, up to and including the entire project area.

The potential carbon ROI are analyzed by one of the crop prediction engine and the mobile application at block **2008**.

At decision diamond **2010**, the previously calculated stopping criteria is evaluated to determine convergence and also whether the min/max boundary conditions have been satisfied. For example, when the min/max limits are reached, the tuning process that determines the potential SOC per project area is stopped. The decision diamond **2010** determines if the ROI tuning loop is complete and if the optimal ROI is acceptable.

If the stopping criteria requirement has been met prior to a maximum number of iterations, the method continues by the crop prediction engine determining the potential payback for a go-no-go decision with the project and identifies polygons for the SOC potential boundaries at block **2012**. The polygonal boundaries of block **2012** are for project areas that have met the estimated SOC-ROI stopping criteria.

Thus, an illustrative polygon satisfies the SOC requirement, i.e., SOC-ROI stopping criteria.

Now referring to FIG. 21, there is shown a flowchart **2100** for determining the SOC project stratification parameters.

At block **2102**, the crop prediction engine generates a high-resolution voxel grid by interpolating the estimated areas where SOC potential exists. A voxel grid is a three-dimensional grid of values that is organized into layers of rows and columns so that each row, column, and layer intersection in the grid is called a voxel or small 3D cube. Voxel grid properties describe the voxel grid as a whole and these properties include, by way of example and not of limitation, name, number of voxels, number of rows, columns, and layers, and other such information.

Block **2102** integrates any measured data with public or private data and includes multispectral satellite data for the region. The voxel grid is generated by one of the crop prediction engine and the mobile application with data fusion in a supervised framework when and/or where measured soil chemistry data is dense in a semi-supervised framework or a self-supervised framework, in which the measured data was “sparse.” For the sparse instance, soil chemistry measurements are applied to one or more SOC maps. In some embodiments, one SOC maps covers an entire SOC project area. The systems internal SOC map is continuously updated as new data becomes available. In regions where there may be only dozens of soil samples, a learning framework based on iterative Random Forests (iRF) is used. In regions where there are at least one thousand soil samples, the method **2100** may use a feed-forward deep learning framework based on dense, multi-scale networks.

At block **2104**, a set of viable stratification boundaries is generated by one of the crop prediction engine and the mobile application, in which soil stratification is defined, in part, as the process step of grouping soil chemistry types that are present or dominant per specific areas of interest. At block **2106**, a number of strata undergo simulation modeling where the number of strata is determined by looping over a large set of potential values. By way of example and not of limitation, minimal values are selected so that no individual strata have an area that is less than 5 acres.

At block **2108**, a selection criteria for determining the number of strata is used to iterate and identify a plurality of strata within a specific area. The illustrative process step **2108** uses ROI as a default objective function along with other implementation risk calculations, e.g., the percentage of originated tons of carbon that may fail to manifest due to stochastic errors. As a default, the initial amount of acceptable risk is controlled to only 2.5%.

At block **2110** a univariate stratification optimization procedure is performed by the crop prediction engine. By way of example and not of limitation, a Zozak’s random search optimization method is used to create the stratification boundaries from the grid-wise SOC estimates.

At block **2112** the selected number of strata are all stratified by the crop prediction engine until complete. The method **2100** continues to block **2114** where the crop prediction engine identifies and stores all the strata parameters (i.e., stratification boundaries associated with SOC areas) into both containerized code, which is associated with MD5-hashing, and human readable tables to support the conclusions of stratification decisions.

Referring to FIG. 22, there is shown a method **2200** that uses iterative or repeated soil sampling simulations to achieve one form of empirical soil sampling distributions within an SOC project. The illustrative method **2200** starts

at block **2202** as an input range of desired soil samples per some number of acres across the project area up to the entire SOC area. Initially, a broad range of soil sample locations are used, again measured in “acres per sample.” More specifically, as the iterative process begins the simulation resolution range is preferably set to one acre per sample. As the iterative process continues, the number may approach a maximum of 250 acres per sample. The higher end of that range may only be applicable to very large aggregation projects, while the smaller number of acres per sample may be needed for single fields or small groups of fields.

At decision diamond **2204**, a determination is made by the crop prediction engine whether the project has strata or if the project is for non-stratified crop or grazing fields. If the analysis is for a non-stratified crop or grazing field, the method proceeds to block **2206**. For example, in the event that an aggregation is too small to admit stratification, process step **2206** conducts simple random samples (i.e., selects a plurality of random samples) and continues with a simple Monte Carlo simulation. The non-stratified smaller parcel or field produces an outcome identical to random sampling. In the illustrative embodiment, the Monte Carlo simulation is performed without using strata as an input, as described in block **2208**.

If the project has strata, as typical for many fields, farms or larger parcels, the method continues to block **2210** where full strata are used as inputs for the Monte Carlo simulation. The Monte Carlo simulation with the full strata increases the sampling efficiency by at least an order of magnitude as compared to prior art use of simple random sampling for entire projects. The resulting increased sampling efficiency is presented in block **2212**.

The Monte Carlo simulation process described in blocks **2210** to **2224** is initiated by defining the strata sampling weights at block **2214**. In the illustrative embodiment, the strata sampling weight uses Wright’s exact algorithm II. Other algorithms include, but are not limited to, Neyman allocations algorithms, which guarantees the strata proportions are created using integer sample values. Default Neyman allocations result in less guarantee of outcome and often require inexact rounding that may introduce bias.

At block **2216**, strata are selected with a sample count to sample for iterations over all candidate strata and sample combinations.

As the iterative process proceeds to blocks **2218**, **2220** and **2222**, random samples within each stratum are selected by the crop prediction engine. The number of random samples within the stratum is determined by the previous sample count calculation at block **2218**. A random sample is taken at each sampling position centered at the center of each voxel. Each voxel sample is then aggregated with other random voxel samples into a single average value containing the mean of the total SOC within each stratum. At block **2220**, the total SOC is aggregated proportionally to the acreage across strata preferably at a defined depth. The process steps illustrated in **2216** then define what may be termed the “carbon container” or in the case of multiple locations the “carbon containers”. The iterative loop completes when the sample count has reached its maximum sample count value as illustrated in **2222**. In some embodiments, the crop prediction engine determines the sample count is complete for one or more selected strata parameter when the iterative loop completes. At block **2224**, the Monte Carlo simulation process generates an empirical sampling distribution outcome. By way of example and not of limitation, the Monte Carlo process described above may be

repeated thousands of times to determine the variance and SOC quantities within the estimated empirical distribution.

Referring to FIG. **23** there is shown a flowchart **2300** for the uncertainty quantification of the project SOC. Uncertainty quantification (UQ) relies on statistical calculations that are based on the Monte Carlo sampling distribution outcome.

At block **2302**, the uncertainty quantification is initiated by requesting the project manager input and/or define an acceptable margin of error for an estimated SOC and a measured SOC. In the illustrative embodiment, the maximum desired margin of error is within 5% of the actual measured value. Note, the analysis and error limit of 5% would be the maximum difference between samples anywhere within the plot of land, project bounds, parcel, farm, field, or the like.

At block **2304**, the UQ process **2300** calculates the number of samples outside the desired error threshold using one of the crop prediction engine and the mobile application. This is accomplished by running a Monte Carlo simulation described above as block **2224**, which generates an empirical sampling distribution outcome, to determine the empirical distribution of total SOC. Block **2304** determines the percent (%) of time the empirical distribution of total SOC deviates from the mean or another predictive analysis measurement of soil chemistry by more than the targeted margin of error.

At block **2306**, the probability of exceedance is calculated. The probability of exceedance is computed as the left-hand or two-tailed empirical probability, where the calculations should not exceed the previously set maximum limit.

Next, the method **2300** calculates the optimal sampling design plan with the crop prediction engine at block **2308**. To determine the optimal sampling design, simulations per strata regime are evaluated and the sample count using the Monte Carlo simulation previously described above in FIG. **22** are utilized.

The simulations are run for each selected strata count in order to identify and select the optimal number of strata. In some embodiments, the number of strata are within a stratification boundary. The optimal number of strata is determined so that the selected sampling design, having a minimal number of samples, satisfies the target risk criteria or maximizes the SOC-ROI potential.

The optimal sampling design is calculated by repeatedly determining that the confidence level of block **2312** has been satisfied. In one illustrative embodiment, at block **2310**, an evaluation of simulations per strata regime and sample count is performed by the crop prediction engine. Once the confidence level is satisfied at block **2312**, the next process step finds the lowest sampling density at block **2314**. Thus, the process selects the minimal sampling density for the project and assumes that the confidence is within the range previously set. Once the lowest sampling density is determined at block **2314**, the method proceeds to block **2316** where the least dense regime that meets the desired margin is output by one of the crop prediction engine and the mobile application.

At block **2318**, the optimal sampling design is calculated by one of the crop prediction engine and the mobile application using the Monte Carlo simulation described above. The method continues by computing the optimal sampling design. Wherein, the assessed and estimated potential ROI for the carbon project may be used to make a go-no-go decision for project initiation.

Referring to FIG. 24 there is shown a flowchart 2400 for the use of QR codes to validate soil collection locations and dates.

At block 2402, QR code labels are printed for each soil sample. The labels may be sent using the U.S. Postal Service mail or the label may be generated by printing a label at an illustrative soil sample collection company. The QR code has both plain text and encoded information about the location that includes the latitude and longitudinal coordinates, and other information related to the identification of the sample number, customer, soil lab ID, and required soil test for the laboratory. The information encoded in the QR code enables the back-end software for all parties to identify the sample and verify the location of the soil collection process. The illustrative soil collection company has a soil collection technician that gathers samples as specified by the illustrative CROPFORCE™ mobile application.

At block 2404, the illustrative CROPFORCE™ platform is used to send the soil sampling location details to the illustrative CROPFORCE™ mobile application disposed on one or more mobile devices. The illustrative mobile application uses GPS to map the soil technician's current location and that of the desired location for collection of the soil sample.

At block 2406, when the soil technician approaches the GPS location for soil collection, the CROPFORCE™ mobile application engages the technician with on-display information so that the technician can take a picture of the collected soil sample and the corresponding QR code, which is represented by block 2408. The illustrative CROPFORCE™ mobile application verifies that the location of the collected soil samples matches the sample location specified in the mobile application.

At block 2408, the technician is instructed to take an image of the QR code. Additionally, a verification step is performed at block 2408, in which verification that the sample was collected at the proper location is accomplished by embedding into the camera image a location, a date, and a time stamp. In some embodiments, the embedded information includes a GPS component.

At block 2410, the QR code attached to the soil sample baggie is transmitted to one or more networks when the mobile device is in range of the one or more networks. Upon transmission to the one or more networks, the images subsequently transmitted to the CROPFORCE™ cloud platform and related remote storage devices. CROPFORCE™ users can track the progress of soil sampling to ensure that both location and collection are carried out successfully.

At block 2412, the collection agency sends the soil sample to one or more labs for soil chemistry analysis. In the illustrative embodiment, the address of the soil lab used by the technician is both on the QR code label and displayed via the CROPFORCE™ mobile application and the CROPFORCE™ platform.

At block 2414, the soil sample chemistry is determined by the soil laboratory with the results being automatically uploaded to the CROPFORCE™ platform via an API handshake. Alternatively, the lab-analysis report may be sent either by mail or electronically for upload to the CROPFORCE™ platform. In the illustrative embodiment, the soil chemistry includes the soil chemical components such as pH and nutrients, and also measurements such as soil bulk density, OM, CEC, and SOC. The CROPFORCE™ platform stores the soil data, which may be indexed by customer account, location, and sample date.

Once soil chemistry and other soil measurements are returned from the lab and transmitted via application inter-

face software to the CROPFORCE™ modeling engines, the system can begin to calculate carbon aggregation and determine yearly estimations for CO₂ and other GHG emission reductions.

Referring now to FIG. 25 there is shown a flowchart 2500 for estimating SOC project carbon offset credit values. Estimations of project carbon or carbon equivalents are important to determine the potential of certain practices to remove CO₂ and to reduce other GHG emissions due to agricultural or rangeland operations. FIG. 25 shows several illustrative remote data storage devices 2502 through 2510 that communicate with the CROPFORCE™ platform. The illustrative remote data storage devices 2502 through 2510 provide various input sources to the CROPFORCE™ platform prediction models. Additionally, there are front-end components for estimating SOC project carbon offset credit values that include user interfaces, report presentation tools, and predictive presentation results.

In the illustrative embodiment, the DayCent model is used for SOC modelling. DayCent is a daily time series biogeochemical model used in agroecosystems to simulate fluxes of carbon and nitrogen. For example, DayCent model inputs may include daily maximum and/or minimum of air temperature, precipitation, surface soil texture class, and land cover and/or use data. DayCent model outputs may include daily fluxes of various nitrogen gases, e.g., N₂O, NO_x, N₂, daily CO₂ flux from heterotrophic soil respiration, soil organic carbon, soil organic nitrogen, daily water and nitrate, i.e., NO₃, leaching. Alternatively, other mathematical models may also be used to estimate SOC.

The illustrative DayCent model is parameterized by the multiple data sources represented by illustrative remote data storage devices 2502 through 2510. Soil strata parameters associated with block 2114, in FIG. 21, are stored or accessed at remote data storage device 2502. Privately measured soil chemistry data associated with block 1914 in FIG. 19 are stored or accessed at remote data storage device 2504. Remote storage device 2506 stores estimated soil sample data and location data associated with block 2414 in FIG. 24. Public data associated with block 1904 in FIG. 19, which can also be used to supplement the measured data in block 1914, is stored or accessed at remote data storage device 2508.

Project information such as boundaries associated with block 2012 in FIG. 20, in which polygonal bounded project areas that have met the estimated SOC-ROI stopping criteria, may be associated with remote storage device 2510. Additionally, uncertainty bounds associated with calculating the optimal sampling design in block 2318 of FIG. 23 may also be associated with remote storage device 2510.

Furthermore, sample counts derived from block 2222, in FIG. 22, which is related to generating a sampling distribution may also be associated with remote storage device 2510. Further still, sample counts associated with block 2310, in FIG. 23, which relates to simulations that include sample counts may also be associated with remote storage device 2510.

At block 2512, the multiple data sources represented by illustrative remote data storage devices 2502 through 2510, are fed into the illustrative DayCent model to determine the SOC potential. By way of example and not of limitation, the outcome of the DayCent model provides a year-by-year estimation of SOC and GHG projections.

The method 2500 for estimating SOC project carbon offset credit values then continues to block 2514 where SOC reports are automatically generated. The generated SOC reports are used by registries and verification organizations

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to assess the sales potential for SOC projects. The system estimates year-on-year carbon changes and assesses the distance between measured and predicted SOC changes. This difference between predicted and measured enables the validation of SOC accumulation for the buyers of SOC credits.

At block 2516, these credits may be defined as offset credits purchased to offset CO₂ and other emissions produced by certain non-green corporations who buy offsets to justify the emissions from their business operations. The method 2500 enables carbon credit producers such as farmers, ranchers, and timberland owners to estimate the financial value of growing SOC within the project area.

Block 2518 indicates that offset credits may be held by SOC producers and/or traded or sold via one or more carbon exchanges. Thus, given the output of at least one mechanistic model that is appropriately validated with measured data, the method 2500 estimates the number of CO₂ equivalent tons in offsets available for origination by the carbon producers.

It is to be understood that the detailed description of illustrative embodiments is provided for illustrative purposes. Thus, the degree of software modularity for the system and method presented above may evolve to benefit from the improved performance and lower cost of the future hardware components that meet the system and method requirements presented. The scope of the claims is not limited to these specific embodiments or examples. Therefore, various process limitations, elements, details, and uses can differ from those just described, or be expanded on or implemented using technologies not yet commercially viable, and yet still be within the inventive concepts of the present disclosure. The scope of the invention is determined by the following claims and their legal equivalents.

What is claimed is:

1. A method for determining optimal sampling parameters, the method comprising:

generating, at a crop prediction engine, a voxel grid for a soil organic carbon (SOC) area, wherein the voxel grid is associated with a map;

identifying, at the crop prediction engine, a plurality of strata parameters, in which each strata parameter includes a stratification boundary associated with the SOC area;

identifying, at the crop prediction engine, a number of strata within the stratification boundary;

inputting, at the crop prediction engine, a plurality of soil samples for the SOC area;

defining, at the crop prediction engine, a plurality of specifications for a plurality of Monte Carlo simulations including:

one or more strata sampling weights, and
a margin of error;

iterating through the plurality of Monte Carlo simulations by increasing a sample count between each iteration, wherein each of the plurality of Monte Carlo simulations includes:

repeatedly sampling, at the crop prediction engine, wherein each sampling includes:

selecting, at the crop prediction engine, at least one strata parameter from the plurality of strata parameters for the Monte Carlo simulation,

selecting, at the crop prediction engine, a sample count for each of the selected at least one strata parameter,

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selecting, at the crop prediction engine, a plurality of random samples for each selected strata parameter,

aggregating, at the crop prediction engine, the plurality of random samples for each selected strata parameter;

determining, at the crop prediction engine, that the aggregated plurality of random samples achieved the sample count for each selected strata parameter, and

generating, at the crop prediction engine, an empirical sampling distribution for each Monte Carlo simulation; and

determining, at the crop prediction engine, an optimal sampling plan from the empirical distribution generated for each of the plurality of Monte Carlo simulations, wherein the optimal sampling plan has a lowest sample count and satisfies the margin of error.

2. The method of claim 1 further comprising exporting, to a mobile device, one or more sampling locations, wherein the optimal sampling plan includes the one or more sampling locations.

3. The method of claim 2 further comprising validating, at the mobile device, a location, a date, and a time for collecting a soil sample from at least one of the one or more sampling locations with a GPS component.

4. The method of claim 3 further comprising storing a plurality of soil sample results, wherein the soil sample results are accessible by the crop prediction engine.

5. The method of claim 4 further comprising analyzing, at the crop prediction engine, the soil sample results with a soil prediction model to generate one or more SOC predictions.

6. The method of claim 5 further comprising estimating, at the crop prediction engine, a greenhouse gas reduction.

7. The method of claim 6 further comprising converting the greenhouse gas reduction to one or more carbon credits that are traded on a carbon exchange.

8. A method for determining optimal sampling parameters, the method comprising:

identifying, at a crop prediction engine, a soil organic carbon (SOC) project area on a map;

gathering, at the crop prediction engine, a plurality of soil data from a soil data source, wherein the soil data is associated with the SOC project area;

determining, at the crop prediction engine, that the soil data is less than optimal determining, at the crop prediction engine, that the soil data is sufficient to generate an optimal sampling plan;

determining, at the crop prediction engine, a potential SOC per unit area for one or more polygons that are within the SOC project area;

identifying the one or more polygons that satisfy an SOC requirement;

generating, at the crop prediction engine, a voxel grid for the polygon that satisfies the SOC requirement;

defining, at the crop prediction engine, a margin of error;

completing, at the crop prediction engine, a plurality of Monte Carlo simulations, which each generate an empirical sampling distribution; and

determining, at the crop prediction engine, the optimal sampling plan, wherein the optimal sampling plan is associated with one Monte Carlo simulation having a lowest sampling density that satisfies the margin of error.

9. The method of claim 8 wherein after generating the voxel grid for the polygon that satisfies the SOC requirement, the method includes identifying, at the crop prediction

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engine, a plurality of strata parameters, in which each strata parameter includes a stratification boundary associated with the SOC area.

10. The method of claim 9 wherein after identifying the plurality of strata parameters, the method further comprises, 5 identifying, at the crop prediction engine, a number of strata within the stratification boundary; and inputting, at the crop prediction engine, a range of soil samples for the SOC area.

11. The method of claim 10 further comprising, 10 defining, at the crop prediction engine, one or more strata sampling weights for each of the plurality of Monte Carlo simulations; and

selecting, at the crop prediction engine, at least one strata parameter and a sample count from the plurality of strata parameters for each of the plurality of Monte Carlo simulations. 15

12. The method of claim 11 further comprising, selecting, at the crop prediction engine, a plurality of random samples for the selected strata parameter for each of the plurality of Monte Carlo simulations; and 20 aggregating, at the crop prediction engine, the random samples for each selected strata parameter for each of the plurality of Monte Carlo simulations.

13. The method of claim 12 further comprising, determining, at the crop prediction engine, when the sample count is completed for each selected strata parameter for each of the plurality of Monte Carlo simulations; and 25

repeatedly selecting, at the crop prediction engine, random samples and aggregating the random samples until the sample count is completed for each selected strata parameter for each of the plurality of Monte Carlo simulations. 30

14. A system for determining optimal sampling parameters, the system comprising: 35

a map that includes a soil organic carbon (SOC) project area;

a crop prediction engine that gathers a plurality of soil data from a soil data source, wherein the soil data is associated with the SOC project area;

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the crop prediction engine determines that the soil data is less than optimal determining, at the crop prediction engine, that the soil data is sufficient to generate an optimal sampling plan;

the crop prediction engine determines a potential SOC per unit area for one or more polygons that are within the SOC project area;

the crop prediction engine identifies the one or more polygons that satisfy an SOC requirement;

the crop prediction engine generates a voxel grid for the polygon that satisfies the SOC requirement;

the crop prediction engine completes a plurality of Monte Carlo simulations, which each generate an empirical sampling distribution; and

the crop prediction engine determines the optimal sampling plan, wherein the optimal sampling plan is associated with one Monte Carlo simulation having a lowest sampling density that satisfies the margin of error.

15. The system of claim 14 further comprising a mobile device that receives one or more sampling locations exported from the crop prediction engine, wherein the one or more sampling locations are associated with the optimal sampling plan.

16. The system of claim 15 wherein the mobile device validates a location, a date, and a time for collecting a soil sample with a GPS component.

17. The system of claim 16 further comprising a plurality of soil sample results that are stored accessible by the crop prediction engine.

18. The system of claim 17 wherein the crop prediction engine analyzes the soil sample results with a soil prediction model to generate one or more SOC predictions.

19. The system of claim 18 wherein the crop prediction engine estimates a greenhouse gas reduction.

20. The system of claim 19 wherein the crop prediction engine converts the greenhouse gas reduction to one or more carbon credits that are traded on a carbon exchange.

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